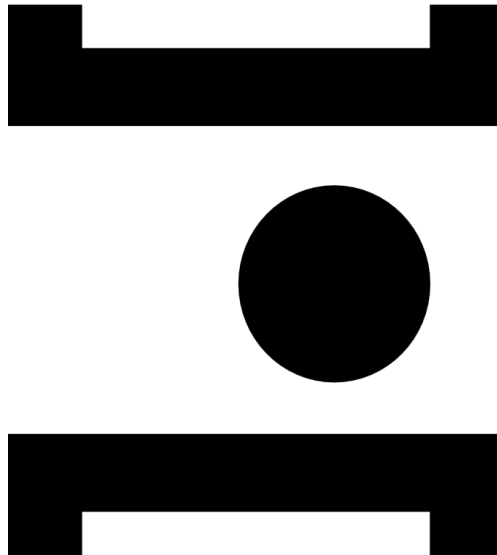


UNIVERSIDADE POLITÉCNICA DE SANTARÉM
Escola Superior de Gestão e Tecnologia de Santarém



**Financial Fraud Detection under Extreme Class Imbalance:
A Leakage-Free Comparative Study of Classical Machine Learning
and Tabular Transformers**

Dissertação

Mestrado em Informática Aplicada

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Declaration on the Use of Generative Artificial Intelligence

Generative artificial intelligence tools were used as assistive technologies during the development and preparation of this dissertation. Google NotebookLM supported the search for and preliminary exploration of academic literature. Anthropic Claude Code and OpenAI Codex assisted with the development, review, debugging, refactoring, and documentation of analysis code; the organisation and review of experimental workflows; and the verification of traceability and internal consistency between saved experimental artefacts and the numerical values, tables, figures, and interpretations reported in the dissertation. These tools also supported the revision of academic English and cross-chapter coherence, LaTeX editing, and the design and refinement of selected TikZ diagrams and aspects of the document layout.

These tools were not treated as scholarly sources, and their outputs were not accepted as scientific evidence. AI assistance did not replace the execution of experiments or the computation of reported metrics. The results presented in this dissertation originate from the saved outputs of the implemented experimental pipeline. Relevant claims, bibliographic references, and interpretations were checked against the original publications or the corresponding project artefacts before inclusion.

All AI-assisted contributions retained in the dissertation or its supporting project, including code, text, analytical suggestions, and visual material, were critically reviewed and revised where necessary by the author. Final decisions concerning the research objectives, methodology, implementation, selection and interpretation of results, and presentation of the work remained with the author, who assumes full responsibility for the accuracy, originality, and academic integrity of this dissertation.

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Deteção de fraude financeira em contexto de desequilíbrio extremo de classes: um estudo comparativo sem fuga de dados entre aprendizagem automática clássica e transformers para dados tabulares

Resumo

Comparam-se seis famílias de modelos em ULB e BAF para detetar fraude sob desequilíbrio extremo, com teste reservado, validação por modelo, sete estratégias de desequilíbrio e quatro regras de limiar. Removem-se perfis duplicados no ULB; a transferência BAF exclui correspondências com dados de desenvolvimento Base. Nas estimativas PR-AUC, LGBM lidera o ULB (0,8205) e CatBoost o BAF Base (0,1796), acima do FT-Transformer (0,1769). Limiares validados melhoram F2 no BAF face a 0,5. Os efeitos dependem dos modelos e dados; SMOTE primário reduz aproximadamente 35% o PR-AUC do FT-Transformer no BAF. LGBM e CatBoost lideram nas cinco variantes filtradas, conforme métrica e corte. SHAP usa os modelos Base congelados: LGBM e CatBoost partilham dez variáveis principais; o conjunto LGBM mantém-se entre variantes. A atenção FT-Transformer é difusa. Resultados descritivos sustentam decisões sensíveis ao limiar, sem demonstrar causalidade arquitetural, ausência completa de fuga ou robustez prospetiva independente.

Palavras-chave

Deteção de fraude financeira; desequilíbrio de classes; avaliação sem fuga de dados; limiares de decisão; aprendizagem automática tabular.

Financial Fraud Detection under Extreme Class Imbalance: A Leakage-Free Comparative Study of Classical Machine Learning and Tabular Transformers

Abstract

Fraud detection under extreme class imbalance demands protected evaluation. This dissertation compares six model families on ULB and BAF using a protected test holdout, model-specific validation, seven imbalance strategies and four threshold rules. ULB predictor duplicates are removed; BAF transfer cohorts exclude Base development matches. LGBM has the highest ULB PR-AUC point estimate (0.8205); CatBoost leads BAF Base (0.1796), ahead of FT-Transformer (0.1769). Validation-selected thresholds improve BAF F2 over fixed 0.5. Imbalance effects depend on model and dataset; primary SMOTE reduces FT-Transformer BAF PR-AUC by approximately 35%. LGBM and CatBoost lead across five filtered BAF Variants, depending on metric and cohort. Pinned SHAP sources describe the frozen BAF baseline models: LGBM and CatBoost share top-feature sets, and LGBM's remains stable across Variants. FT-Transformer attention is diffuse. These descriptive findings support threshold-aware model selection without proving architectural causality, complete leakage exclusion or independent prospective robustness.

Keywords

Financial fraud detection; class imbalance; leakage-free evaluation; decision thresholds; tabular machine learning.

Contents

List of Tables	xv
List of Figures	xvii
Abbreviations	xix
1 Introduction	1
1.1 Context	1
1.2 Motivation	2
1.3 Objectives and Research Questions	2
1.4 Research Approach	3
1.5 Document Structure	5
2 State of the Art	7
2.1 Fraud Detection in Financial Transactions	7
2.1.1 The Financial Fraud Problem	7
2.1.2 Benchmark Datasets	8
2.2 Evaluation of Fraud Detection Systems under Class Imbalance	9
2.2.1 The Accuracy Paradox and Metric Selection	9
2.2.2 Threshold Selection and Probability Calibration	10
2.2.3 Class Imbalance Correction Strategies	10
	ix

2.3	Classical Machine Learning for Fraud Detection	10
2.3.1	Linear Models and Ensemble Methods	10
2.3.2	Gradient-Boosted Decision Trees	11
2.3.3	Cost-Sensitive and Anomaly Detection Approaches	11
2.4	Deep Learning for Fraud Detection	11
2.5	Transformers for Tabular Fraud Detection	12
2.5.1	The Transformer Architecture	12
2.5.2	Tabular Transformer Architectures	13
2.5.3	The GBDT vs. Deep Learning Debate on Tabular Data	13
2.6	Explainable Artificial Intelligence in Fraud Detection	14
2.6.1	Motivation: Compliance, Trust, and Operational Constraints	14
2.6.2	SHAP: Shapley Additive Explanations	15
2.6.3	Interpretability Consistency Across Model Families	15
2.7	Synthesis, Limitations of Existing Work, and Research Gaps	16
2.7.1	Recurring Methodological Limitations	16
2.7.2	Research Gaps and Positioning of This Work	17
3	Theoretical Background	19
3.1	Model Families	19
3.2	Class-Imbalance Handling Strategies	21
3.3	Evaluation Metrics	23
3.4	Decision-Threshold Selection	24
3.5	Hyperparameter Optimisation	25
3.6	Feature Attribution with SHAP	26
3.7	Chapter Summary	27

4	Methodology	29
4.1	Research Objective and Experimental Design	29
4.2	Datasets	32
4.3	Models	34
4.4	Imbalance Handling Strategies	35
4.5	Leakage-Free Evaluation Protocol	36
4.5.1	Outer Split (Holdout Test Set)	36
4.5.2	Inner Validation (Model Selection and Thresholding)	37
4.5.3	Final Training and Test Evaluation	37
4.6	Threshold Selection Study	39
4.7	Hyperparameter Tuning Strategy	39
4.7.1	Classical Models	40
4.7.2	FT-Transformer	40
4.7.3	General Notes	41
4.8	Cross-Domain Generalisation	41
4.9	Evaluation Metrics	45
4.10	Computational Cost and Reproducibility	46
4.11	Interpretability	47
4.12	Summary	49
5	Results and Discussion	51
5.1	Baseline Model Comparison (No Imbalance Strategy)	51
5.1.1	ULB Credit Card 2013	52
5.1.2	BAF Base	53
5.2	Threshold Sensitivity Analysis	55
5.2.1	ULB Credit Card 2013	56

5.2.2	BAF Base	58
5.3	Full Factorial Analysis — Imbalance Strategy $\times$ Model	59
5.3.1	ULB Credit Card 2013	60
5.3.2	BAF Base	62
5.3.3	Interaction Heatmaps	63
5.4	FT-Transformer Sensitivity to Imbalance Interventions	65
5.4.1	Predefined BAF Sensitivity Controls	67
5.5	Cross-Domain Generalisation (BAF Internal)	70
5.6	Interpretability Analysis (SHAP)	72
5.6.1	Global Feature Importance	72
5.6.2	Cross-Model Feature Consistency	75
5.6.3	Cross-Variant Feature Stability	75
5.6.4	Local Explanations	77
5.7	FT-Transformer Attention Diagnostics	78
5.7.1	Aggregate Attention Distribution	79
5.7.2	Individual Case Attention Heatmaps	80
5.8	Consolidated Summary	81
5.8.1	PR Curves — All Models Overlaid (Baseline)	82
5.8.2	Decision Framework	83
5.8.3	Bootstrap Confidence Intervals	84
5.8.4	Computational Cost	85
5.9	Synthesis and Threats to Validity	87
6	Conclusion	91
6.1	Summary of Contributions	91
6.2	Answers to Research Questions	93

6.3	Limitations	95
6.4	Future Work	97
6.5	Closing Remarks	98
A	Appendix	99
A.1	Selected Hyperparameter Configurations	99
A.2	FT-Transformer Attention Diagnostics — Individual Case Bar Charts	102
	Bibliography	109

List of Tables

4.1	Alignment between research questions, experimental operations, and Results sections.	30
4.2	Documented numerical absence codes in the supplied BAF Base partitions, before preprocessing. DEV contains 800 000 applications and TEST 200 000; no actual NaN values occur in these six fields.	33
4.3	BAF transfer cohorts after excluding exact common-predictor matches to Base DEV. Prevalence refers to the retained TEST population; fraud counts in parentheses refer to excluded rows.	43
5.1	Baseline metrics on ULB 2013	52
5.2	Baseline operational metrics on ULB 2013	52
5.3	Baseline metrics on BAF Base	54
5.4	Baseline operational metrics on BAF Base	54
5.5	F_2 by threshold rule on ULB 2013	56
5.6	F_1 by threshold rule on ULB 2013	56
5.7	Recall by threshold rule on ULB 2013	57
5.8	Alert Rate by threshold rule on ULB 2013	57
5.9	F_2 by threshold rule on BAF Base	58
5.10	F_1 by threshold rule on BAF Base	58
5.11	Recall by threshold rule on BAF Base	58
5.12	Alert Rate by threshold rule on BAF Base	59

5.13 PR-AUC factorial on ULB 2013	60
5.14 ROC-AUC factorial on ULB 2013	60
5.15 F_2 factorial on ULB 2013	61
5.16 PR-AUC factorial on BAF Base	62
5.17 ROC-AUC factorial on BAF Base	62
5.18 F_2 factorial on BAF Base	62
5.19 FT-Transformer and CatBoost sensitivity on ULB 2013	66
5.20 FT-Transformer and CatBoost sensitivity on BAF Base	66
5.21 BAF absence-code sensitivity with paired intervals	68
5.22 BAF categorical-sampling sensitivity with paired intervals	69
5.23 Cross-domain PR-AUC on BAF Variants	70
5.24 Cross-domain F_2 on BAF Variants	70
5.25 Cross-domain ROC-AUC on BAF Variants	71
5.26 Cross-model SHAP top-10 Jaccard agreement	75
5.27 Cross-variant SHAP top-10 stability for LGBM	76
5.28 Descriptive decision framework	83
5.29 Bootstrap PR-AUC and ROC-AUC CIs on ULB 2013	84
5.30 Bootstrap PR-AUC and ROC-AUC CIs on BAF Base	84
5.31 Computational cost on ULB 2013	85
5.32 Computational cost on BAF Base	86
A.1 Selected hyperparameters on ULB 2013	100
A.2 Selected hyperparameters on BAF Base	101

List of Figures

4.1	End-to-end experimental pipeline for fraud detection	31
4.2	Leakage-free inner cross-validation protocol	38
4.3	Controlled BAF Base-to-Variant transfer protocol	44
5.1	Baseline PR-AUC by model on ULB 2013	53
5.2	Baseline PR-AUC by model on BAF Base	54
5.3	Δ PR-AUC heatmap on ULB 2013	64
5.4	ΔF_2 heatmap on ULB 2013	64
5.5	Δ PR-AUC heatmap on BAF Base	64
5.6	ΔF_2 heatmap on BAF Base	65
5.7	Top-15 original-feature SHAP importance by model on BAF Base . .	73
5.8	SHAP beeswarm plot for LGBM on BAF Base	74
5.9	Additive local SHAP waterfalls for selected LGBM cases	77
5.10	Mean per-token attention on BAF Base (FT-Transformer)	79
5.11	Attention matrix for a borderline false-positive case	80
5.12	Precision–Recall curves for all models on ULB and BAF	82
A.1	Per-feature attention for false-positive case 1	103
A.2	Per-feature attention for false-positive case 2	104
A.3	Per-feature attention for false-positive case 3	105

A.4	Per-feature attention for false-negative case 1	106
A.5	Per-feature attention for false-negative case 2	107
A.6	Per-feature attention for false-negative case 3	108

Abbreviations

AI Artificial Intelligence

ML Machine Learning

DL Deep Learning

MLP Multi-Layer Perceptron

XAI Explainable Artificial Intelligence

LR Logistic Regression

RF Random Forest

GBDT Gradient-Boosted Decision Tree

LGBM LightGBM — Light Gradient Boosting Machine

CB CatBoost

SVM Support Vector Machine

OCSVM One-Class Support Vector Machine

FT-T FT-Transformer — Feature Tokenizer + Transformer

ULB Université Libre de Bruxelles (Credit Card Fraud dataset, 2013)

BAF Bank Account Fraud dataset suite (NeurIPS 2022)

PCA Principal Component Analysis

AUC Area Under the Curve

PR-AUC Precision-Recall Area Under the Curve (Average Precision)

ROC-AUC Receiver Operating Characteristic Area Under the Curve

TP True Positive

FP False Positive

TN True Negative

FN False Negative

RUS Random Undersampling

ROS Random Oversampling

SMOTE Synthetic Minority Oversampling Technique

SMOTEENN SMOTE with Edited Nearest Neighbours

CV Cross-Validation

HPO Hyperparameter Optimisation

TPE Tree-structured Parzen Estimator

SHAP SHapley Additive exPlanations

LIME Local Interpretable Model-agnostic Explanations

GDPR General Data Protection Regulation

1

Introduction

1.1 Context

Financial fraud in electronic payment systems is a persistent and evolving threat, and the expansion of digital payment channels increases the attack surface [1]. Automated fraud detection systems trained on historical transaction data have therefore become an important component of financial institutions' defence infrastructure.

The machine learning problem underlying fraud detection is characterised by three structural challenges. First, class imbalance is extreme in the evaluated benchmarks: fraud accounts for approximately 0.173% of the raw ULB transactions and approximately 1.1% of BAF Base applications. A majority-only classifier would therefore exceed 98% accuracy without detecting a single fraud, illustrating why accuracy is inadequate in this setting [2]. Second, the distribution of fraud patterns is non-stationary, as fraudsters adapt their behaviour in response to deployed detection systems. Third, the operational context imposes asymmetric error costs — a missed fraud carries direct financial and reputational losses, while excessive false positives burden human investigators and degrade customer experience.

Within this landscape, gradient-boosted decision tree (GBDT) models have demonstrated strong performance on tabular benchmarks [3]. Transformer-based architectures for tabular data [4] have opened the question of whether attention-based models can challenge that position on fraud detection benchmarks. Characterising when the observed performance gap changes, and examining evidence that may help explain it, has direct implications both for practitioners selecting models and for the research community studying the inductive biases of tabular learners. This thesis addresses that question through a controlled, leakage-free experimental study spanning two benchmark datasets, six model families, and a factorial design over

imbalance handling strategies.

1.2 Motivation

Despite the volume of published work on fraud detection, a review of the literature reveals recurring methodological limitations that can undermine reported conclusions and hinder fair comparison between model families.

One consequential problem is data leakage: applying oversampling before the train/test split, or without isolating resampling to the training portion of each cross-validation fold, can inflate reported performance and compromise comparisons [5]. A second limitation is an incomplete operational evaluation: under extreme class imbalance, a high area under the receiver operating characteristic curve (ROC-AUC) can coexist with poor precision–recall performance [6]. Third, operating-point choice must be separated from score ranking and probability calibration, a distinction also examined by Luzio et al. [7]. A fixed threshold of 0.5 is a legitimate reference but need not satisfy the intended operational trade-off; selecting a threshold on TEST instead introduces selection bias. Fourth, conclusions based on one dataset have a narrower evidential scope than conclusions supported across distinct benchmarks.

These limitations can compound: inappropriate metrics, unjustified threshold rules, and leakage can undermine a comparison, while a single dataset limits its scope. The studies reviewed in Chapter 2 provide limited directly comparable evidence for the combined model, dataset, imbalance, and threshold design considered here. The contribution is this documented combination of evaluations and diagnostics, not a new learning algorithm or an exhaustive claim that no related comparison exists. This thesis therefore combines controlled comparisons with post-hoc diagnostics, while treating the latter as descriptive evidence rather than proof of a causal mechanism.

1.3 Objectives and Research Questions

This thesis conducts a controlled, leakage-free comparative study of classical machine learning models and FT-Transformer for financial fraud detection under extreme class imbalance across two structurally distinct benchmark datasets. The work is organised around five research questions.

RQ1 — Comparative performance under a leakage-free protocol. Under a shared leakage-free outer evaluation framework with documented model-specific validation paths, how do classical supervised models (Logistic Regression, Random Forest, LightGBM, CatBoost) and FT-Transformer compare on financial fraud detection, with OCSVM as an unsupervised reference? Does the observed performance gap depend on the dataset?

RQ2 — Impact of threshold selection strategy. How does the choice of decision threshold strategy — fixed at 0.5, maximising F_1 , maximising F_2 , or constrained by a minimum precision target — affect model rankings and operational metrics?

RQ3 — Factorial interaction of imbalance strategies and model type. What is the effect of seven class imbalance handling strategies on model performance across datasets, and how do these effects interact with model family? Do resampling strategies affect GBDT and transformer models differently?

RQ4 — Cross-domain generalisation under distribution shift. Do models trained on BAF Base generalise to the controlled group-size, prevalence, separability, and temporal conditions represented by the BAF Variants without retraining, and which model families retain the strongest performance?

RQ5 — Feature attribution consistency and transformer attention diagnostics. To what extent do the top-feature sets produced by Shapley additive explanations (SHAP) agree across the linear and tree-based models, and does the LGBM top-feature set remain stable across BAF Variants? What descriptive evidence do FT-Transformer attention patterns provide about how the architecture processes the BAF features?

1.4 Research Approach

To address the five research questions, this thesis uses a structured experimental pipeline across two benchmark datasets: the ULB Credit Card Fraud dataset [8, 9] and the Bank Account Fraud (BAF) suite [10, 11]. Six model families are evaluated: Logistic Regression, Random Forest, LightGBM, CatBoost, One-Class SVM, and FT-Transformer. The five supervised models participate in the full factorial comparison of seven class imbalance handling strategies, whereas OCSVM is evaluated separately as an unsupervised anomaly-detection baseline.

1. Introduction

All experiments use a leakage-free protocol in which an outer stratified 80/20 hold-out separates the test set from training, tuning, resampling, and threshold selection. Exact ULB predictor repetitions are removed before partitioning; the resulting evaluation concerns distinct recorded profiles, not necessarily independent real-world entities. The primary ranking metric is average precision, reported as PR-AUC, and threshold selection is treated as a separate experimental variable. The seven named imbalance strategies are evaluated as implemented conditions, not assumed to produce seven distinct training populations: the sampler audit checks whether the Tomek cleaner changes the SMOTE output. Predefined missingness and categorical-aware resampling controls examine representation sensitivity without selecting a policy retrospectively on TEST. OCSVM provides a one-class optimisation reference, but its pipeline is not entirely label-free: labels identify legitimate training observations and support validation-only threshold selection.

Cross-domain generalisation is assessed by applying models trained on BAF Base to test cohorts from the five BAF Variants without retraining or recalibration, after excluding exact common-predictor matches to Base DEV. This filter protects against observed-profile reuse from development data; it does not establish entity-level independence or mutual independence among the test distributions. RQ4 retains the suite’s motivating shift conditions, but its answer is bounded by the shared 30-feature projection: injected separability features are excluded and months are pooled, so the experiment does not isolate the full separability intervention or demonstrate prospective temporal robustness. SHAP analysis uses separately pinned sources for the same frozen BAF Base fitted models used in the predictive comparison: Logistic Regression, LightGBM, and CatBoost [12]. FT-Transformer attention is examined separately on the evaluated BAF Base checkpoint as a descriptive architectural diagnostic. Model-specific preprocessing and validation, and the self-contained transformer implementation, limit any architecture-only interpretation of the comparison; these choices are documented in Chapter 4.

To support transparency and inspection, the research code, configuration metadata, and aggregate results accompanying this study are available in the public project repository: <https://github.com/Olavo200100274/fraud-imbalance-classical-vs-transformers>.

Two companion research articles based on this study have been prepared for journal submission.

The first article, *Tree Ensembles and Tabular Transformers for Financial Fraud Detection: Imbalance Sensitivity, Controlled Transfer and Interpretation Boundaries*, examines imbalance sensitivity, controlled transfer and interpretability. Its dedicated repository is available at <https://github.com/Applied-Intelligence-Hub/transformers-vs-ml-fraud-detection>.

The second article, *Threshold Selection as a Critical Design Choice in Financial Fraud Detection: A Systematic Comparison of Classical Machine Learning Models and Tabular Transformers*, focuses on decision thresholds and alert workload, with its own repository available at <https://github.com/Applied-Intelligence-Hub/tabular-ml-dl-threshold-benchmark>.

These articles share experimental evidence from the thesis rather than representing independent replications.

1.5 Document Structure

The remainder of this thesis is organised as follows. Chapter 2 (p. 7) reviews fraud-detection datasets, evaluation practice, model families, explainability, and the gaps motivating this work. Chapter 3 (p. 19) defines the algorithms, imbalance interventions, metrics, threshold rules, optimisation procedure, and SHAP concepts used later. Chapter 4 (p. 29) specifies the datasets, models, leakage-free protocol, tuning and threshold procedures, BAF transfer study, and interpretability analyses. Chapter 5 (p. 51) presents and discusses the evidence in research-question order, including the consolidated decision framework and threats to validity. Chapter 6 (p. 91) answers the research questions, summarises the contributions and limitations, and identifies future work.

2

State of the Art

The detection of fraud in financial transactions is a challenging problem at the intersection of machine learning, class imbalance, and operational deployment constraints. This chapter surveys five areas that define the present study: the fraud detection problem and its benchmark datasets, evaluation under extreme class imbalance, classical machine learning, deep learning and transformer-based models for tabular data, and explainability. It concludes by synthesising recurring methodological limitations and identifying the research gaps that motivate the experimental design.

2.1 Fraud Detection in Financial Transactions

2.1.1 The Financial Fraud Problem

Financial fraud in electronic payment systems causes substantial economic and operational harm as digital payment channels expand [1]. The wider financial-fraud and financial-crime literature covers behaviours such as card-not-present fraud, account takeover, synthetic identity fraud, and money laundering, with distinct detection problems [13]. Common machine learning challenges include extreme class imbalance, concept drift as adversaries adapt, and differing error consequences: missed fraud can incur financial loss, whereas false alerts create investigative workload and customer friction [14]. Their relative monetary cost is context-dependent and is not measured by the present benchmarks.

Rule-based expert systems were historically the dominant detection paradigm, offering interpretability and direct encoding of domain knowledge at the cost of continuing manual maintenance and limited generalisation to novel attack patterns [1]. Supervised machine learning uses historical labelled transactions to model feature

interactions and can be updated as new labelled data become available without requiring every decision rule to be specified explicitly. Its practical limitation is data availability: fraud labels are costly to produce, may arrive with delay, and the underlying distribution shifts as fraudsters respond to deployed models [15]. These constraints motivate the use of benchmark datasets to standardise evaluation and the assessment of model behaviour under distribution shift.

2.1.2 Benchmark Datasets

Reproducible comparative studies benefit from publicly available datasets with documented characteristics. The following two benchmark collections are central to this thesis.

The raw **ULB Credit Card Fraud dataset** [8, 9] comprises 284 807 transactions made by European cardholders over two days in September 2013, with 492 fraudulent transactions (fraud rate 0.1727%). Sensitive features were transformed through principal component analysis (PCA), yielding 28 anonymised numerical components (V1–V28) alongside the elapsed **Time** and transaction **Amount**. The published representation limits disclosure of the original variables while retaining a numerical feature space for modelling. Its principal limitations are temporal and semantic: the two-day window constrains temporal generalisation studies, and the transformed features limit domain-specific interpretation and feature engineering. Despite these constraints, ULB remains a widely used benchmark in academic fraud detection research [14, 16]. The distinct-profile population used in this thesis is defined by the pre-split integrity check in Section 4.2.

The **Bank Account Fraud (BAF)** dataset suite [10, 11], introduced at NeurIPS 2022, provides a synthetic yet realistic alternative designed to overcome several limitations of ULB. BAF Base contains one million bank account opening applications with thirty modelling features — twenty-five numerical and five categorical (`payment_type`, `employment_status`, `housing_status`, `source`, `device_os`) — and a fraud rate of approximately 1.1%. Unlike ULB, BAF preserves semantically meaningful feature names, enabling interpretation of model decisions in domain terms. The suite includes five controlled variants: Variant I increases group-size disparity while equalising group prevalence; Variant II increases group-wise prevalence disparity while equalising group sizes; Variant III introduces group-wise separability differences through two synthetic features; Variant IV changes prevalence disparity

over time; and Variant V changes separability over time through the same synthetic features. This structure supports evaluation under controlled bias and distributional conditions without requiring access to production environments.

These two datasets span different problem regimes. ULB is a dense, low-dimensional binary classification problem with extreme imbalance in a temporally constrained snapshot. BAF is a large-scale, semantically rich multi-variant benchmark designed to expose robustness limitations. Evaluating the same model families with a shared outer holdout and reporting protocol across both datasets reduces dependence on any single data regime, as argued in the context of multi-dataset fraud benchmarks [17].

2.2 Evaluation of Fraud Detection Systems under Class Imbalance

2.2.1 The Accuracy Paradox and Metric Selection

Under extreme class imbalance, a classifier that predicts every transaction as legitimate can achieve high accuracy without detecting a single fraud. For example, the positive-class prevalence is approximately 0.173% in raw ULB and approximately 1.1% in BAF Base. This *accuracy paradox* makes overall accuracy alone an inadequate performance summary and motivates the use of metrics that directly measure performance on the minority class [2].

ROC-AUC is widely reported in the fraud detection literature, yet under extreme imbalance a small false-positive rate can still correspond to a large absolute number of false alerts [6]. ROC-AUC summarises ranking across all operating points; it does not directly report alert precision or the workload at the selected threshold. It is therefore informative as a ranking measure but insufficient on its own as an operational summary; formal metric definitions are provided in Section 3.3.

The precision–recall curve directly represents the trade-off between precision and recall for the positive class, making its summary area an appropriate primary measure for imbalanced classification [6]. Its no-skill precision level is class prevalence, whereas the no-skill ROC-AUC reference is 0.5 regardless of prevalence. Average precision, the PR summary used here, is therefore interpreted alongside the evaluated population’s prevalence and complemented by the operating-point measures defined in Section 3.3.

2.2.2 Threshold Selection and Probability Calibration

Threshold-dependent metrics require an explicit operating rule. Selecting τ on TEST introduces selection bias, whereas retaining a predefined default does not itself leak information but may give an unsuitable precision–recall trade-off. Luzio et al. [7] examine the separation between classifier calibration and business-level decisions; under-sampling can also distort posterior probabilities and complicate cost-based interpretation [18]. These distinctions motivate validation-only threshold selection and separate assessment of score ranking, probability calibration, and operational decisions. Section 3.4 defines the threshold rules used in this thesis; their systematic comparison across datasets and model families remains an identified research gap.

2.2.3 Class Imbalance Correction Strategies

The literature covers data-level resampling, including SMOTE [19] and cleaning extensions such as SMOTE+Tomek and SMOTEENN [20], as well as algorithm-level remedies such as class weighting. Survey evidence [2, 21] cautions that no single strategy consistently dominates across datasets and model families. A critical implementation constraint is that resampling must remain within the training portion of each validation fold: applying it before the split introduces leakage that can inflate reported performance [5]. Section 3.2 defines the interventions; the State-of-the-Art implication is that their interaction with model family and input representation must be evaluated empirically.

2.3 Classical Machine Learning for Fraud Detection

2.3.1 Linear Models and Ensemble Methods

Logistic Regression remains a standard, interpretable reference in binary classification studies, while Random Forest provides a non-linear bagged-tree baseline; both continue to appear in comparative imbalanced-learning benchmarks [22]. Their inclusion establishes reference points of different capacity without assuming calibration or distribution-shift robustness. Section 3.1 gives the corresponding algorithmic definitions.

2.3.2 Gradient-Boosted Decision Trees

Gradient-boosted decision trees (GBDTs) have repeatedly achieved strong performance on tabular data [3]. LightGBM [23] and CatBoost [24] represent distinct efficient and categorical-aware implementations, respectively, and are relevant to BAF fields such as `device_os` and `employment_status`. Their mechanisms are defined in Section 3.1; here the important literature finding is their persistent competitiveness against tabular deep learning.

In Leevy et al.’s ULB comparison, CatBoost has the highest mean AUPRC among the five evaluated binary classifiers [22]; that study does not evaluate LightGBM. Singh et al. [25] consider a broader ensemble comparison that includes LightGBM and highlights XGBoost and CatBoost in its reported results. Tree ensembles’ practical characteristics include limited sensitivity to feature scale, implementation-level support for missing values, and efficient exact attribution under the TreeSHAP formulation [26]. Probability calibration remains data- and configuration-dependent and should be evaluated rather than assumed.

2.3.3 Cost-Sensitive and Anomaly Detection Approaches

Cost-sensitive learning [27] offers an alternative to resampling by directly incorporating the asymmetric misclassification costs into the model’s objective function. Rather than modifying the training data distribution, cost-sensitive methods specify a loss matrix and optimise expected cost rather than expected error rate. This approach avoids the distributional distortion introduced by oversampling and is particularly natural for fraud detection, where the cost ratio between false negatives and false positives can be quantified from operational data.

One-class classification treats fraud detection as an anomaly-detection problem and is attractive when labelled fraud examples are scarce. OCSVM provides a principled support-estimation formulation for this setting [28], but, when reliable labels are available, supervised methods can exploit information that a one-class objective deliberately excludes [22, 29]. Its mechanism is defined in Section 3.1.

2.4 Deep Learning for Fraud Detection

Deep learning methods have been applied to fraud detection across a range of architectures, with uneven results. A recent review [16] surveys the deployment of

convolutional networks, recurrent models, and autoencoders across credit card fraud benchmarks.

Fully connected networks (MLPs) offer greater nonlinear capacity than logistic regression but require regularisation and hyperparameter tuning to control overfitting on imbalanced datasets. Unlike decision trees, they do not encode split-based inductive biases that often suit heterogeneous tabular variables [30].

Sequence models — LSTMs and GRUs — are natural when transactions form meaningful entity-level sequences with state-dependent fraud patterns. ULB provides an elapsed timestamp but no cardholder identifiers from which such sequences can be reconstructed, whereas BAF represents individual account-opening applications rather than transaction streams. Recurrent models are therefore not a natural match for the structures available in these two benchmarks.

Autoencoder-based anomaly detection trains a reconstruction network on legitimate transactions, deriving anomaly scores from reconstruction error. This objective differs from directly optimising supervised classification, so relative performance must be established under the specified evaluation conditions rather than inferred from the model designation. A challenge for deep learning approaches in fraud detection is sensitivity to extreme class imbalance. Both GBDT and neural implementations can incorporate loss weighting, while data-level resampling changes the training distribution for either family. The interaction between model family and imbalance strategy is therefore treated as an empirical variable in this thesis rather than assumed in advance.

2.5 Transformers for Tabular Fraud Detection

2.5.1 The Transformer Architecture

The Transformer architecture [31] demonstrated sequence modelling without recurrence. Its success in language and vision motivated extensions to tabular data, despite the absence of an inherent sequential or spatial ordering among features; Section 3.1 provides the self-attention definition used in this thesis.

2.5.2 Tabular Transformer Architectures

TabTransformer [32] was among the first applications of self-attention to tabular data. Its design applies column-wise attention to categorical feature embeddings, while numerical features bypass the attention blocks and are concatenated with the contextual embeddings before prediction. Consequently, numerical features — all predictors in ULB and most predictors in BAF — receive no attention-based contextual transformation.

SAINT (Self-Attention and Intersample Attention Transformer) [33] extends attention along two axes: column-wise attention across features within a row, and row-wise attention across samples within a batch. The dual attention mechanism is designed to capture inter-sample dependencies, which may be relevant in fraud detection when groups of transactions share coordinated patterns. SAINT reports competitive performance against GBDT models on several tabular benchmarks, though results are inconsistent across datasets and tasks.

FT-Transformer (Feature Tokenizer + Transformer) [4] extends attention to both categorical and numerical feature tokens and motivates the tabular transformer used in this thesis. The evaluated model is a self-contained PyTorch implementation inspired by this design, not the authors' official implementation; its specific choices are documented in Section 4.3. Subsequent work reports gains from richer numerical embeddings on some predominantly numerical datasets [34]. Whether this flexibility provides an advantage over the inductive bias of GBDTs on fraud data remains an empirical question rather than an architectural certainty.

2.5.3 The GBDT vs. Deep Learning Debate on Tabular Data

The empirical comparison of deep learning and GBDTs on tabular data motivates the transformer-GBDT question but does not always test transformers directly. Shwartz-Ziv and Armon [3] compare XGBoost with TabNet, NODE, DNF-Net, and a 1D-CNN across eleven datasets; XGBoost is generally competitive or stronger, and combined ensembles can improve performance. That benchmark does not evaluate FT-Transformer and does not establish a self-attention-specific explanation. Borisov et al. [30] survey a broader range of tabular deep learning methods, including transformers, and document the importance of dataset and evaluation conditions

rather than a universally superior model family.

In a published financial-fraud conference study, Thimonier et al. [29] report higher LightGBM performance than the evaluated anomaly-detection baselines on credit card fraud data. On BAF, Sun et al. [35] examine objective-function choices for LR, SVM, RF, LightGBM, and GRU, illustrating that training objectives and operating-point criteria matter in a model comparison. Their study does not evaluate tabular transformers and is not direct evidence about FT-Transformer’s relative performance.

Several hypotheses may explain why GBDTs often retain an advantage on financial tabular data. Their split-based inductive bias can represent irregular decision boundaries efficiently, whereas attention-based models must estimate a larger set of continuous parameters and may require more data and regularisation. Probability calibration can also differ across model families and datasets, although it is not guaranteed by either architecture. Finally, synthetic oversampling may interact differently with tree partitioning and gradient-based optimisation, particularly when numerical and categorical variables are represented jointly. These are plausible mechanisms to be evaluated cautiously; aggregate performance alone cannot identify which mechanism is responsible.

Whether these theoretical considerations translate into consistent empirical gaps on fraud-specific benchmarks is an open question that this thesis addresses empirically. The accompanying diagnostics are used to characterise fitted-model behaviour, not to identify a causal mechanism from aggregate performance alone.

2.6 Explainable Artificial Intelligence in Fraud Detection

2.6.1 Motivation: Compliance, Trust, and Operational Constraints

The deployment of machine learning systems in financial decision-making is increasingly subject to transparency, governance, and accountability expectations. Regulations such as the European General Data Protection Regulation (GDPR), together with national banking supervisory frameworks, motivate documented oversight of automated decisions, although the precise legal obligations depend on the decision context and system use [36]. Beyond regulatory compliance, operational teams benefit from explanations when auditing model decisions, identifying systematic errors,

and maintaining justified confidence in automated systems.

The trade-off between predictive accuracy and interpretability is well-documented [37]. Logistic regression and shallow decision trees permit relatively direct inspection of coefficients or splits, but may be less expressive than ensemble and deep models. Post-hoc explanation methods address this tension by providing local or global attributions of black-box model decisions without requiring inspection of model internals.

2.6.2 SHAP: Shapley Additive Explanations

SHAP [12] is a widely adopted post-hoc explanation framework in fraud detection research. TreeSHAP [26] provides an efficient tree-specific formulation, whereas neural explanations are commonly approximate and more implementation-dependent. This asymmetry matters when comparing explanations across architectures; the formal attribution framework and explainer distinctions are defined in Section 3.6.

Attention and SHAP answer different questions in this study. Attention coefficients describe token mixing within a particular transformer layer, whereas SHAP assigns additive output contributions under an explicit coalition-value formulation [12]. The head-averaged, last-layer [CLS] attention reported here is therefore labelled as a descriptive architectural diagnostic, not computed or interpreted as a Shapley attribution or a causal explanation.

2.6.3 Interpretability Consistency Across Model Families

Whether different model families identify the same features as important — cross-model interpretability consistency — is understudied in the fraud detection literature. Walauskis and Khoshgoftaar [38] apply SHAP-based feature selection to large-scale tabular data but do not compare SHAP ranking consistency across model types. Models with similar aggregate PR-AUC may still generate different feature attributions; such disagreement is relevant evidence for model governance, although attribution agreement alone cannot establish that either model has learned the correct reasoning process.

LIME [39] provides an alternative post-hoc explanation approach: for a given instance, it fits a locally linear approximation to the model’s predictions in a neighbourhood of that instance. LIME attributions are approximate by construction

and can vary with the sampled neighbourhood. SHAP is commonly selected when additive attributions and a consistent comparison framework are desired, but its interpretation still depends on the explainer, background distribution, and feature-dependence assumptions [37].

2.7 Synthesis, Limitations of Existing Work, and Research Gaps

2.7.1 Recurring Methodological Limitations

The literature reviewed in this chapter motivates four methodological concerns that can compromise the reliability or scope of a model comparison.

Data leakage via incorrect resampling. A consequential error is the application of oversampling before the train/test split, or within cross-validation without isolating resampling to the training portion of each fold. Hayat and Magnier [5] demonstrate performance inflation under problematic preprocessing and evaluation practices. Such leakage compromises the validity of a comparison even when its reported metrics appear strong.

Incomplete evaluation summaries. Using ROC-AUC as the sole primary metric, as detailed in Section 2.2, leaves alert precision and operational workload unspecified under extreme class imbalance [6]. Studies that report only ROC-AUC or accuracy therefore provide an incomplete characterisation of fraud detection capability; this does not make ROC-AUC itself an invalid ranking measure.

Insufficiently specified operating points. A predefined $\tau = 0.5$ is a valid evaluation rule but may not match an operational precision–recall target or error-cost ratio. Optimising τ on TEST instead introduces selection bias. Probability calibration is a separate issue when scores are interpreted as posterior risks or linked to explicit costs. A reproducible comparison should therefore distinguish the score model, calibration assumptions, and validation-only operating-point selection rather than treating all threshold choices as interchangeable.

Single-dataset studies without robustness assessment. An evaluation restricted to ULB cannot establish performance on different financial-fraud populations. Its two-day temporal window, PCA-transformed features, and specific imbalance ratio limit the scope of conclusions drawn from it. BAF provides a distinct

application-level benchmark and controlled Variant conditions, although its shared synthetic generating process does not substitute for validation on independent production datasets.

2.7.2 Research Gaps and Positioning of This Work

The preceding limitations identify specific gaps that this thesis addresses through six experimental contributions.

Gap 1: Limited joint, leakage-free comparison of GBDTs and tabular transformers on fraud data. The reviewed literature provides limited directly comparable evidence for the joint ULB–BAF model and validation design considered here. The tabular benchmarks of Shwartz-Ziv and Armon [3] and Borisov et al. [30] suggest GBDTs remain competitive, but fraud datasets differ structurally from general tabular benchmarks through extreme imbalance, regulated use, and behavioural and device features. The present contribution is a documented benchmark comparison, not a claim to introduce a new model or to establish priority over every related study.

Gap 2: Limited directly comparable threshold-rule studies. Systematic comparisons of fixed, $\max\text{-}F_1$, $\max\text{-}F_2$, and precision-constrained rules across the same model families and both benchmark datasets are limited in the reviewed studies. Evaluating these rules on saved validation and TEST scores separates operating-point choice from refitting.

Gap 3: Factorial interactions between imbalance strategy and model type are understudied. The reviewed literature does not provide a common evidence base for the full combination of seven selected strategies, five supervised models, and ULB–BAF datasets examined here. The factorial design documents interactions within the implemented preprocessing and sampling paths; it cannot isolate architecture from input representation.

Gap 4: Cross-domain generalisation on BAF variants is insufficiently characterised across model families. Understanding how GBDTs, Random Forest, linear, one-class, and transformer models respond to the BAF suite’s controlled conditions motivates a transfer comparison. The present evaluation is narrower: exact matches with Base DEV are excluded, the shared predictor projection omits the injected separability columns, and pooled months do not constitute a prospective temporal test. These limits bound the answer to RQ4 rather than being treated as evidence

of unrestricted robustness.

Gap 5: SHAP consistency across compatible model explainers and dataset variants is insufficiently documented. Whether linear and tree-based models that achieve similar PR-AUC identify overlapping top-feature sets, and whether the Base-trained LGBM set remains stable across BAF Variant test distributions, is insufficiently documented.

Gap 6: FT-Transformer attention patterns on fraud data are insufficiently characterised. In particular, the extent to which the model's last-layer attention is concentrated or diffuse across BAF feature tokens has received limited analysis. Attention is treated here as a descriptive architectural diagnostic, not as a faithful substitute for feature attribution.

Chapter 3 (p. 19) defines the concepts used by the six experimental contributions presented in Chapter 4 (p. 29) to address these gaps.

3

Theoretical Background

The State of the Art chapter (Chapter 2) surveyed the *literature landscape* around fraud detection under extreme class imbalance: which datasets are standard, which evaluation pitfalls recur, which debates remain open. The Methodology chapter (Chapter 4) will describe how the present study was actually conducted: which datasets, which splits, which hyperparameter budget, which leakage-free protocol. Sitting between the two, this chapter introduces the *theoretical toolkit* that the methodology draws on. Each tool — learning algorithm, imbalance intervention, evaluation metric, threshold rule, optimiser, attribution estimator — is defined here once, in general terms, so that the Methodology chapter can focus exclusively on project-specific design choices and refer back to this chapter whenever a definition or mechanism needs to be recalled.

The exposition deliberately stops short of comparative analysis. Which combinations perform best on the present datasets is an empirical question answered in the Results and Discussion chapter (Chapter 5). The narrower purpose here is to provide a self-contained reference for the terminology and mechanisms used in the remainder of the thesis.

3.1 Model Families

The study covers four supervised approaches — a linear classifier, bagged trees, gradient-boosted trees, and an attention-based deep network — together with a one-class anomaly-detection baseline. This section recalls their defining mechanisms; the rationale for selecting these families is given in the Methodology chapter.

Logistic Regression. Logistic Regression is the canonical linear probabilistic classifier. It models the log-odds of the positive class as a linear combination of

3. Theoretical Background

the input features, $\log \frac{P(y=1|\mathbf{x})}{P(y=0|\mathbf{x})} = \mathbf{w}^\top \mathbf{x} + b$, and recovers a probability through the sigmoid link $\sigma(z) = (1 + e^{-z})^{-1}$. Parameters are typically learned by minimising the regularised cross-entropy loss; the elastic-net penalty $\lambda(\alpha \|\mathbf{w}\|_1 + (1 - \alpha) \|\mathbf{w}\|_2^2/2)$ combines ℓ_1 and ℓ_2 regularisation and acts as a unified sparsity/shrinkage control. Logistic Regression is interpretable through its coefficients, can be well calibrated when the modelling assumptions are adequate, and serves as a simple parametric reference against which more flexible models are compared.

Random Forest. A Random Forest is a bagged ensemble of decision trees. Each tree is fitted on a bootstrap sample of the training data and, at every split, considers only a random subset of features; the ensemble prediction is the average of the per-tree class probabilities. The combination of bootstrap aggregation and feature subsampling is designed to reduce correlation among trees and lower ensemble variance relative to a single tree. Tree splits are also relatively insensitive to monotonic feature scaling.

Gradient-boosted decision trees. Gradient boosting builds an additive ensemble $F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \nu h_m(\mathbf{x})$ in which each new weak learner h_m is fitted to the negative gradient of a chosen loss with respect to the current prediction, scaled by a learning rate ν . Two modern implementations are used in this work. **LightGBM** [23] adopts a *leaf-wise* (best-first) tree growth strategy, which splits the single leaf with the largest loss reduction at each step rather than expanding the tree level by level, and bins continuous features into discrete histograms so that split evaluation becomes a constant-time operation per bin. **CatBoost** [24] introduces *ordered boosting*, which computes the residual for each training example using a model constructed without that example, thereby reducing the prediction-shift bias that can arise when the same data are used to compute gradients and fit the next learner. CatBoost also handles categorical features natively via permutation-aware target statistics, removing the need for one-hot encoding of high-cardinality categorical inputs. These are capabilities of the algorithm family, not evidence that every option is active in a particular implementation. Section 4.3 documents the preprocessing and boosting configuration actually used in this thesis.

One-Class SVM. The One-Class Support Vector Machine [28] is an unsupervised anomaly-detection model. Given training examples drawn predominantly from a single class, it maps the observations into a kernel feature space and learns a

maximum-margin hyperplane that separates most training observations from the origin. Test points on the anomalous side of the boundary receive a negative decision-function value. Its optimisation does not use a supervised fraud-versus-legitimate loss. This distinction does not make every one-class pipeline label-free: in this study, labels identify legitimate training observations and are used for validation-only threshold selection. OCSVM is therefore a reference for a different learning objective, not a controlled experiment in which all access to labels has been removed.

FT-Transformer. The FT-Transformer (Feature Tokenizer + Transformer) [4] adapts the transformer architecture [31] to tabular inputs. Each input feature — numerical or categorical — is first embedded as a d -dimensional token through a feature-specific tokeniser (numerical features are passed through a learned linear-plus-bias map; each categorical level has its own embedding vector). A learnable [CLS] token is prepended to the sequence of feature tokens, and the resulting sequence of length $T=n_{\text{features}}+1$ is processed by a stack of transformer blocks. Each block applies multi-head scaled dot-product self-attention, $\text{Attn}(Q,K,V) = \text{softmax}(QK^\top/\sqrt{d_k})V$, followed by a position-wise feed-forward network and residual connections with layer normalisation. The final embedding of the [CLS] token is passed to a linear classification head. Self-attention provides every feature with a soft, learned weighting over every other feature, providing a mechanism for modelling feature interactions. This paragraph describes the general design; the self-contained implementation evaluated here uses its own PyTorch encoder, token-position bias, and regularisation choices, detailed in Section 4.3. Performance of that implementation does not establish the performance of every FT-Transformer implementation.

3.2 Class-Imbalance Handling Strategies

Under extreme class imbalance, the minority class contributes relatively few examples to the empirical training distribution, and minimising an unmodified loss can favour the majority class. The literature on imbalance handling [2, 21] distinguishes *data-level* interventions that modify the empirical training distribution, *algorithm-level* interventions that modify the loss. Both are compared with an unmodified baseline.

Random under- and over-sampling. *Random Under-Sampling* (RUS) removes majority examples uniformly at random until a target ratio is reached. *Random Over-Sampling* (ROS) duplicates minority examples uniformly at random until the same ratio is reached. RUS discards information and is therefore most useful when the majority class is large enough that the resulting training set still provides a meaningful empirical estimate; ROS does not discard information but, by duplicating exact examples, can amplify overfitting to the specific minority realisations present in the training set.

SMOTE and its variants. The Synthetic Minority Oversampling Technique [19] generates synthetic minority examples by interpolation in feature space: for each minority point $\mathbf{x}_i$, one of its k nearest minority neighbours $\mathbf{x}_j$ is selected at random and a new example $\mathbf{x}_i + \lambda(\mathbf{x}_j - \mathbf{x}_i)$ is created with $\lambda \sim \mathcal{U}(0,1)$. Two cleaning-augmented variants [20] post-process the SMOTE output to reduce noise near the decision boundary: *SMOTE+Tomek Links* identifies pairs of mutual nearest neighbours from opposite classes and removes one or both members according to its cleaning policy. *SMOTEENN* applies an Edited Nearest Neighbour rule, with the removal criterion and eligible classes depending on the configured voting and sampling policies. How many examples the cleaning step removes depends on the geometry of the resampled training data. A cleaning stage is not guaranteed to remove any samples. Equal downstream metrics for a cleaning variant and plain SMOTE do not, by themselves, reveal whether the resampled training sets were identical. Ordinary SMOTE's interpolation also does not preserve categorical membership when applied to ordinal codes or one-hot columns; categorical-aware controls are distinguished from the primary sampling paths in Section 4.4.

Cost-sensitive learning. The cost-sensitive (or *class weighting*) family modifies the training objective rather than the training data. A common balanced weighting rule assigns weights inversely proportional to class frequency, increasing the relative contribution of minority-class errors. The empirical training distribution is left unchanged, which avoids the distributional distortions introduced by oversampling, but the model is encouraged to allocate capacity to the minority class through the reweighted gradient.

3.3 Evaluation Metrics

Under extreme class imbalance, the choice of evaluation metric is non-trivial: several widely used scalars are dominated by the majority class and become nearly insensitive to the minority. This section defines the metric family used throughout this thesis.

Confusion-matrix primitives. For a binary classifier and a fixed decision rule, the confusion matrix has four counts: true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). Precision and recall are defined as

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}.$$

Neither expression contains the true-negative count. Precision quantifies the fraction of alerts that are fraudulent, whereas recall quantifies the fraction of fraudulent cases detected. They therefore expose positive-class behaviour that overall accuracy can obscure under extreme imbalance.

The F_β -score. The F_β -score is the weighted harmonic mean of precision and recall,

$$F_\beta = (1 + \beta^2) \frac{\text{Precision} \cdot \text{Recall}}{\beta^2 \text{Precision} + \text{Recall}},$$

with β controlling the relative weight assigned to recall: $\beta=1$ recovers the symmetric F_1 , while $\beta=2$ (F_2) assigns recall four times the weight of precision through the β^2 term. In fraud detection F_2 provides a recall-sensitive summary when missed fraud is given greater importance than a false alert. The value $\beta=2$ is a reporting choice rather than a direct monetary cost model.

PR-AUC and ROC-AUC. The precision–recall (PR) curve traces the trade-off between precision and recall as the decision threshold sweeps the full range of model scores. This thesis reports average precision under the conventional label PR-AUC as a threshold-independent summary. The receiver operating characteristic (ROC) curve traces the true-positive rate against the false-positive rate over the same threshold sweep, and the area under it is ROC-AUC. ROC-AUC is invariant to prevalence when the class-conditional score distributions are fixed. Under extreme imbalance, however, a small false-positive rate can still correspond to many false alerts, and the ROC curve does not expose precision directly. The no-skill precision

level in PR space is positive prevalence π , and a constant-score classifier has average precision equal to π . An empirical random ordering need not produce exactly that average precision in a finite sample. PR-AUC summarises positive-class ranking and is therefore the primary ranking metric used here, but its dependence on prevalence must be considered when comparing different test populations.

Calibration. A probabilistic classifier is well calibrated if its output scores match empirical conditional positive frequencies: among examples assigned a score of, say, 0.05, the fraction that are fraudulent should be approximately 0.05. The Brier score $\frac{1}{N} \sum_i (\hat{p}_i - y_i)^2$ is the mean squared error of the predicted probability against the binary label and is a standard proper scoring rule for probabilistic accuracy; lower is better. Because it reflects both calibration and resolution, it is not a pure calibration measure. Reliability curves assess calibration more directly by plotting empirical positive frequency against predicted score within score bins.

3.4 Decision-Threshold Selection

A classifier produces a score $s(\mathbf{x})$ for each input; for probabilistic models this score is commonly in $[0,1]$, whereas an anomaly detector may produce an uncalibrated decision value. Converting a score into a binary decision requires a *threshold rule* that maps scores to predictions, the canonical form being $\hat{y} = \mathbb{1}\{s(\mathbf{x}) \geq \tau\}$. The choice of τ determines where on the precision–recall curve the system operates, and under extreme imbalance different defensible threshold rules can materially alter threshold-dependent metrics. Several classes of threshold rule appear in the literature.

Fixed default ($\tau=0.5$). The most common choice in software defaults is $\tau=0.5$. For a calibrated posterior probability, this threshold corresponds to a decision rule with equal false-positive and false-negative costs; it does not imply equal class prevalence. Under asymmetric operational costs or model-dependent score calibration, 0.5 need not be an appropriate operating point. Validation data should therefore be used to determine whether the default achieves the required precision–recall trade-off.

Validation-optimal F_β thresholds. A second family selects τ by maximising a chosen F_β score on held-out validation data, $\tau^* = \arg \max_\tau F_\beta(\tau)$. Setting $\beta=1$

gives precision and recall equal weight; $\beta=2$ assigns greater weight to recall and is suitable when missed positives receive greater emphasis. The selected value does not, by itself, encode a monetary cost ratio.

Precision-constrained thresholds. A third family selects τ to enforce a lower bound on validation precision, $\tau^* = \min\{\tau : \text{Precision}(\tau) \geq P_{\min}\}$. The resulting operating point targets a minimum positive predictive value and therefore influences the fraction of false alerts. It does not directly fix the number of alerts, and the realised precision may differ on future data. If no validation operating point meets the target, the constraint is infeasible; issuing no alerts does not demonstrate attainment of the required precision.

Alert-budget thresholds. A fourth family targets the top- k scoring instances using a quantile of the validation score distribution, which amounts to fixing the operational alert volume independently of any performance criterion. This rule represents settings in which investigation throughput is constrained by a fixed review capacity rather than by a target precision. Ties at the boundary require an explicit tie-breaking policy to obtain exactly k alerts; a score threshold alone may flag more than k . Applying a validation-derived quantile to another score distribution does not guarantee the same alert volume. This conceptual rule is distinct from the four rules compared in RQ2.

To remain leakage-free, any of these rules must be computed exclusively on data selected without access to the test set. The model-specific validation procedures used to select τ and apply it to the held-out test set are described in Section 4.6 (p. 39) of the Methodology chapter.

3.5 Hyperparameter Optimisation

Hyperparameter optimisation (HPO) selects values for the parameters that control a learning algorithm but are not learned from data — regularisation strength, tree depth, number of attention heads, learning rate, and so on. Brute-force grid search becomes expensive in high-dimensional configuration spaces, while random search does not use information from earlier trials. Modern HPO frameworks can instead adopt *sequential model-based optimisation* (SMBO), in which a probabilistic surrogate model of the objective is updated after every completed trial and the

next configuration is selected to maximise an acquisition criterion such as expected improvement.

Tree-structured Parzen Estimator. The Tree-structured Parzen Estimator (TPE), used as the default sampler in Optuna [40], is one realisation of SMBO. Rather than modelling the objective $p(y|\mathbf{x})$ directly, TPE models the two conditional distributions, in a minimisation convention, $p(\mathbf{x}|y < y^*)$ and $p(\mathbf{x}|y \geq y^*)$ of "good" and "bad" configurations (split at a quantile y^* of the observed objective values; the favourable direction is reversed for maximisation) and proposes the configuration that maximises the ratio of these two densities. This heuristic concentrates exploration in regions of the configuration space where the surrogate estimates that good outcomes are more likely than bad ones.

Pruning of unpromising trials. Many HPO objectives produce intermediate values during a single trial — a cross-validation fold's metric, an epoch's validation loss — which a *pruner* can compare against the trajectory of completed trials. A median pruner in a maximisation study compares a trial's best intermediate value so far with the median of completed trials' values at the same step, subject to its startup and warm-up conditions. A value below that reference can terminate the trial, allowing the optimiser to spend less computation on unpromising regions. Pruning reduces the cost of the search but can disadvantage configurations whose intermediate performance improves slowly.

3.6 Feature Attribution with SHAP

SHAP (*SHapley Additive exPlanations*) [12] attributes a model's prediction $f(\mathbf{x})$ on a specific instance to its input features by treating the features as cooperating players in a coalitional game whose payoff is the prediction itself. The SHAP value ϕ_j assigned to feature j is the cooperative-game-theoretic Shapley value of that feature: the average marginal contribution of feature j to the prediction across all possible orderings in which features can be revealed to the model,

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} \left[f_{S \cup \{j\}}(\mathbf{x}_{S \cup \{j\}}) - f_S(\mathbf{x}_S) \right].$$

Within the additive explanation framework, the formulation is characterised by three properties: *local accuracy* (the attributions sum to the prediction relative to

a baseline), *missingness* (a feature absent from the simplified explanation input receives zero attribution), and *consistency* (an attribution cannot decrease when that feature's marginal contribution increases across coalitions). These properties provide a common vocabulary for model explanation. Missingness in this sense does not require zero attribution for a CSV missing value or an absence code such as a BAF sentinel. Comparisons across models still depend on compatible output scales, background data, feature grouping, and explainer assumptions.

Architecture-specific explainers. Naive model-agnostic enumeration evaluates the coalition-value function over all $2^{|F|}$ feature subsets, which is intractable for non-trivial feature counts. Architecture-specific explainers exploit model structure to make the computation efficient. *LinearExplainer* uses the linear model form and a specified background distribution. *TreeSHAP* [26] computes exact values for its tree-specific SHAP formulation in polynomial time. *GradientExplainer* approximates expected gradients for differentiable models using a background sample; its output depends on both the approximation and the selected background data.

Comparing attributions across models. Once per-feature SHAP values are available, a model's global feature importance is summarised as the mean absolute SHAP value across a held-out sample. Two models can then be compared on the agreement between their top- k feature sets through the *Jaccard similarity* $|A \cap B|/|A \cup B|$, which equals one when the two sets coincide and zero when they are disjoint. In this thesis, Jaccard agreement is used only to compare membership of the top- k global feature sets. It does not compare rank order, attribution magnitude, local explanations, or whether different models have learned the same decision process.

3.7 Chapter Summary

This chapter has defined the model families, imbalance interventions, evaluation measures, threshold rules, optimisation procedure, and SHAP quantities used in the empirical study. Chapter 4 now translates these general concepts into the project-specific datasets, preprocessing paths, validation procedures, and analysis plan.

4

Methodology

This chapter describes the experimental methodology adopted in the thesis. Chapter 3 (p. 19) defines the learning algorithms, imbalance interventions, evaluation metrics, threshold rules, hyperparameter optimisation procedure, and feature-attribution estimators. The present chapter explains how those elements were configured and applied in this study.

Section 4.1 links the research questions to the overall experimental design. Sections 4.2–4.3 introduce the datasets and models, while Sections 4.4–4.5 describe the imbalance interventions and leakage-free evaluation protocol. Sections 4.6–4.7 specify threshold selection and hyperparameter optimisation. Section 4.8 defines the BAF transfer experiment, and Sections 4.9–4.11 document the reported metrics, reproducibility provisions, and interpretability analyses.

4.1 Research Objective and Experimental Design

This thesis investigates fraud detection in financial transactions under extreme class imbalance. Its five research questions are operationalised through six complementary contributions; RQ5 is divided into a SHAP consistency analysis and a separate FT-Transformer attention diagnostic:

1. **Unified model comparison.** Compare classical machine learning models and a transformer-based tabular model across distinct fraud datasets using a shared leakage-free outer holdout, common metrics, and model-appropriate validation procedures.
2. **Threshold selection as a first-class experimental variable.** Examine how the decision-threshold rule affects model rankings and operational metrics using validation-only selection.

3. **Full factorial analysis of imbalance handling.** Evaluate seven imbalance strategies $\times$ five supervised model families $\times$ two datasets in a structured design, documenting how each intervention changes the observed metrics.
4. **Cross-domain generalisation under controlled BAF conditions.** Assess robustness by training on BAF Base and evaluating, without retraining, on disjoint test cohorts from the five BAF Variants under the common-feature projection.
5. **SHAP feature-set consistency.** Compare top-feature sets from architecture-compatible SHAP explainers for Logistic Regression, LightGBM, and CatBoost, and assess the stability of the Base-trained LGBM top-feature set across BAF Variant test distributions.
6. **FT-Transformer attention diagnostics.** Characterise the concentration of last-layer, head-averaged [CLS] attention across BAF feature tokens as a descriptive diagnostic that is separate from SHAP feature attribution.

Table 4.1 shows how the five research questions are operationalised and where their evidence is reported.

Table 4.1: Alignment between research questions, experimental operations, and Results sections.

RQ	Experimental operationalisation	Primary evidence
RQ1	Baseline comparison of the six model families under the shared outer holdout and model-specific inner validation.	Section 5.1
RQ2	Comparison of a fixed threshold with $\max-F_1$, $\max-F_2$, and precision-constrained thresholds selected within DEV.	Section 5.2
RQ3	Factorial comparison of seven imbalance strategies across five supervised models and two datasets, followed by a focused FT-Transformer–CatBoost comparison.	Sections 5.3 and 5.4
RQ4	Evaluation of fixed BAF Base-trained models and thresholds on Variant test distributions without retraining.	Section 5.5
RQ5	SHAP top-feature-set comparisons for compatible explainers and a separate descriptive FT-Transformer attention diagnostic.	Sections 5.6 and 5.7

Figure 4.1 provides a high-level overview of the end-to-end experimental pipeline.

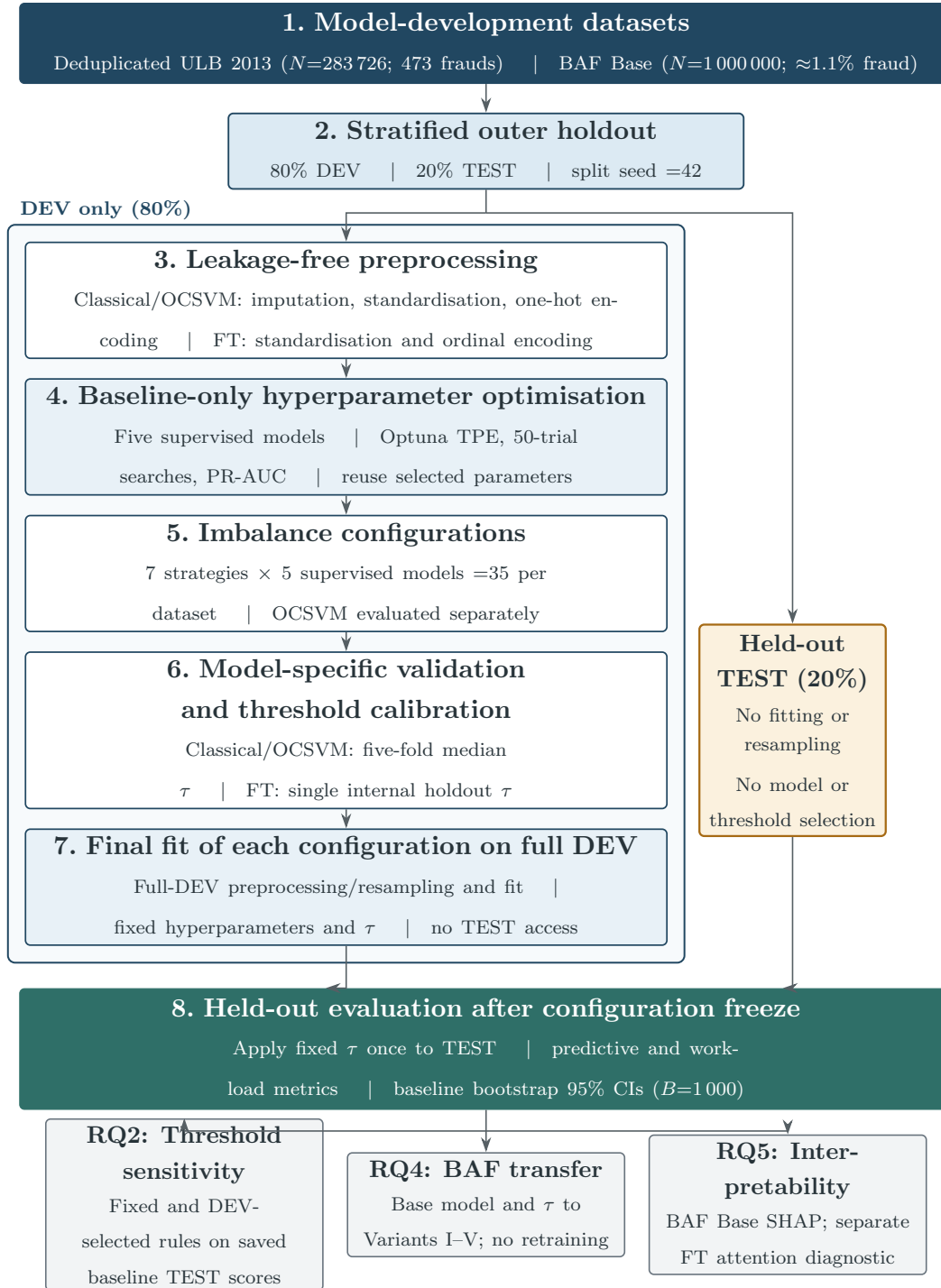


Figure 4.1: Leakage-free experimental pipeline. Model development and threshold calibration remain within DEV, and TEST is accessed only after configuration freeze. The factorial study covers 35 supervised configurations per dataset plus OCSVM; downstream analyses examine thresholds, BAF transfer, and interpretability.

4.2 Datasets

Credit Card Fraud Detection 2013 (ULB). The raw ULB dataset [8,9] contains 284 807 European credit card transactions, including 492 frauds (0.1727%), with anonymised features (PCA components V1–V28, plus **Time** and **Amount**) and a binary target indicating fraud. Exact duplicates across all 30 predictors are removed before any partition is created. The first occurrence in the original file is retained, and a predictor profile associated with conflicting labels would cause the procedure to fail rather than be assigned an arbitrary label. In the supplied file, this removes 1 081 repeated rows, including 19 fraud-labelled rows, leaving 283 726 profiles and 473 frauds. This deterministic integrity check does not select profiles using model performance. Original row indices are retained for auditing but never used as predictors.

Bank Account Fraud (BAF) Suite (NeurIPS 2022). The BAF suite [10,11] contains a Base dataset and five variants (Variant I–V) designed to stress-test models under controlled bias and distributional conditions. Each dataset comprises $\approx 1\,000\,000$ instances with mixed numerical and categorical variables (5 categorical features: `payment_type`, `employment_status`, `housing_status`, `source`, `device_os`) and a binary label `fraud_bool` ($\approx 1.1\%$ fraud in Base). The raw Base file contains 32 columns, including the target and the `month` field. The pipeline excludes both fields from the predictors, yielding 30 modelling features. Variants III and V additionally contain `x1` and `x2`; these synthetic columns are also excluded so that all Base-to-Variant evaluations use the same 30-feature input.

Cross-dataset compatibility note. The ULB and BAF datasets have incompatible feature spaces (PCA-anonymised vs. semantic features). Therefore, cross-dataset transfer between ULB and BAF is not feasible. Cross-domain generalisation is studied *within* the BAF suite, where a shared modelling feature space enables Base-trained models to be evaluated on each controlled variant.

BAF absence codes and primary preprocessing policy. The official datasheet [41] defines `-1` as an absence code for `prev_address_months_count`, `current_address_months_count`, `bank_months_count`, `session_length_in_minutes`, and the device distinct-email count (`device_distinct_emails_8w` in the supplied CSV). It also identifies negative `intended_balcon_amount` values as

absent information. These are numerical codes, not CSV NaN values. The primary experiment preserves them: mean imputation handles actual NaN values only, and standardisation includes the codes in the training-partition mean and variance. Thus a tree may separate absence through a split, whereas a linear or neural model receives a coded numerical magnitude. This is an explicit representation choice, not recovery of an unobserved measurement. Other negative values, including those in `credit_risk_score` and `velocity_6h`, are retained as measurements rather than indiscriminately recoded.

Predefined missing-value sensitivity analysis. A supplementary BAF baseline comparison converts only the documented absence codes to NaN, imputes numerical means fitted on each training partition, adds missingness indicators, and then standardises. It holds the original split and baseline hyperparameters fixed and selects thresholds within DEV; OCSVM retains its fixed parameter configuration. For FT-Transformer, both representations use the same revised validation and training schedule. Paired TEST comparisons quantify the consequences of this preprocessing intervention. The primary policy is specified in advance and is not replaced by whichever policy obtains the higher TEST score; the sensitivity analysis is not a second hyperparameter search.

Table 4.2 reports the observed code counts before preprocessing. For example, the previous-address field contains an absence code in 71.32% of DEV applications and the initial-transfer field in 74.23%. Their prevalence makes the representation policy a substantive modelling assumption rather than a minor cleaning detail.

Table 4.2: Documented numerical absence codes in the supplied BAF Base partitions, before preprocessing. DEV contains 800 000 applications and TEST 200 000; no actual NaN values occur in these six fields.

CSV field	Code	DEV count	TEST count
<code>prev_address_months_count</code>	-1	570 525	142 395
<code>current_address_months_count</code>	-1	3 391	863
<code>bank_months_count</code>	-1	202 604	51 031
<code>session_length_in_minutes</code>	-1	1 623	392
<code>device_distinct_emails_8w</code>	-1	290	69
<code>intended_balcon_amount</code>	< 0	593 868	148 655

4.3 Models

Six model families are evaluated: five supervised classifiers — Logistic Regression (LR), Random Forest (RF), LightGBM (LGBM) [23], CatBoost (CB) [24], and the Feature Tokenizer Transformer (FT-Transformer) [4] — together with One-Class SVM (OCSVM) as an unsupervised anomaly-detection baseline. Definitions of each architecture are given in Section 3.1.

The selection covers four supervised approaches — linear classification (LR), bagged trees (RF), gradient-boosted trees (LGBM and CB), and an attention-based deep network (FT-Transformer) — together with the OCSVM reference. The two gradient-boosted implementations are retained separately: LightGBM provides a computationally efficient tabular model, whereas CatBoost uses a distinct boosting procedure. OCSVM uses a one-class fitting objective and is therefore excluded from the imbalance-strategy and class-weighting comparisons; labels nevertheless define its legitimate fitting subset and calibrate its validation thresholds, so the complete evaluation pipeline is not label-free. Including FT-Transformer alongside the four classical supervised families allows their observed responses to the implemented imbalance interventions to be compared without assuming that any difference is inherent to the architecture.

Implementation scope. The neural model is a self-contained feature-tokeniser transformer inspired by Gorishniy et al. [4], implemented with PyTorch `Transformer Encoder`; it is not the authors’ official `rtdl` implementation. It uses per-feature numerical linear embeddings, categorical embeddings, a learnable `[CLS]` token and token-position bias, pre-normalised encoder blocks, and a normalised linear logit head. The implementation and its regularisation choices, rather than every FT-Transformer implementation, are the object of comparison.

Model-specific preprocessing. The classical models and OCSVM use a shared `ColumnTransformer`: numerical variables are mean-imputed and standardised, while categorical variables are mode-imputed and one-hot encoded. Consequently, although CatBoost supports native categorical processing in general, that capability is not used in this experiment. FT-Transformer follows a separate input path in which numerical variables are mean-imputed and standardised and categorical variables are ordinally encoded as integer embedding indices. Each preprocessor is fitted only on the relevant training partition.

4.4 Imbalance Handling Strategies

To address extreme class imbalance, seven strategies are evaluated for the five supervised model families (LR, RF, LGBM, CB, and FT-Transformer): an unmodified baseline (**None**); random undersampling (**RUS**) and random oversampling (**ROS**); **SMOTE** [19] and its two cleaning-augmented variants, **SMOTE+Tomek Links** and **SMOTEENN** [20]; and cost-sensitive **Class Weights**. The selection covers undersampling, oversampling, and cost-sensitive adjustment [2, 21], together with the unmodified reference needed to measure changes associated with each intervention. Section 3.2 defines the algorithms.

OCSVM is evaluated separately (no resampling or class weighting), given its one-class anomaly-detection paradigm. The full factorial design yields 7 strategies $\times$ 5 supervised models = 35 configurations per dataset (plus OCSVM), enabling a structured analysis of the *strategy* $\times$ *model* $\times$ *dataset* interaction.

Configured interventions. RUS reduces the majority population to the minority count; ROS and the SMOTE stage increase the minority population to the majority count. The target ratio is therefore 1:1 before any subsequent cleaning. All samplers use seed 42, and SMOTE uses five minority neighbours. The composite defaults clean both classes: Tomek removes both members of detected links, while ENN uses three neighbours and retains a point only when all three agree with its class. Cleaning can change the final ratio. Class weighting uses inverse-frequency balanced weights for LR, RF, and LGBM and `auto_class_weights=Balanced` for CatBoost. FT-Transformer uses binary cross-entropy with positive weight $N_{\text{legitimate}}/N_{\text{fraud}}$, computed on the relevant fitting partition. No class-weight configuration resamples the training data.

Resampling representation. The primary samplers operate after model-specific preprocessing. For the classical models, SMOTE therefore interpolates the standardised numerical variables and the one-hot-encoded categorical columns. For FT-Transformer, the sampler receives concatenated standardised numerical values and ordinal category codes; synthetic categorical values are subsequently converted to integer embedding indices. Neither primary path uses a categorical-aware sampler. The resulting SMOTE comparisons describe the behaviour of this implemented pipeline and cannot isolate an architecture-only effect from the representation used during synthetic resampling. Supplementary, fixed-hyperparameter

SMOTENC controls for BAF CatBoost and FT-Transformer preserve membership of the five nominal categorical features during synthesis: CatBoost uses one-hot encoding after this categorical-aware sampling, while FT-Transformer uses the resulting integer codes directly. Binary indicators stored as numerical features remain in the interpolated numerical block, so these controls do not impose all possible discrete-domain constraints. The controls are reported separately from the predefined seven-strategy factorial matrix.

For recorded validation and final-DEV resampling stages, sampler audits retain class counts and hashes before and after resampling. For SMOTE+Tomek and SMOTEENN, these records also identify the post-SMOTE population and cleaning removals. An identical SMOTE and SMOTE+Tomek score does not establish a dispatch error: if the configured Tomek stage removes no samples, the two fitted training populations may be identical. Equality is assessed using saved diagnostics and explicitly labelled sampler reconstructions; reconstruction does not certify unrecorded fold behaviour. The sampling rule is not changed after inspecting TEST to force different results.

4.5 Leakage-Free Evaluation Protocol

A central methodological concern in fraud detection is preventing data leakage: information from the test set must never influence model selection, preprocessing, resampling, or threshold calibration. This section formalises the leakage-free protocol adopted throughout this thesis. Figure 4.2 illustrates one fold and the subsequent aggregation across all five folds.

4.5.1 Outer Split (Holdout Test Set)

For each dataset, an outer stratified split creates an 80% development partition (DEV) and a 20% holdout test partition (TEST), with split seed 42. On deduplicated ULB this gives 226 980 DEV profiles (378 frauds) and 56 746 TEST profiles (95 frauds). Exact predictor overlap is verified to be zero across the outer split, all five classical inner folds, and the FT-Transformer internal holdout. BAF Base retains 800 000 DEV and 200 000 TEST applications; its outer partitions contain no exact common-predictor overlap. The BAF split pools months and is not a prospective chronological evaluation: unlike the temporal recommendation in the dataset documentation [41], it does not train exclusively on earlier months and test on later

months. TEST is not used to fit preprocessing, resample data, select hyperparameters, or select decision thresholds. Post-hoc analyses that use TEST are conducted only after the predictive evaluation has been fixed. Exact profile disjointness is an observable integrity condition, not proof of independence between unobserved entities.

4.5.2 Inner Validation (Model Selection and Thresholding)

For the classical supervised models and OCSVM, selection occurs exclusively within the DEV partition via stratified k -fold cross-validation ($k = 5$). For each fold:

1. Fit the preprocessor on the fold's training subset only.
2. Apply the imbalance strategy only on the fold's training subset.
3. Fit the model.
4. Evaluate on the fold's validation subset: compute PR-AUC and obtain predicted scores.
5. Select a decision threshold τ by maximising the chosen criterion (see Section 4.6) on the validation subset.
6. Using τ , compute F_1 , F_2 , and confusion matrix counts $\{TN, FP, FN, TP\}$.

For OCSVM, the model is fitted only on legitimate examples from each training fold, while the corresponding validation fold retains both classes for threshold selection and evaluation. FT-Transformer replaces five-fold cross-validation with one stratified 80/20 holdout inside DEV. Its preprocessing, early stopping, hyperparameter selection, and F_2 threshold selection use only this internal training/validation split. This model-specific difference reduces computational cost but means that fold dispersion and median thresholds are available only for the classical models and OCSVM.

4.5.3 Final Training and Test Evaluation

For each model-strategy-dataset configuration, the final model is fitted on the full DEV partition. For the classical models and OCSVM, the decision threshold τ applied to TEST is the median of the five validation-fold thresholds. For FT-Transformer, it is the threshold selected on the single internal validation split. In both paths, TEST remains untouched until the final predictive evaluation.

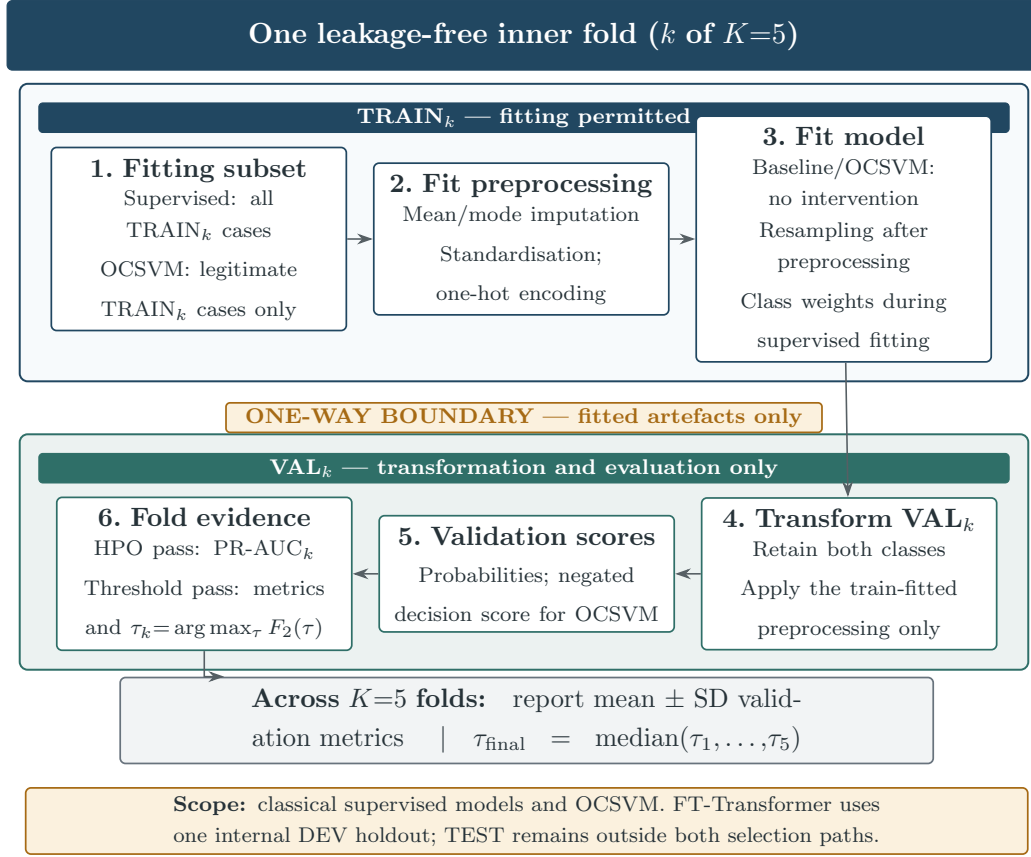


Figure 4.2: Leakage-free five-fold inner validation for the classical supervised models and OCSVM. Fitting and imbalance interventions remain within TRAIN_k; the fitted pipeline is applied to VAL_k. Fold thresholds are aggregated by their median. FT-Transformer uses a separate internal DEV holdout.

4.6 Threshold Selection Study

A predefined threshold is not selected using TEST, whereas optimising it on TEST introduces selection bias. This thesis compares four explicit rules: the fixed default $\tau=0.5$; the validation-optimal F_1 threshold ($\tau^* = \arg \max_{\tau} F_1(\tau)$), which gives precision and recall equal weight; the validation-optimal F_2 threshold ($\tau^* = \arg \max_{\tau} F_2(\tau)$), which assigns greater weight to recall; and a precision-constrained rule that selects the smallest τ for which $\text{Precision}(\tau) \geq 0.5$ on validation. The last rule represents a minimum validation-precision target, not a guarantee that test precision will remain above 0.5 under sampling variation or distribution shift. If no validation threshold satisfies the target, the implementation records $\tau = \infty$; this no-alert convention produces no positive TEST predictions and therefore zero alert rate, recall, and F scores. Section 3.4 defines these rules, and Section 3.3 defines F_{β} .

For OCSVM, these rules operate on the negated decision function rather than a calibrated probability. Its fixed value of 0.5 is retained as a numerical sensitivity reference but is not equivalent to a 50% posterior-probability decision.

For the classical models and OCSVM, each rule is evaluated separately on the five saved validation folds and the median of the five full-precision thresholds is applied to the saved TEST scores. The precision constraint is checked within each fold; the median threshold is not itself re-optimised on pooled validation scores. For FT-Transformer, all four rules use the scores retained from the same selected internal-validation checkpoint that produced the baseline threshold; no additional validation model is reconstructed for this study. For max- F_1 and max- F_2 , a tie between optimal validation thresholds is resolved in favour of the smallest threshold. TEST labels are never used to select a threshold. The study's max- F_2 threshold and resulting confusion matrix must reproduce the baseline operating point exactly, apart from displayed rounding; a mismatch causes the reporting procedure to fail. This separates changes due to the decision rule from changes in fitting or random initialisation.

4.7 Hyperparameter Tuning Strategy

Hyperparameter tuning is performed only for the baseline setting (`strategy=None`) to keep the factorial study computationally feasible and internally consistent. The baseline-selected hyperparameters are reused for every imbalance strategy within

the same model and dataset. This fixes the configured hyperparameters while the planned imbalance intervention changes; it does not guarantee identical realised tree complexity or selected FT-Transformer epoch counts. The comparison remains conditional on the baseline search result.

4.7.1 Classical Models

Tuning uses **Optuna** [40] with the Tree-structured Parzen Estimator (TPE) sampler and a **MedianPruner** that terminates unpromising trials on the basis of per-fold intermediate results; Section 3.5 describes both methods. Each baseline search has a budget of 50 attempted trials, including pruned trials, and optimises PR-AUC through five-fold stratified cross-validation. BAF evaluation retains the parameters selected by its baseline searches; validation refitting does not introduce another search. During hyperparameter search only, LightGBM and CatBoost use early stopping with a patience of 50 rounds, monitoring their default binary-loss metric; average precision remains the external Optuna objective. The median best-iteration count from the selected trial’s folds fixes the number of trees for subsequent threshold-selection refits and the final full-DEV fit, neither of which uses early stopping. The sampler seed fixes the recorded search sequence. Logistic Regression uses `solver=saga` with `l1_ratio` controlling the Elastic Net mixture under scikit-learn 1.8: both C and the mixture are effective parameters. The ULB search omits inactive LightGBM `subsample` and CatBoost `min_data_in_leaf` dimensions under the configured tree policies. The BAF search sampled these dimensions, but they are not interpreted as effective interventions. Appendix A.1 (p. 99) reports the selected configurations and these implementation qualifications.

4.7.2 FT-Transformer

For FT-Transformer, five-fold cross-validation is replaced by a **single holdout validation split** (80/20 within DEV), because fitting a deep model for every fold would multiply the cost of each trial. This holdout-based choice is consistent with the evaluation design used in the FT-Transformer literature [4], but it provides a different estimate of selection variability from the five-fold classical-model path.

Tuning also uses Optuna TPE with a budget of 50 attempted trials, including pruned trials, and optimises PR-AUC on the validation split. Each trial trains the model with early stopping (patience = 15 epochs, monitoring validation PR-AUC). The

hyperparameter search space includes the number of transformer blocks (2–4), token dimension in $\{64, 128, 192, 256\}$, attention-head count in $\{4, 8\}$, and FFN dimension multiplier from 1.33 to 2.67. Attention, FFN, and additional token residual dropout are searched over $[0, 0.3]$, $[0, 0.4]$, and $[0, 0.2]$, respectively; the additional residual dropout is applied to the encoder output before the [CLS] representation is extracted for the prediction head. Learning rate is searched over 10^{-5} – 10^{-3} , weight decay over 10^{-6} – 10^{-3} , and batch size over $\{256, 512, 1024\}$. The decision threshold τ is selected by maximising F_2 on the validation split (rather than the median across folds).

For ULB, each trial is initialised deterministically with seed 42 + trial number. After hyperparameter selection, a validation refit with seed 42 determines the best checkpoint, threshold, and epoch count using maximum validation average precision. The final full-DEV model is freshly initialised with seed 42 and trained for exactly this selected number of epochs. AdamW and cosine learning-rate scheduling use the same 200-epoch horizon in validation and final fitting, even if the selected fit stops earlier. The BAF FT-Transformer experiments retain the baseline-selected BAF hyperparameters and use this validation/final-fitting procedure without a new hyperparameter search. Validation scores, row indices, preprocessors, and the selected checkpoint are saved so that threshold and attention analyses can use the exact fitted artefacts.

4.7.3 General Notes

For the BAF transfer study, tuning and fitting are performed on Base only. The saved Base-trained model, selected hyperparameters, and decision threshold are then applied unchanged to every Variant test distribution.

An equal trial budget does not imply equal computational work across families: pruning, early stopping, five-fold versus single-holdout validation, and model-specific fit costs differ. The search budget controls the number of attempted configurations, not an equal wall-clock allocation.

4.8 Cross-Domain Generalisation

To assess robustness beyond the Base test distribution, the study performs transfer within the BAF suite. Variant I changes group-size disparity, Variant II changes group-wise prevalence disparity, Variant III changes group-wise separability, Variant IV changes prevalence disparity over time, and Variant V changes separability

over time. The `target` and `month` fields are excluded from modelling, as are the additional `x1` and `x2` columns in Variants III and V, so every model receives the same 30-feature input.

The protocol is as follows:

1. **Base reference:** Train and evaluate each model on BAF Base using the standard holdout protocol.
2. **Disjoint Variant cohorts:** Start from each Variant’s original stratified 200 000-row TEST partition. Exclude any row whose 30 common predictors exactly match a row in Base DEV, using numeric canonicalisation and element-wise confirmation of hash candidates. Variant labels are not used to choose exclusions. Retain original row indices and report removed and retained class counts.
3. **Variant transfer:** Apply the saved Base-trained model to each retained Variant I–V cohort *without retraining or re-tuning*. Use the same exclusion rule for every model.
4. **Threshold transfer:** Apply the decision threshold τ selected on Base validation unchanged to all Variant test sets.
5. **Descriptive comparison:** Report PR-AUC, ROC-AUC, F_1 , and F_2 on Base and each Variant. Differences are measured relative to the Base reference, not to separately trained in-domain Variant models.

Table 4.3 reports the evaluated populations after the common-predictor overlap check. The Base TEST partition remains unchanged; Variant exclusions are defined against Base DEV, rather than against Base TEST. Every original Variant TEST partition contains 2 206 fraud cases. The same retained row indices are used for all six models.

This is a controlled stress test of a fixed Base-trained model under the common 30-feature projection. The injected separability variables `x1/x2` are excluded from Variants III/V, so those evaluations do not isolate the suite’s full separability intervention. Months are pooled; the experiment therefore does not establish prospective temporal robustness or reproduce transfer between institutions. Because filtering changes the evaluated population and may change prevalence, retained cohort sizes and prevalence accompany the transfer metrics. Exact separation from Base DEV is verified for the retained cohorts; neither unobserved entity independence nor mutual

Table 4.3: BAF transfer cohorts after excluding exact common-predictor matches to Base DEV. Prevalence refers to the retained TEST population; fraud counts in parentheses refer to excluded rows.

Cohort	Original TEST rows	Excluded rows (fraud cases)	Retained TEST rows	Retained fraud cases	Prevalence (%)
Base	200 000	0 (0)	200 000	2 206	1.1030
Variant I	200 000	22 474 (262)	177 526	1 944	1.0951
Variant II	200 000	22 469 (250)	177 531	1 956	1.1018
Variant III	200 000	21 282 (245)	178 718	1 961	1.0973
Variant IV	200 000	21 350 (246)	178 650	1 960	1.0971
Variant V	200 000	21 172 (228)	178 828	1 978	1.1061

independence among the TEST distributions is inferred from this check. Figure 4.3 summarises this fixed Base-to-Variant protocol.

Note on ULB $\leftrightarrow$ BAF. Cross-dataset transfer between ULB 2013 and the BAF suite is not feasible due to incompatible feature spaces (PCA-anonymised vs. semantic features). The BAF internal transfer setting instead provides controlled Variant test distributions with a matched 30-feature modelling space.

4. Methodology

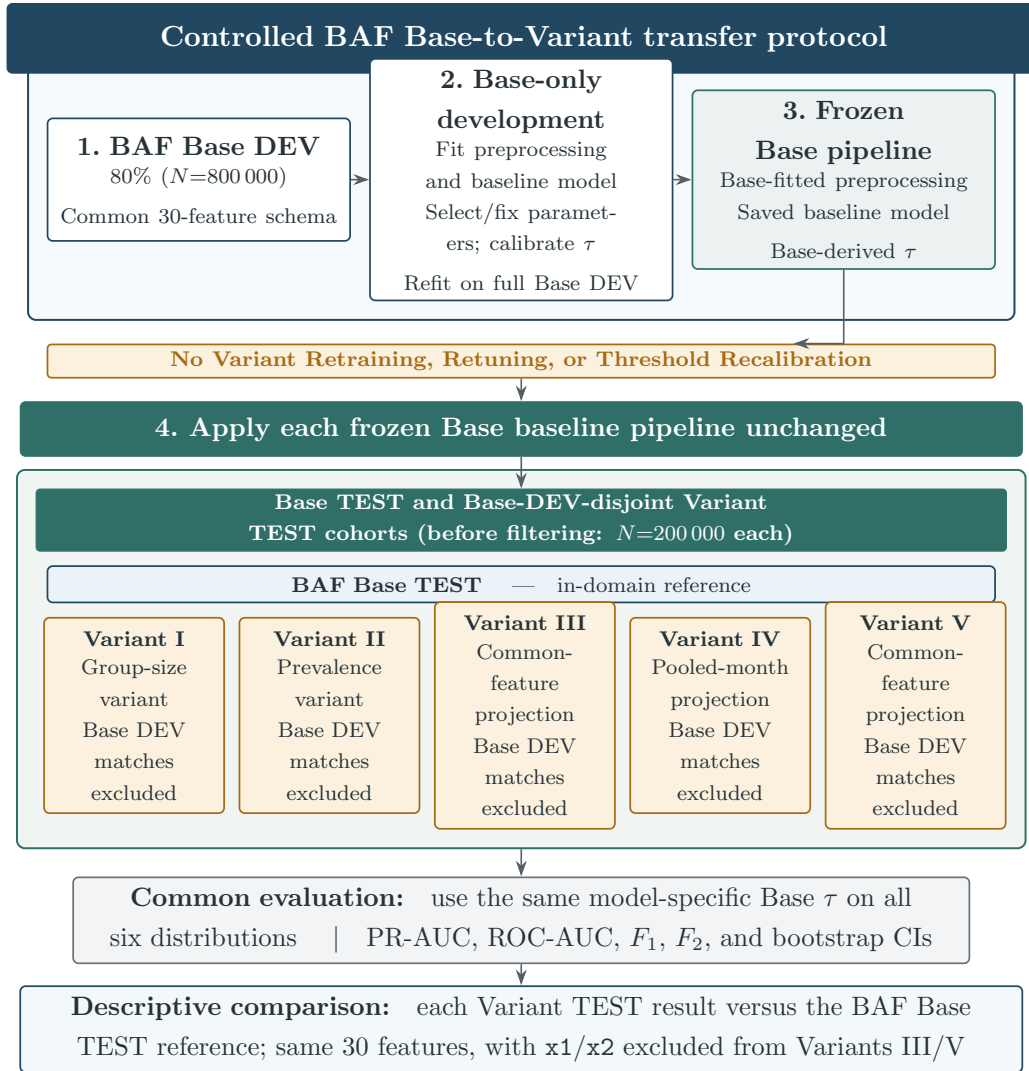


Figure 4.3: Controlled BAF Base-to-Variant transfer. Base-fitted preprocessing, model, and threshold are frozen and applied to Base TEST and disjoint Variant I–V TEST cohorts without retraining or recalibration. Exact common-predictor matches with Base DEV are excluded before Variant evaluation. All evaluations use the aligned 30-feature input.

4.9 Evaluation Metrics

The formal definitions of the metrics summarised below are given in Section 3.3; this section states which metrics are reported and the specific operational choices made for their computation.

Primary and secondary summary metrics. The primary threshold-independent summary metric is average precision, computed with `average_precision_score` and reported as PR-AUC. ROC-AUC is a secondary ranking metric that supports comparison with prior work. The Brier score is reported for models that produce probability outputs as a summary of probabilistic accuracy, not as an isolated calibration measure. No post-hoc probability calibrator is fitted: selecting a decision threshold changes the operating rule, not the calibration of the score distribution. OCSVM produces an uncalibrated decision function rather than probabilities, so its clipped implementation value is neither reported nor interpreted as a Brier score.

Operating-point metrics. The primary operating-point metric is F_2 , which assigns recall four times the weight of precision through the β^2 term and is the criterion used for threshold selection (Section 4.6). F_1 is reported alongside F_2 for the balanced-criterion view. Precision, recall, and the four confusion-matrix counts $\{TN, FP, FN, TP\}$ are reported in full to expose the underlying trade-offs.

Operational workload proxies. Two proxies for human review workload are reported: the alert rate $\frac{TP+FP}{N}$ (fraction of observations flagged for review) and the ratio $\frac{FP}{TP}$ (false alerts per true fraud caught, recorded as infinity when $TP = 0$). These quantify alert volume and yield, not measured investigation time or financial cost. Precision@ k and Recall@ k complement them by selecting exactly the highest-scoring $k = \max(1, \lfloor 0.005N \rfloor)$ observations under a fixed capacity budget. Tied scores at the boundary can make membership of this selected set dependent on score-array ordering.

Confidence intervals. Bootstrap 95% confidence intervals ($B=1\,000$ iterations) are computed on the held-out test set for PR-AUC, ROC-AUC, and F_2 in the baseline and transfer comparisons. A random generator initialised with seed 42 draws N TEST rows with replacement per replicate; the bounds are the 2.5th and

97.5th percentiles, and a replicate lacking either class is discarded. The fitted model and selected threshold remain fixed. The primary factorial interventions are reported as point estimates, without a separate bootstrap for every configuration. Supplementary predefined sensitivity/control comparisons use paired TEST resampling where indicated, applying the same row draw to both configurations. The resampling unit is an observed TEST row, not a customer or entity cluster. These intervals describe test-sample uncertainty conditional on each fitted model and split; they do not capture training-seed, split, or hyperparameter-search variability, and overlap between separate marginal intervals is not a formal paired significance test. They are descriptive per-comparison intervals rather than simultaneous guarantees across the complete experimental programme.

Reporting precision. Baseline and cross-domain tables retain four decimal places where close point estimates must be distinguished. Factorial and threshold tables, figures, and running text generally use three decimal places for readability. Derived summaries use metrics before table-level rounding; where full precision is required, ranking metrics are recomputed from saved score arrays and operating-point metrics from the integer confusion counts. Saved metric JSON files may themselves contain six-decimal rounding. Apparent discrepancies of approximately 0.001 can therefore arise from presentation rounding rather than different experimental observations.

4.10 Computational Cost and Reproducibility

All experiments are conducted on a single machine with an Intel Core i5-13600KF (14 cores), 16 GB RAM, and an NVIDIA RTX 3070 (8 GB VRAM). Classical ML models run on CPU via scikit-learn, LightGBM, and CatBoost; the FT-Transformer runs on GPU via PyTorch.

The saved execution provenance includes Python 3.12.7 and Python 3.12.14 on Windows 11, with package versions recorded per run. These records identify the software context of each fitted or reused artefact; the experiments are not presented as an invariant software environment or as a controlled timing comparison.

Each run logs its configuration (dataset identifier and SHA-256 hash, split seed, model parameters, preprocessing specification, imbalance strategy, and full-precision threshold), training and inference time, and hardware/software context. The revi-

sion additionally retains DEV/TEST and validation row indices, split hashes, validation scores, Optuna trial records, and sampler diagnostics. An explicit manifest pins every reporting source; an incomplete run is not selected merely because its directory is the newest. Original results and submitted PDFs are preserved separately from revised artefacts. Seed 42 controls data splitting, Optuna sampling, validation refitting, and final model initialisation; FT hyperparameter trials use the deterministic offset described above. These controls do not substitute for repeated-seed evaluation. Recorded final-fit and inference times are descriptive measurements on this machine, not isolated architectural cost estimates.

4.11 Interpretability

Interpretability is analysed through SHAP attributions [12] on BAF Base, whose features have semantic names. ULB 2013 features (PCA components V1–V28) are not interpreted in domain terms, so the substantive SHAP analysis is restricted to BAF. Section 3.6 defines SHAP, LinearExplainer, TreeExplainer, and the Jaccard set-agreement measure.

The primary SHAP comparison covers LR, LGBM, and CatBoost using explicitly pinned baseline-model artefacts; source-model and attribution hashes are retained separately from predictive reporting. Conclusions refer to those inspected fitted models rather than an implicitly substituted refit. LinearExplainer is used for LR and TreeExplainer [26] for LGBM and CatBoost. LR uses the empirical preprocessed Base TEST attributes as its explainer reference, which the SHAP masker may summarise; the tree explainers use their tree-path-dependent reference. This post-hoc use of unlabelled TEST attributes does not fit the predictive model or select its operating point, but the differing reference distributions preclude an absolute cross-model attribution-magnitude comparison. For categorical features that are one-hot encoded during preprocessing, global importance aggregates absolute SHAP values across the corresponding columns; local explanations instead sum signed contributions to preserve additivity at the original-feature level. A GradientExplainer output was also generated for FT-Transformer using a 500-instance background and a 2,000-instance test sample. However, the explainer wrapper converts categorical inputs from floating-point tensors to integer embedding indices during its forward pass. This non-differentiable conversion prevents reliable gradient attribution to those categorical inputs. The FT-Transformer SHAP output is therefore retained only

as an exploratory artefact and is excluded from the cross-model feature-consistency conclusions.

The analysis comprises four components. For LR, LGBM, and CatBoost, global feature importance is summarised as mean $|\text{SHAP}|$ across the full BAF Base test set, identifying the top- k ($k = 15$) features per model. Local explanations are produced for selected LGBM true-positive, false-positive, and false-negative cases. Cross-model consistency is quantified by pairwise Jaccard similarity between the top-10 feature *sets* of LR, LGBM, and CatBoost; this statistic measures membership overlap, not rank order or attribution magnitude. Cross-variant stability uses the frozen Base-trained LGBM model on the full Base TEST set and on all retained observations in each duplicate-filtered, common-30-feature Variant cohort. Native exact TreeSHAP contributions are obtained in raw log-odds space with the tree-path-dependent reference; their equivalence to the saved Base TreeExplainer attributions is checked at 128 uniformly spaced Base TEST positions, with an absolute tolerance of 10^{-10} . No predictive model is refitted and no operating threshold is changed. The resulting top-10 sets characterise global set membership for this one fixed model on these cohorts, not rank stability, causal feature effects, or the validity of every local explanation under shift.

FT-Transformer attention diagnostic. Attention is analysed separately for the baseline BAF Base FT-Transformer. The diagnostic extracts the [CLS] query row from the final transformer layer, with PyTorch averaging attention weights across heads, and aggregates those weights over the 200 000 TEST observations. Entropy includes all 31 tokens, including [CLS], and is normalised by $\log(31)$; categorical-versus-numerical attention compares the mean weight per token, not total group mass. These summaries and [CLS] self-attention describe how concentrated this specific last-layer average is. The diagnostic forward pass reproduces the saved baseline TEST scores and threshold decisions. Head averaging may conceal head-specific structure, earlier layers are not included, and attention weights are not treated as faithful output attributions.

4.12 Summary

The methodology establishes a shared outer holdout, common reporting metrics, controlled imbalance comparisons, validation-only threshold selection, a fixed Base-to-Variant transfer protocol, and architecture-compatible SHAP conclusions. The classical models and FT-Transformer use different inner-validation and preprocessing paths, and these differences remain explicit. Guided by the mapping in Table 4.1, Chapter 5 (p. 51) now presents the evidence in research-question order and interprets it within the documented boundaries.

5

Results and Discussion

This chapter presents and discusses the empirical findings in research-question order. The baseline comparison addresses RQ1; threshold sensitivity addresses RQ2; the factorial and focused FT-Transformer analyses address RQ3; BAF Base-to-Variant transfer addresses RQ4; and the SHAP and attention analyses address RQ5. The consolidated summary then brings together predictive performance, operational priorities, sampling uncertainty, and computational cost, before the chapter closes with a synthesis and threats to validity.

All interpretations follow the procedures and evidence boundaries defined in Chapter 4 (p. 29). Formal metric definitions appear in Section 3.3 (p. 23), and the study-specific reporting choices appear in Section 4.9 (p. 45).

5.1 Baseline Model Comparison (No Imbalance Strategy)

The baseline addresses RQ1 by establishing a reference for each model without resampling or cost-sensitive weighting. All ULB models are refitted after exact predictor-profile deduplication. Their supervised baseline searches use Optuna TPE with 50 trials: five-fold stratified validation for the four classical models and one internal DEV holdout for FT-Transformer. On BAF Base, the original selected hyperparameters are retained; the classical final models are preserved, their validation scores are recovered, and FT-Transformer is refitted using the recorded seed and checkpoint-selection protocol. OCSVM uses a fixed configuration without hyperparameter search. Every TEST operating point uses a threshold selected within DEV. Section 5.2 applies the four predefined rules to the same saved validation and TEST scores, with its max- F_2 rule reproducing the baseline threshold and confusion matrix.

5.1.1 ULB Credit Card 2013

Tables 5.1 and 5.2 report the predictive and operational metrics on ULB, and Figure 5.1 visualises the PR-AUC ranking.

Table 5.1: Baseline comparison on ULB 2013 (strategy = None, threshold = max- F_2). Brier score is omitted for OCSVM because its decision function is not a probability.

Model	PR-AUC	ROC-AUC	F_1	F_2	Precision	Recall	τ	Brier
LR	0.6921	0.9567	0.7912	0.7709	0.8276	0.7579	0.0550	0.00076
RF	0.7928	0.9285	0.8202	0.7883	0.8795	0.7684	0.2600	0.00048
LGBM	0.8205	0.9799	0.8523	0.8134	0.9259	0.7895	0.1581	0.00045
CatBoost	0.8196	0.9802	0.8208	0.7751	0.9103	0.7474	0.1279	0.00045
FT-Trans.	0.8015	0.9670	0.8000	0.7609	0.8750	0.7368	0.2737	0.00051
OCSVM	0.2593	0.9222	0.3593	0.4847	0.2510	0.6316	2.2391	—

Table 5.2: Operational metrics on ULB 2013 (strategy = None, threshold = max- F_2).

Model	Alert Rate	FP/TP	P@ k	R@ k	k
LR	0.00153	0.2083	0.2756	0.8211	283
RF	0.00146	0.1370	0.2721	0.8105	283
LGBM	0.00143	0.0800	0.2898	0.8632	283
CatBoost	0.00137	0.0986	0.2862	0.8526	283
FT-Trans.	0.00141	0.1429	0.2792	0.8316	283
OCSVM	0.00421	2.9833	0.2191	0.6526	283

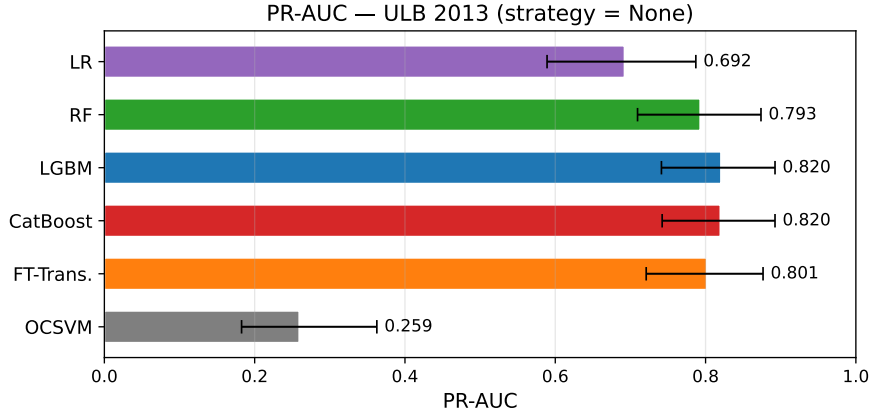


Figure 5.1: PR-AUC for all models on deduplicated ULB 2013 under the baseline setting. This ranking metric is independent of the validation-selected operating threshold. OCSVM is not tuned.

LGBM has the highest ULB baseline PR-AUC point estimate at 0.8205, closely followed by CatBoost (0.8196), FT-Transformer (0.8015), and RF (0.7928). Logistic Regression records 0.6921, whereas the OCSVM reference records 0.2593. At the validation-selected max- F_2 thresholds, LGBM also has the highest supervised precision (0.9259), recall (0.7895), and F_2 (0.8134). Table 5.2 shows alert rates between 0.137% and 0.153% for the five supervised models. LGBM has the lowest FP/TP ratio (0.0800), followed by CatBoost (0.0986); OCSVM produces approximately three false alerts per detected fraud. Under the fixed top- k review budget ($k = 283$, the integer floor of 0.5% of TEST), LGBM has the highest Precision@ k (0.2898) and Recall@ k (0.8632). These ranking and operating-point comparisons are descriptive, not paired tests of model differences.

5.1.2 BAF Base

BAF Base produces substantially lower PR-AUC values than ULB for the leading models (Tables 5.3 and 5.4; Figure 5.2).

5. Results and Discussion

Table 5.3: Baseline comparison on BAF Base (strategy = None, threshold = max- F_2). Brier score is omitted for OCSVM because its decision function is not a probability.

Model	PR-AUC	ROC-AUC	F_1	F_2	Precision	Recall	τ	Brier
LR	0.1432	0.8745	0.1871	0.2868	0.1185	0.4447	0.0527	0.01009
RF	0.1588	0.8778	0.2048	0.2941	0.1360	0.4148	0.0631	0.01007
LGBM	0.1768	0.8985	0.2224	0.3159	0.1490	0.4388	0.0636	0.00986
CatBoost	0.1796	0.8980	0.2106	0.3143	0.1359	0.4678	0.0573	0.00984
FT-Trans.	0.1769	0.8965	0.1808	0.3041	0.1079	0.5580	0.0584	0.00996
OCSVM	0.0190	0.6279	0.0449	0.0805	0.0258	0.1714	-40.26	—

Table 5.4: Operational metrics on BAF Base (strategy = None, threshold = max- F_2).

Model	Alert Rate	FP/TP	P@ k	R@ k	k
LR	0.04140	7.4393	0.2720	0.1233	1000
RF	0.03365	6.3552	0.3120	0.1414	1000
LGBM	0.03249	5.7128	0.3420	0.1550	1000
CatBoost	0.03798	6.3605	0.3420	0.1550	1000
FT-Trans.	0.05707	8.2721	0.3290	0.1491	1000
OCSVM	0.07321	37.7328	0.0300	0.0136	1000

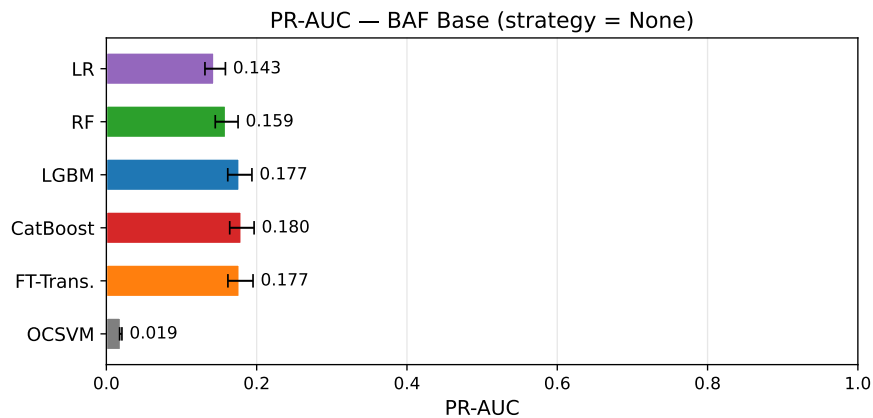


Figure 5.2: PR-AUC for all models on BAF Base under the baseline setting. This ranking metric is independent of the validation-selected operating threshold. OCSVM is not tuned.

Three models cluster at the top — CatBoost (0.1796), FT-Transformer (0.1769), and LGBM (0.1768) — within 0.003 PR-AUC. Their operating points differ: LGBM re-

cords $F_2 = 0.3159$, CatBoost 0.3143, and FT-Transformer 0.3041. BAF contains approximately 3.5 times as many observations as the deduplicated ULB population, but data volume is not isolated here. Feature structure, prevalence, and model-selection procedures also differ across datasets, so the smaller transformer-GBDT gap cannot be assigned to any of these untested mechanisms. Relative to the highest point estimate, Random Forest is approximately 12% lower and Logistic Regression approximately 20% lower in PR-AUC. OCSVM remains weak at 0.019. The operational metrics show a heavier review burden than on ULB: LGBM and CatBoost produce approximately 5.7–6.4 false alerts per detected fraud at alert rates of 3.25% and 3.80%. FT-Transformer achieves higher recall (55.8%) but lower precision (10.8%), raising its alert rate to 5.71% and FP/TP to 8.27. Under the top-0.5% review budget ($k = 1\,000$), Recall@ k is approximately 15% for all three.

Together, the two baselines answer RQ1 at the point-estimate level: LGBM and CatBoost lead ULB ranking performance, while CatBoost, FT-Transformer, and LGBM have closely grouped BAF Base PR-AUC values. FT-Transformer does not establish a predictive advantage over the boosted trees on either dataset, and similar BAF PR-AUC does not imply equal alert workload or threshold-dependent performance. Both comparisons use the None strategy and validation-selected max- F_2 operating points under the model-specific procedures described in Chapter 4. The following sections examine how threshold and imbalance choices affect these reference results.

5.2 Threshold Sensitivity Analysis

This section addresses RQ2 by applying the four rules introduced in Section 4.6 — fixed $\tau = 0.5$, max- F_1 , max- F_2 , and the lowest validation threshold satisfying Precision ≥ 0.5 — to the saved baseline TEST scores. For the classical models and OCSVM, thresholds are medians from the same five saved validation folds. For FT-Transformer, all rules use the validation scores retained from the exact selected baseline validation checkpoint; no additional model is fitted for the threshold study. The max- F_2 rule reproduces each baseline operating point, so differences between rules concern the decision threshold rather than changes in model fitting. TEST labels remain excluded from threshold selection throughout. The analysis reports changes in F_2 , F_1 , recall, alert rate, and model ordering.

5.2.1 ULB Credit Card 2013

Tables 5.5–5.8 report F_2 , F_1 , Recall, and Alert Rate as the threshold strategy changes, holding everything else constant. Because max- F_2 is selected on validation rather than optimised on TEST, its TEST-set F_2 need not exceed the value obtained with the fixed threshold; FT-Transformer illustrates this distinction (0.761 vs. 0.766).

Table 5.5: Threshold sensitivity on ULB 2013 — F_2 by threshold strategy (strategy = None).

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.618	0.771	0.771	0.708
RF	0.776	0.782	0.788	0.713
LGBM	0.765	0.771	0.813	0.730
CatBoost	0.784	0.784	0.775	0.740
FT-Trans.	0.766	0.762	0.761	0.780
OCSVM	0.375	0.481	0.485	0.433

Table 5.6: Threshold sensitivity on ULB 2013 — F_1 by threshold strategy (strategy = None).

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.688	0.791	0.791	0.612
RF	0.843	0.840	0.820	0.622
LGBM	0.831	0.828	0.852	0.627
CatBoost	0.845	0.845	0.821	0.645
FT-Trans.	0.814	0.821	0.800	0.798
OCSVM	0.213	0.391	0.359	0.395

Table 5.7: Threshold sensitivity on ULB 2013 — Recall by threshold strategy (strategy = None).

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.579	0.758	0.758	0.789
RF	0.737	0.747	0.768	0.789
LGBM	0.726	0.737	0.789	0.821
CatBoost	0.747	0.747	0.747	0.821
FT-Trans.	0.737	0.726	0.737	0.768
OCSVM	0.758	0.568	0.632	0.463

Table 5.8: Threshold sensitivity on ULB 2013 — Alert Rate (%) by threshold strategy (strategy = None).

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.11%	0.15%	0.15%	0.26%
RF	0.13%	0.13%	0.15%	0.26%
LGBM	0.13%	0.13%	0.14%	0.27%
CatBoost	0.13%	0.13%	0.14%	0.26%
FT-Trans.	0.14%	0.13%	0.14%	0.16%
OCSVM	1.02%	0.32%	0.42%	0.23%

On ULB, threshold choice has a moderate effect. Logistic Regression gains most among the supervised models: F_2 rises from 0.618 under the fixed rule to 0.771 under max- F_2 , approximately a 25% relative improvement. The tree ensembles gain less, or decline slightly, because their fixed-threshold F_2 values are already closer to their study-specific maxima; this observation alone does not establish probability calibration. The Precision ≥ 0.5 rule produces the highest recall and alert rate for each supervised model in these TEST results. Its lowest-feasible validation threshold therefore trades additional review workload for recall rather than acting as the lowest-workload rule. Numerical threshold scales depend on the score distributions and are not a calibration test. Model ordering is not preserved: CatBoost has the highest fixed-rule and max- F_1 F_2 (0.784), LGBM leads under max- F_2 (0.813), and FT-Transformer leads under the precision-constrained rule (0.780). The selected rule can change which model has the highest F_2 even without refitting.

5.2.2 BAF Base

Threshold effects are substantially larger on BAF Base. Tables 5.9–5.12 report the same four metrics across the four rules.

Table 5.9: Threshold sensitivity on BAF Base — F_2 by threshold strategy (strategy = None).

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.015	0.245	0.287	0.032
RF	0.001	0.263	0.294	0.045
LGBM	0.042	0.260	0.316	0.056
CatBoost	0.037	0.274	0.314	0.058
FT-Trans.	0.048	0.293	0.304	0.133
OCSVM	0.031	0.071	0.081	0.000

Table 5.10: Threshold sensitivity on BAF Base — F_1 by threshold strategy (strategy = None).

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.023	0.219	0.187	0.050
RF	0.001	0.237	0.205	0.068
LGBM	0.065	0.240	0.222	0.084
CatBoost	0.057	0.244	0.211	0.086
FT-Trans.	0.074	0.235	0.181	0.177
OCSVM	0.031	0.046	0.045	0.000

Table 5.11: Threshold sensitivity on BAF Base — Recall by threshold strategy (strategy = None).

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.012	0.267	0.445	0.026
RF	0.000	0.285	0.415	0.036
LGBM	0.034	0.275	0.439	0.046
CatBoost	0.030	0.299	0.468	0.047
FT-Trans.	0.039	0.349	0.558	0.114
OCSVM	0.030	0.108	0.171	0.000

Table 5.12: Threshold sensitivity on BAF Base — Alert Rate (%) by threshold strategy (strategy = None).

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.02%	1.59%	4.14%	0.06%
RF	0.00%	1.55%	3.36%	0.08%
LGBM	0.07%	1.43%	3.25%	0.10%
CatBoost	0.05%	1.59%	3.80%	0.10%
FT-Trans.	0.07%	2.17%	5.71%	0.32%
OCSVM	1.03%	4.06%	7.32%	0.00%

On BAF, the fixed $\tau = 0.5$ rule yields $F_2 < 0.05$ for every supervised model, and Random Forest produces almost no positive predictions ($F_2 = 0.001$, Recall = 0.0005). Few cases receive scores above 0.5, so the default threshold misses most positives. For the three leading models, max- F_2 produces approximately 7.5-fold, 8.5-fold, and 6.3-fold increases for LGBM (0.042 to 0.316), CatBoost (0.037 to 0.314), and FT-Transformer (0.048 to 0.304), respectively. The max- F_1 rule has lower recall than max- F_2 , while the precision-constrained rule raises fewer alerts and misses more frauds. OCSVM cannot satisfy the validation precision target, represented by $\tau = \infty$, and therefore produces no TEST alerts under that rule. Moving from fixed to max- F_2 raises the supervised alert rate from at most 0.07% to 3.25–5.71%, making the review-workload trade-off explicit.

Threshold rules also reorder models on BAF. FT-Transformer has the highest fixed-rule F_2 (0.048), max- F_1 -rule F_2 (0.293), and precision-constrained F_2 (0.133), whereas LGBM leads under max- F_2 (0.316). For balanced F_1 , CatBoost under the max- F_1 rule has the highest baseline TEST value (0.244). Together with the ULB changes, this shows that model selection based on a threshold-dependent metric must specify the operating rule. On BAF, the observed within-model F_2 changes are substantially larger than the baseline differences among the leading supervised models; this conclusion is limited to threshold-dependent metrics and does not affect PR-AUC.

5.3 Full Factorial Analysis — Imbalance Strategy $\times$ Model

This section addresses RQ3 through the full grid of seven imbalance handling strategies $\times$ five supervised models $\times$ two datasets, evaluated under the

primary max- F_2 rule. Hyperparameters are fixed at their baseline-selected values (strategy = None), while the planned imbalance intervention changes. This controls the configured hyperparameters, not necessarily realised tree complexity or validation-selected FT-Transformer epoch counts. The conclusions remain conditional on those baseline searches and the model-specific input representations. OCSVM is omitted because it fits only legitimate examples; labels are nevertheless used to select those training examples and calibrate its threshold.

5.3.1 ULB Credit Card 2013

Tables 5.13–5.15 report PR-AUC, ROC-AUC, and F_2 for every (model, strategy) cell on ULB. The best unrounded value per row is highlighted in bold; equal displayed values need not be exact ties.

Table 5.13: PR-AUC by model $\times$ strategy on ULB 2013.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.692	0.619	0.669	0.672	0.672	0.683	0.670
RF	0.793	0.698	0.807	0.814	0.814	0.803	0.810
LGBM	0.820	0.632	0.811	0.814	0.814	0.811	0.813
CatBoost	0.820	0.637	0.811	0.798	0.798	0.793	0.802
FT-Trans.	0.801	0.641	0.784	0.785	0.785	0.797	0.756

Table 5.14: ROC-AUC by model $\times$ strategy on ULB 2013.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.957	0.963	0.966	0.962	0.962	0.962	0.965
RF	0.929	0.976	0.934	0.964	0.964	0.954	0.934
LGBM	0.980	0.973	0.977	0.980	0.980	0.980	0.977
CatBoost	0.980	0.974	0.976	0.967	0.967	0.968	0.977
FT-Trans.	0.967	0.951	0.901	0.927	0.927	0.923	0.962

Table 5.15: F_2 by model $\times$ strategy on ULB 2013.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.771	0.732	0.778	0.772	0.772	0.778	0.780
RF	0.788	0.774	0.812	0.805	0.805	0.753	0.821
LGBM	0.813	0.759	0.787	0.766	0.766	0.789	0.767
CatBoost	0.775	0.763	0.808	0.788	0.788	0.779	0.776
FT-Trans.	0.761	0.600	0.763	0.692	0.692	0.698	0.774

For LGBM, the corrected baseline PR-AUC is 0.8205. ROS, SMOTE, SMOTE+Tomek, SMOTEENN and Class Weights produce 0.8112, 0.8144, 0.8144, 0.8106 and 0.8133, respectively, whereas RUS produces 0.6319. Similar ranking metrics do not imply identical scores: ROS and Class Weights have different saved prediction arrays. SMOTE and SMOTE+Tomek are instead bitwise identical for this model. Their recorded pre-cleaning and post-cleaning arrays confirm zero Tomek removals in all five validation folds and the final DEV fit; both final resampled sets contain 453 204 examples. Here, the cleaning step makes no change to the realised training input, rather than merely producing equal metrics after rounding.

The other models do not follow one common response. RF improves PR-AUC from 0.793 to 0.814 under SMOTE/SMOTE+Tomek and attains the highest factorial F_2 with Class Weights (0.821). LR, LGBM, CatBoost, and FT-Transformer instead have their highest PR-AUC under None. Increases in F_2 without corresponding PR-AUC gains reflect changes in the score distribution and the independently validation-selected operating point, not necessarily improved ranking over the full range.

LightGBM score-resolution audit. The seven corrected frozen models reproduce their saved probabilities exactly. None has scores equal to zero or one. In the five rebalanced or weighted conditions listed above, each of the five highest distinct-score groups contains only fraud cases, with groups of one to eight rows. These inspected groups therefore do not exhibit mixed-label ties at the top of the ranking. ROC-AUC nevertheless remains high across the corrected conditions, including RUS, showing why ROC-AUC alone does not describe precision at the upper end of the ranking.

Exact probability ties can arise when observations share the same raw margin and complete tree-leaf vector, without scores being rounded or saturated at zero or one. Mixed-label groups high in the ranking can reduce average precision while

involving a small fraction of the positive–negative comparisons used by ROC-AUC. The frozen-model audit checks score resolution directly rather than inferring it from similar aggregate metrics. It does not isolate the total causal effect of an imbalance intervention, nor constitute a controlled ablation of deduplication, hyperparameter selection, or changes in the evaluated population.

5.3.2 BAF Base

Tables 5.16–5.18 report the same three metrics on BAF Base, again marking row maxima before display rounding.

Table 5.16: PR-AUC by model $\times$ strategy on BAF Base.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.143	0.137	0.137	0.135	0.135	0.136	0.122
RF	0.159	0.135	0.149	0.120	0.120	0.124	0.146
LGBM	0.177	0.171	0.181	0.146	0.146	0.158	0.181
CatBoost	0.180	0.178	0.185	0.154	0.154	0.165	0.184
FT-Trans.	0.177	0.163	0.166	0.115	0.115	0.124	0.176

Table 5.17: ROC-AUC by model $\times$ strategy on BAF Base.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.875	0.875	0.876	0.874	0.874	0.875	0.855
RF	0.878	0.882	0.885	0.874	0.874	0.876	0.885
LGBM	0.898	0.896	0.897	0.882	0.882	0.886	0.897
CatBoost	0.898	0.897	0.899	0.887	0.887	0.889	0.899
FT-Trans.	0.896	0.893	0.892	0.862	0.862	0.867	0.895

Table 5.18: F_2 by model $\times$ strategy on BAF Base.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.287	0.283	0.283	0.280	0.280	0.282	0.267
RF	0.294	0.288	0.303	0.280	0.280	0.280	0.296
LGBM	0.316	0.318	0.319	0.293	0.293	0.300	0.320
CatBoost	0.314	0.317	0.316	0.303	0.303	0.309	0.317
FT-Trans.	0.304	0.301	0.310	0.257	0.257	0.269	0.313

On BAF, no strategy increases CatBoost or LGBM by more than 0.01 PR-AUC relative to baseline. ROS produces their highest PR-AUC values (0.185 and 0.181), while LGBM with Class Weights has the highest factorial F_2 (0.320). The gains are small at the point-estimate level and do not establish a generally preferable strategy. SMOTE and SMOTE+Tomek reduce CatBoost PR-AUC from 0.1796 to 0.1543 (approximately 14%) and FT-Transformer from 0.1769 to 0.1149 (approximately 35%). Class Weights produces FT-Transformer PR-AUC of 0.176, less than 1% below baseline, and increases its F_2 from 0.304 to 0.313.

The SMOTE/SMOTE+Tomek equality is supported by sampler and score evidence, not only the rounded tables. Across all newly fitted ULB supervised pairs and the BAF FT-Transformer pair, the cleaner removes zero rows and the resampled inputs and TEST score arrays are identical. The BAF FT-Transformer final DEV sets each contain 1 582 354 examples. The preserved BAF classical pairs have identical saved TEST scores for LR, RF, and CatBoost; LGBM differs only at numerical precision (maximum absolute difference approximately 1.1×10^{-13}), with unchanged reported metrics. A reconstruction of the historical full-DEV classical preprocessing and sampler also removes zero rows and produces identical 1 582 354-row sets. This reconstruction is not an original execution log for the historical validation folds, whose sampler records are unavailable. These observations explain why Tomek contributes no measured change here; they do not establish algorithmic equivalence on other datasets or sampling settings.

5.3.3 Interaction Heatmaps

The four heatmaps in Figures 5.3–5.6 visualise the strategy $\times$ model interactions as deltas relative to the baseline (None); green denotes improvement, red denotes degradation. The deltas are computed from unrounded saved metrics and displayed to three decimal places, whereas the accompanying tables show rounded cell values.

5. Results and Discussion

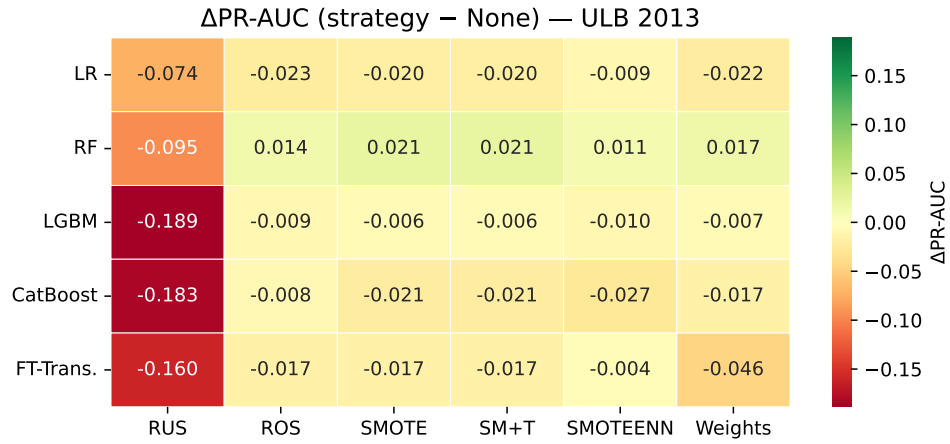


Figure 5.3: Δ PR-AUC (strategy – None) per model on ULB 2013.

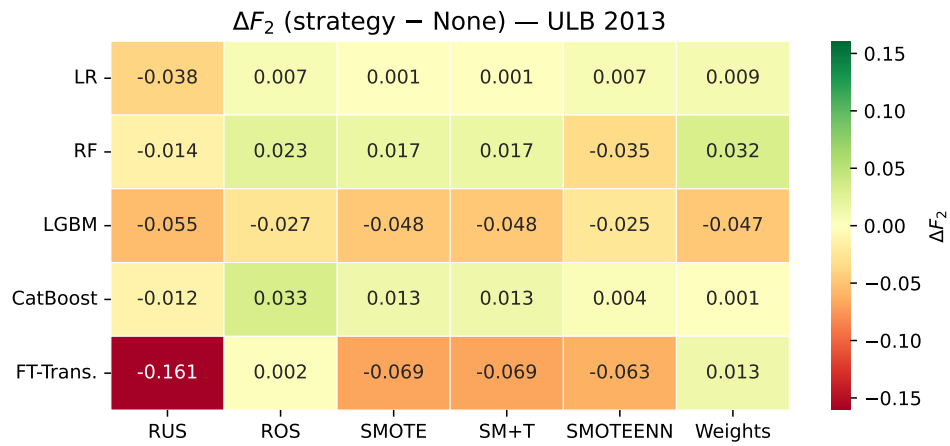


Figure 5.4: ΔF_2 (strategy – None) per model on ULB 2013.

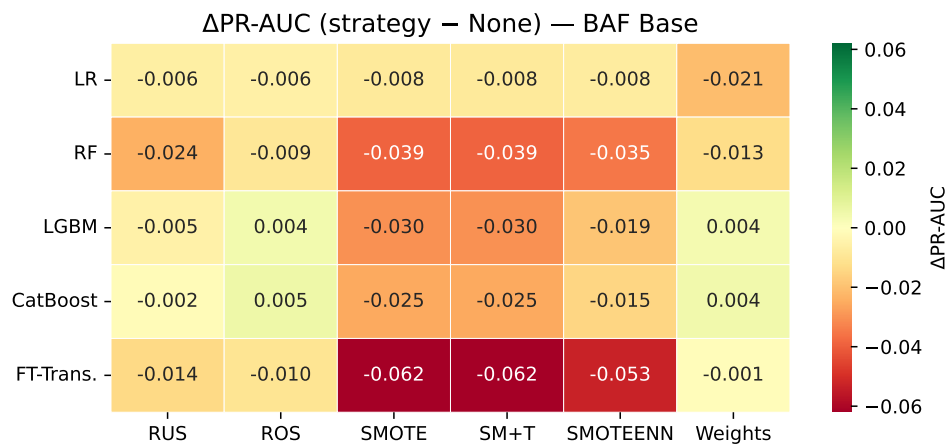


Figure 5.5: Δ PR-AUC (strategy – None) per model on BAF Base.

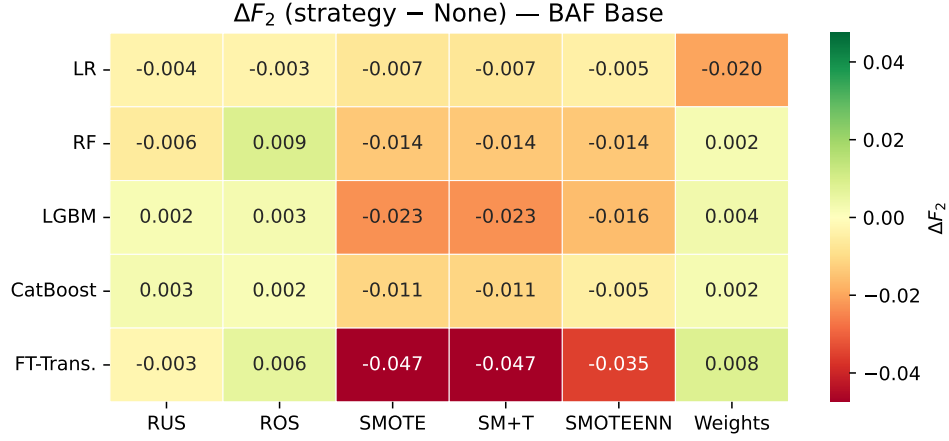


Figure 5.6: ΔF_2 (strategy – None) per model on BAF Base.

Three patterns become visible. First, RUS reduces PR-AUC for every model on both datasets, although its BAF F_2 values include small increases for LGBM and CatBoost after validation-based threshold selection. Second, the identical reported SMOTE and SMOTE+Tomek columns are consistent with the sampler-identity audit above. Third, the BAF PR-AUC deltas for LGBM and CatBoost are mostly near zero or negative. None remains a defensible reference within the evaluated setting, but the small positive ROS/weighting deltas and RF’s different ULB response preclude a universal recommendation against imbalance handling.

5.4 FT-Transformer Sensitivity to Imbalance Interventions

To complete the answer to RQ3, this section compares FT-Transformer and CatBoost directly across the implemented imbalance interventions. It describes their empirical spread and their different responses to SMOTE without treating those differences as architecture-only effects.

Table 5.19: FT-Transformer and CatBoost performance across imbalance strategies on ULB 2013. The final row is the population standard deviation across the seven complete strategy values, computed from unrounded metrics.

Strategy	FT-Transformer		CatBoost	
	PR-AUC	F_2	PR-AUC	F_2
None	0.801	0.761	0.820	0.775
RUS	0.641	0.600	0.637	0.763
ROS	0.784	0.763	0.811	0.808
SMOTE	0.785	0.692	0.798	0.788
SM+Tomek	0.785	0.692	0.798	0.788
SMOTEENN	0.797	0.698	0.793	0.779
Weights	0.756	0.774	0.802	0.776
<i>Pop. std. dev.</i>	<i>0.052</i>	<i>0.057</i>	<i>0.059</i>	<i>0.013</i>

Table 5.20: FT-Transformer and CatBoost performance across imbalance strategies on BAF Base. The final row is the population standard deviation across the seven complete strategy values, computed from unrounded metrics.

Strategy	FT-Transformer		CatBoost	
	PR-AUC	F_2	PR-AUC	F_2
None	0.177	0.304	0.180	0.314
RUS	0.163	0.301	0.178	0.317
ROS	0.166	0.310	0.185	0.316
SMOTE	0.115	0.257	0.154	0.303
SM+Tomek	0.115	0.257	0.154	0.303
SMOTEENN	0.124	0.269	0.165	0.309
Weights	0.176	0.313	0.184	0.317
<i>Pop. std. dev.</i>	<i>0.027</i>	<i>0.023</i>	<i>0.012</i>	<i>0.006</i>

On ULB both architectures follow a similar PR-AUC profile: RUS produces the largest reduction, while the remaining strategies are closer to baseline. CatBoost has higher PR-AUC in every condition except RUS and SMOTEENN, where FT-Transformer records 0.641 versus 0.637 and 0.797 versus 0.793, respectively. The population standard deviations across the seven complete strategy values are 0.052 versus 0.059 for PR-AUC and 0.057 versus 0.013 for F_2 (FT-Transformer versus CatBoost). Thus, similar ranking-metric spread does not imply similar operating-

point sensitivity. These are descriptive measures across interventions, not estimates of run-to-run variability or statistical stability.

On BAF, the population standard deviations are 0.027 versus 0.012 for PR-AUC and 0.023 versus 0.006 for F_2 , indicating greater descriptive spread for FT-Transformer across the evaluated interventions. The models have close, but not equal, baseline PR-AUC values. Their observed SMOTE responses differ: FT-Transformer declines by approximately 35% and CatBoost by 14%. Class Weights is the closest non-baseline FT-Transformer intervention on BAF (0.176), but it is not a consistent default across datasets: on ULB, it reduces PR-AUC from 0.801 to 0.756. The primary FT-Transformer resampling path interpolates ordinal category codes and then converts them to integers. The SMOTE reduction is therefore sensitivity under this specific preprocessing-sampling pipeline, not evidence of an inherent transformer weakness. The predefined categorical-aware control below does not recover baseline PR-AUC, so ordinal interpolation is not established as the sole cause. The baseline attention diagnostic in Section 5.7 does not test SMOTE-trained attention and is not used to explain the reduction.

5.4.1 Predefined BAF Sensitivity Controls

The following controls complete RQ3’s representation-sensitive interpretation but remain separate from the primary factorial matrix. They compare predefined alternatives on the same original BAF Base TEST observations, with hyperparameters held fixed within each pair and each member’s max- F_2 threshold frozen from DEV. AP denotes average precision, reported elsewhere as PR-AUC. Positive differences favour the second member only for the displayed metric; no policy is selected using TEST.

5. Results and Discussion

Table 5.21: Predefined BAF absence-code sensitivity: preserved codes versus NaN conversion, mean imputation, and missingness indicators. Differences are NaN-and-indicators minus preserve; intervals use 1 000 row-paired bootstrap resamples.

Model	Metric	Preserve	NaN + indic- ators	Δ [paired 95% CI]
Logistic gression	AP	0.1432	0.1477	+0.0045 [+0.0017, + 0.0074]
	F_2	0.2868	0.2910	+0.0042 [−0.0008, + 0.0096]
Random Forest	AP	0.1588	0.1596	+0.0008 [−0.0015, + 0.0030]
	F_2	0.2941	0.2931	−0.0010 [−0.0061, + 0.0044]
LGBM	AP	0.1768	0.1759	−0.0009 [−0.0029, + 0.0011]
	F_2	0.3159	0.3173	+0.0015 [−0.0039, + 0.0062]
CatBoost	AP	0.1796	0.1790	−0.0005 [−0.0023, + 0.0015]
	F_2	0.3143	0.3126	−0.0017 [−0.0063, + 0.0031]
FT- Transformer	AP	0.1769	0.1732	−0.0038 [−0.0082, + 0.0006]
	F_2	0.3041	0.3026	−0.0016 [−0.0068, + 0.0034]
OCSVM	AP	0.0190	0.0162	−0.0028 [−0.0035, − 0.0024]
	F_2	0.0805	0.0679	−0.0126 [−0.0166, − 0.0087]

Table 5.21 does not show a uniform benefit from replacing absence codes. LR gains 0.0045 AP, whereas OCSVM loses 0.0028 AP and 0.0126 F_2 . The remaining AP paired intervals include zero. All supervised F_2 difference intervals include zero. This describes the tested combined conversion–imputation–indicator intervention, not the optimal treatment of missing values or a causal effect of any one component. It also does not establish equivalence for models with small observed differences. The primary preserve policy remains predefined rather than chosen from these TEST comparisons.

Table 5.22: Predefined BAF categorical-sampling controls: primary SMOTE versus SMOTE-NC. Differences are SMOTE-NC minus primary SMOTE; intervals use 1 000 row-paired bootstrap resamples.

Model	Metric	SMOTE	SMOTE-NC	Δ [paired 95% CI]
CatBoost	AP	0.1543	0.1433	−0.0110 [−0.0147, − 0.0074]
	F_2	0.3029	0.2856	−0.0173 [−0.0233, − 0.0108]
FT-Transformer	AP	0.1149	0.1157	+0.0008 [−0.0068, + 0.0076]
	F_2	0.2567	0.2577	+0.0010 [−0.0088, + 0.0111]

The nominal-aware control in Table 5.22 does not recover FT-Transformer’s baseline AP: 0.1157 remains well below 0.1769, with a paired change of only +0.0008 relative to primary SMOTE. Its alert rate falls from 6.11% to 4.22% under the independently DEV-selected thresholds, so similar AP and F_2 do not imply unchanged workload. CatBoost instead has lower AP and F_2 under this control. The intervention changes the sampling representation and nominal-aware resampling jointly, not an isolated architectural component. CatBoost still receives one-hot encoded classifier inputs rather than native categorical inputs. Thus, respecting nominal categories in this specific control is not sufficient to undo the observed SMOTE reduction, and neither a universal SMOTE-NC benefit nor an architecture-only explanation follows.

The reported 95% intervals describe TEST-row uncertainty conditional on the paired fitted models, their frozen thresholds, and the shared split, not training-seed variability. Single-class resamples are excluded; all 1 000 requested resamples were valid for these controls. Points, differences, and interval bounds are rounded to four decimal places. An interval containing zero does not establish equivalence, and the comparisons are not adjusted for the complete set of tested metrics and controls. For primary CatBoost SMOTE, the archived six-decimal threshold is retained and its saved TEST confusion counts are reproduced; a more precise historical threshold is not assumed. That historical primary run did not save explicit TEST row indices. Pairing therefore uses verified raw-file identity, the documented seed-42 split, and exactly matched ordered TEST labels, without inventing a historical index artefact.

5.5 Cross-Domain Generalisation (BAF Internal)

This section addresses RQ4 by evaluating each BAF Base-trained baseline model on the corrected holdout cohorts of Variants I–V *without retraining*. Every Variant TEST profile matching any Base DEV profile on the common 30 predictors is excluded before scoring, removing 21 172–22 474 observations per Variant and retaining 177 526–178 828. The same predeclared cohorts are used for Variant SHAP. The Base column repeats the full in-domain TEST performance for reference; each model’s Base DEV-selected max- F_2 threshold is frozen for all Variants. The additional **x1** and **x2** columns in Variants III and V are excluded so that every model receives the same 30-feature input. Exact-profile separation removes this verified Base DEV overlap, but does not establish entity-level independence or constitute prospective temporal evaluation.

Table 5.23: Cross-domain PR-AUC of Base-trained models without retraining: full Base TEST and Base-DEV-profile-disjoint Variant cohorts under the common 30-feature projection. Bold marks each row’s highest value, not the best model per Variant.

Model	Base	Var I	Var II	Var III	Var IV	Var V
LR	0.1432	0.1066	0.1442	0.1039	0.1334	0.1005
RF	0.1588	0.1342	0.1615	0.1193	0.1477	0.1093
LGBM	0.1768	0.1483	0.1803	0.1328	0.1646	0.1185
CatBoost	0.1796	0.1472	0.1822	0.1329	0.1668	0.1207
FT-Trans.	0.1769	0.1462	0.1751	0.1327	0.1589	0.1168
OCSVM	0.0190	0.0164	0.0220	0.0178	0.0205	0.0165

Table 5.24: Cross-domain F_2 of Base-trained models without retraining, using frozen Base DEV-selected thresholds: full Base TEST and Base-DEV-profile-disjoint Variant cohorts. Bold marks each row’s highest value.

Model	Base	Var I	Var II	Var III	Var IV	Var V
LR	0.2868	0.2391	0.2656	0.2370	0.2680	0.2284
RF	0.2941	0.2459	0.2840	0.2522	0.2796	0.2438
LGBM	0.3159	0.2693	0.3035	0.2760	0.3008	0.2620
CatBoost	0.3143	0.2701	0.2994	0.2697	0.2991	0.2588
FT-Trans.	0.3041	0.2661	0.2655	0.2489	0.2722	0.2391
OCSVM	0.0805	0.0662	0.0925	0.0749	0.0870	0.0690

Table 5.25: Cross-domain ROC-AUC of Base-trained models without retraining: full Base TEST and Base-DEV-profile-disjoint Variant cohorts under the common 30-feature projection. Bold marks each row’s highest value.

Model	Base	Var I	Var II	Var III	Var IV	Var V
LR	0.8745	0.8488	0.8786	0.8517	0.8819	0.8450
RF	0.8778	0.8533	0.8851	0.8586	0.8880	0.8461
LGBM	0.8985	0.8820	0.9004	0.8809	0.9062	0.8710
CatBoost	0.8980	0.8812	0.8994	0.8797	0.9042	0.8712
FT-Trans.	0.8965	0.8785	0.8964	0.8760	0.9009	0.8681
OCSVM	0.6279	0.5959	0.6616	0.6144	0.6456	0.5907

The boosted trees have the strongest transferred point estimates, rather than Random Forest. LGBM has the highest PR-AUC on Variant I (0.1483), while CatBoost leads on Variants II–V (0.1822, 0.1329, 0.1668, and 0.1207). CatBoost has the highest Variant I F_2 (0.2701), while LGBM leads on Variants II–V (0.3035, 0.2760, 0.3008, and 0.2620). RF’s PR-AUC is below its Base value on Variants I, III, IV, and V, and slightly above it on Variant II (0.1615 versus 0.1588). A transferred ranking metric need not fall on every changed population after verified profile-overlap removal: the retained population and score ordering can differ in either direction. This modest Variant II increase does not establish leakage or a general transfer advantage. All five supervised models have lower F_2 on every Variant than on Base at their frozen thresholds.

LGBM, CatBoost, and FT-Transformer show similar broad point-estimate profiles across the controlled variants. Variant II, which increases group-wise prevalence disparity while equalising group sizes, remains close to or slightly above Base PR-AUC. Variant I changes group-size disparity and Variant IV changes prevalence disparity over time; both produce smaller changes for the three leading Base models than Variants III and V. The latter were constructed to alter group-wise and temporal separability, respectively, and produce PR-AUC reductions of approximately 25–34% for these models relative to Base. However, this experiment omits their additional separability variables and filters exact profiles, so it does not isolate a causal separability effect. Within the common-feature protocol, FT-Transformer shows no point-estimate transfer advantage over LGBM or CatBoost. Because PR-AUC is prevalence-sensitive, ROC-AUC and F_2 are considered alongside it rather than interpreting every difference as architecture-only degradation.

OCSVM provides a third, consistently lower profile, with PR-AUC below 0.023 on every Variant. Its one-class objective does not approach the supervised point estimates on either the Base or Variant distributions. Supervised Variant ROC-AUC remains approximately 0.845–0.906 despite larger relative PR-AUC changes. ROC-AUC is insensitive to a class-prior change when class-conditional score distributions are held fixed, but those distributions can also change across Variants; it therefore provides a complementary ranking view rather than proof that prevalence explains the observed transfer differences.

5.6 Interpretability Analysis (SHAP)

This section addresses the SHAP component of RQ5 on BAF Base, whose features have semantic names. The primary comparison is restricted to LR, LGBM, and CatBoost using architecture-compatible LinearExplainer and TreeExplainer outputs (Section 4.11). The explanation artefacts are explicitly pinned to the same frozen final BAF baseline models retained for primary predictive reporting; validation-score recovery does not refit them. They describe these fitted functions, not the newly fitted sensitivity-control models. The archived FT-Transformer GradientExplainer artefact is not displayed or included in the comparisons: its wrapper converts categorical inputs to integer embedding indices, preventing reliable gradient attribution to those inputs.

5.6.1 Global Feature Importance

The global view uses all 200 000 BAF Base TEST observations and computes importance for all 30 original features before selecting each model’s own top 15. Numerical features use their mean absolute attribution; categorical features use the mean of the per-observation sum of absolute one-hot contributions. The latter can exceed the absolute value of the corresponding signed sum because opposite contributions may cancel. All three explainers operate in log-odds, but their different reference distributions preclude treating absolute attribution magnitudes as a cross-model performance or importance scale.

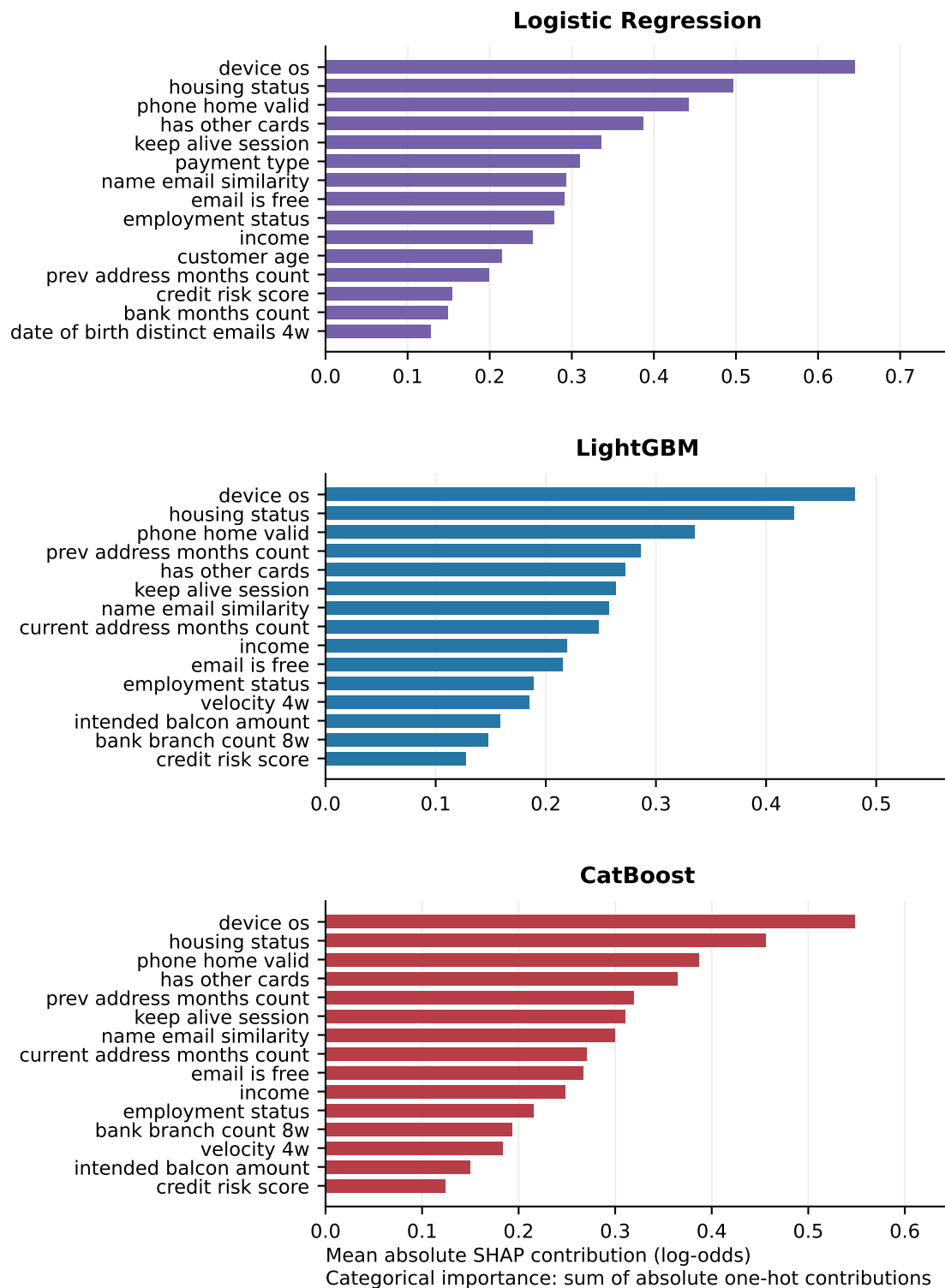


Figure 5.7: Each model's own top-15 original-feature SHAP importance on BAF Base, in log-odds. Categorical importance sums absolute one-hot contributions before averaging.

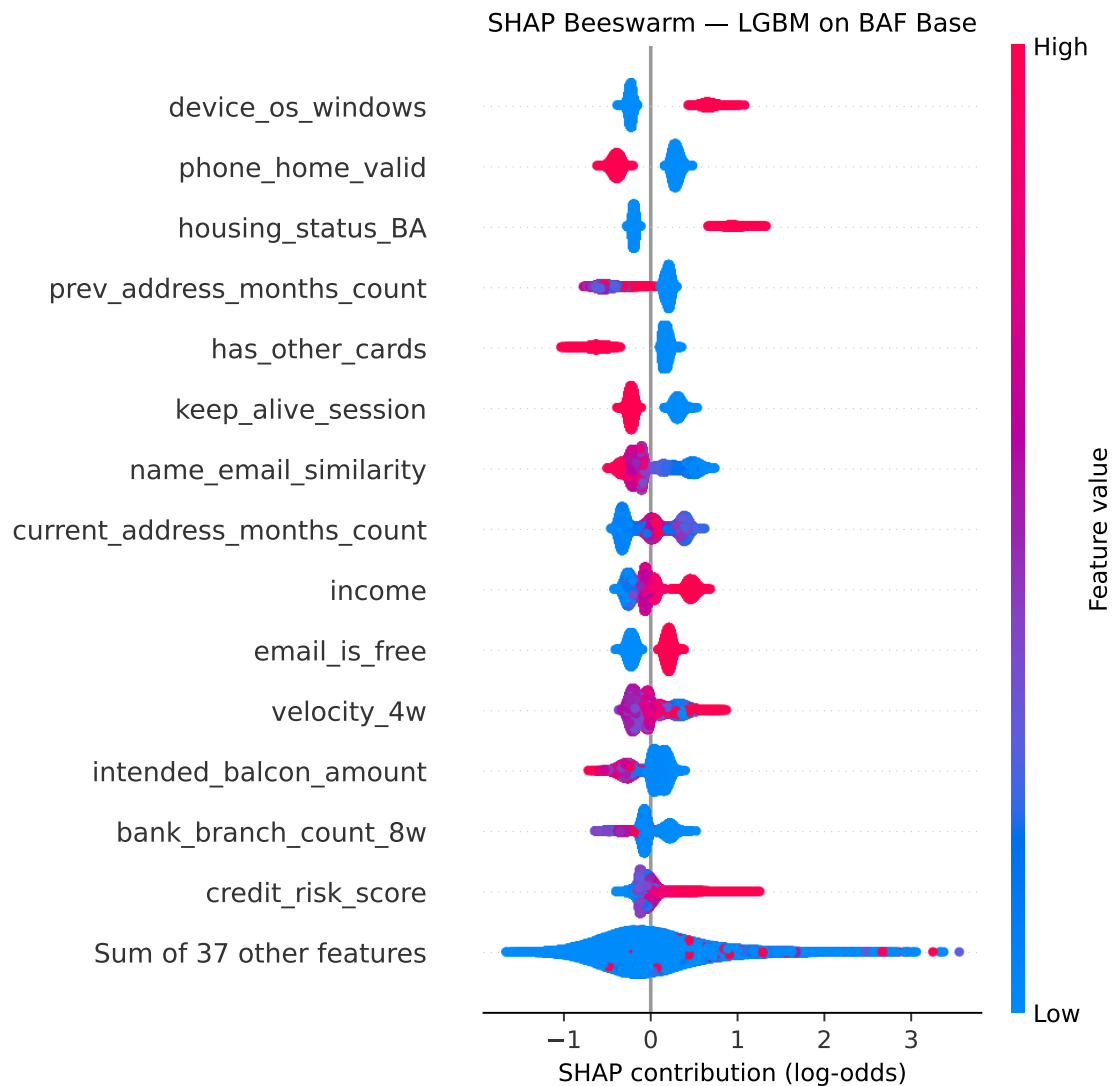


Figure 5.8: LGBM SHAP beeswarm in log-odds: the 14 leading encoded columns and the signed sum of the remaining 37. Dots are TEST observations; colour indicates the transformed feature value (red = high, blue = low).

Figure 5.7 shows that LR, LGBM, and CatBoost share the same three leading original features — `device_os`, `housing_status`, and `phone_home_valid`. The LGBM beeswarm in Figure 5.8 instead displays encoded-column contributions, with the remaining columns combined into a signed aggregate rather than assigned zero importance. Low values of `phone_home_valid` are associated with positive SHAP contributions, while the one-hot `device_os` columns show distinct contribution patterns. These are associations in the fitted model’s explanations, not causal effects or independently validated fraud indicators.

5.6.2 Cross-Model Feature Consistency

The Jaccard similarity between top-10 feature sets quantifies membership overlap; it does not compare the order within each set.

Table 5.26: Jaccard similarity of top-10 original-feature sets from the explicitly pinned LR, LGBM, and CatBoost baseline explanations on BAF Base. Agreement concerns set membership, not ranking order.

	LR	LGBM	CatBoost
LR	—	0.67	0.67
LGBM	0.67	—	1.00
CatBoost	0.67	1.00	—

LGBM and CatBoost have the same top-10 feature set (Jaccard = 1.00), although the order within that set is not identical. Each shares eight of ten features with Logistic Regression, corresponding to $\text{Jaccard} = 8/(10 + 10 - 8) = 0.67$. This agreement supports a common global candidate set across the fitted linear and boosted-tree models, but it does not prove identical reasoning, equal attribution magnitudes, or causal importance.

5.6.3 Cross-Variant Feature Stability

The same Jaccard analysis is applied to Base TEST and the five corrected Variant TEST cohorts under the common 30-feature projection. Every Variant profile matching any Base DEV profile is excluded before explanation, using the same predeclared populations as the transfer experiment. The retained Variant cohorts contain 177 526–178 828 observations; Base retains its 200 000 TEST observations. All retained observations are explained with the frozen Base-trained LGBM, without refitting the model or selecting a threshold on a Variant. Exact native tree-path-dependent TreeSHAP reproduces the saved Base explainer and uses the same raw log-odds and original-feature grouping throughout.

Table 5.27: Jaccard similarity of the frozen LGBM top-10 original-feature sets on Base TEST and the corrected, Base-DEV-profile-disjoint Variant cohorts.

	Base	Var. I	Var. II	Var. III	Var. IV	Var. V
Base	—	1.00	1.00	1.00	1.00	1.00
Var. I	1.00	—	1.00	1.00	1.00	1.00
Var. II	1.00	1.00	—	1.00	1.00	1.00
Var. III	1.00	1.00	1.00	—	1.00	1.00
Var. IV	1.00	1.00	1.00	1.00	—	1.00
Var. V	1.00	1.00	1.00	1.00	1.00	—

The Base-trained LGBM produces the same top-10 feature *set* on all six evaluated populations (Jaccard = 1.00 for every pairwise comparison). This result is recovered from the complete attribution matrices on the corrected cohorts, not inherited from the original overlapping Variant TEST populations. It records stable membership at the chosen value of k for this fixed model. The ordering is not identical: `has_other_cards` and `prev_address_months_count` exchange fourth and fifth positions between Base/Variant I and Variants II–V. Attribution magnitudes also vary across the populations. Set agreement therefore does not establish stable magnitudes, unchanged local explanations, invariance of the feature–target relationship, or unchanged predictive performance. The conclusion is further bounded by the common-feature projection and pooled-month evaluation described in Section 4.8. Explanations should still be monitored under distribution shift even when global top-10 membership is unchanged.

5.6.4 Local Explanations

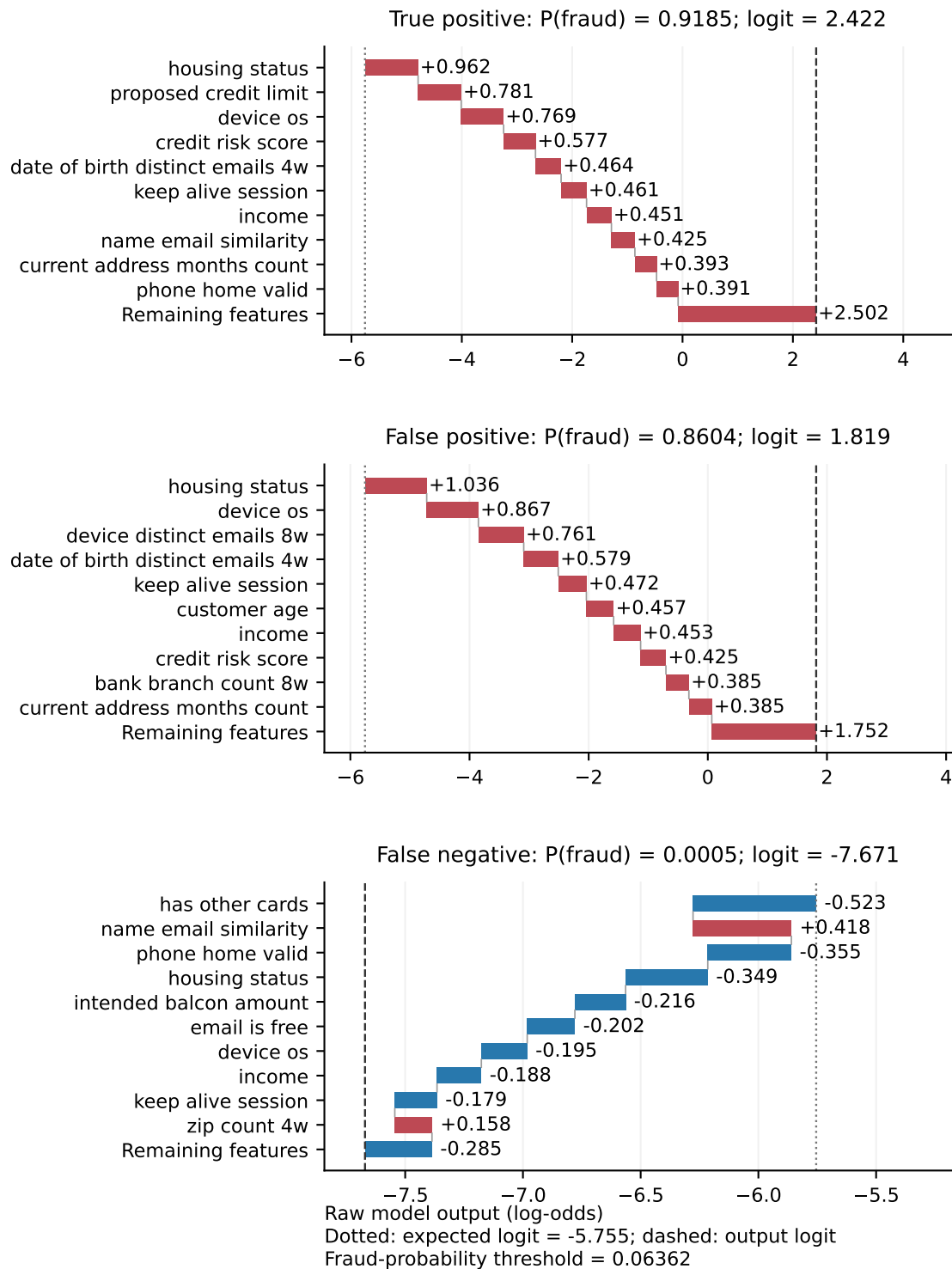


Figure 5.9: Additive LGBM log-odds waterfalls for the highest-score TP and FP and lowest-score FN. Ten original-feature contributions plus the residual connect the expected and case outputs.

The waterfall plots in Figure 5.9 preserve the complete additive decomposition of the LGBM raw output: the expected log-odds plus all signed contributions equals the individual output, which is converted to the displayed fraud score. Categorical one-hot contributions are summed with their signs intact. Red contributions increase the output towards fraud; blue contributions decrease it. The residual bar contains every feature outside the ten largest absolute original-feature contributions, so no attribution is silently omitted.

These cases are deliberately selected extremes: the highest-scoring true positive and false positive, and the lowest-scoring false negative, under the fixed validation-derived threshold. In the true-positive case, positive contributions combine to place the score above that threshold. The false-positive case contains positive contributions from a similar subset of features but is legitimate, illustrating that an explanation describes the fitted model rather than validating its prediction. Negative contributions outweigh positive contributions in the false-negative case. These examples are illustrative local decompositions, not representative estimates of the corresponding error classes.

5.7 FT-Transformer Attention Diagnostics

This section addresses the attention component of RQ5. The exploratory diagnostic describes the final-layer, head-averaged [CLS] attention distribution; it is not a faithful feature-attribution method and does not identify the cause of predictive performance. The analysis examines overall concentration through normalised entropy, [CLS] self-attention, the balance between categorical and numerical tokens, and selected case-level matrices.

The quantities are computed on all 200 000 BAF Base TEST samples using the baseline FT-Transformer configuration ($\tau = 0.058$). The model operates on 31 tokens: 25 numerical features, 5 categorical features, and the [CLS] classification token. Only the final transformer layer is aggregated, and PyTorch averages the weights across attention heads.

5.7.1 Aggregate Attention Distribution

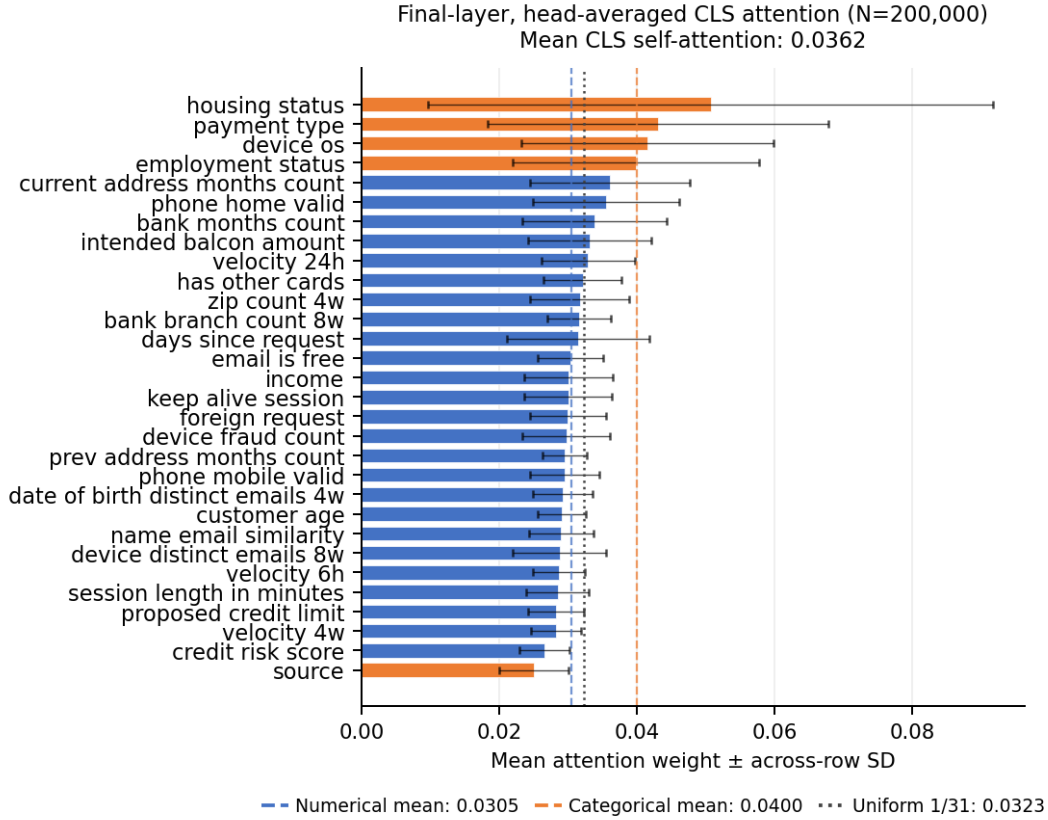


Figure 5.10: Mean attention weight per feature token aggregated over all 200 000 BAF Base test samples. Categorical features are shown in orange; numerical features in blue. Error bars represent standard deviation across samples. A perfectly uniform distribution would assign $1/31 \approx 3.2\%$ to each token.

Four statistics summarise the aggregate distribution shown in Figure 5.10. The mean normalised entropy is $H_{\text{norm}} = 0.985$: the observed mean entropy (3.381 nats) is 98.5% of the theoretical maximum for 31 tokens ($\ln 31 = 3.434$ nats), including [CLS]. Mean [CLS] self-attention is 3.6%, compared with the uniform reference of $1/31 \approx 3.2\%$. Mean attention per categorical and numerical token is 4.0% and 3.1%, respectively; their unrounded ratio is $1.311\times$. The three largest mean feature-token weights are **housing_status** (5.1%), **payment_type** (4.3%), and **device_os** (4.2%). Housing status and device operating system also appear among the leading SHAP features, but this descriptive overlap does not make attention an attribution method. The final-layer, head-averaged [CLS] distribution is diffuse rather than strongly concentrated; individual queries, heads, and earlier layers need not follow the same pattern.

5.7.2 Individual Case Attention Heatmaps

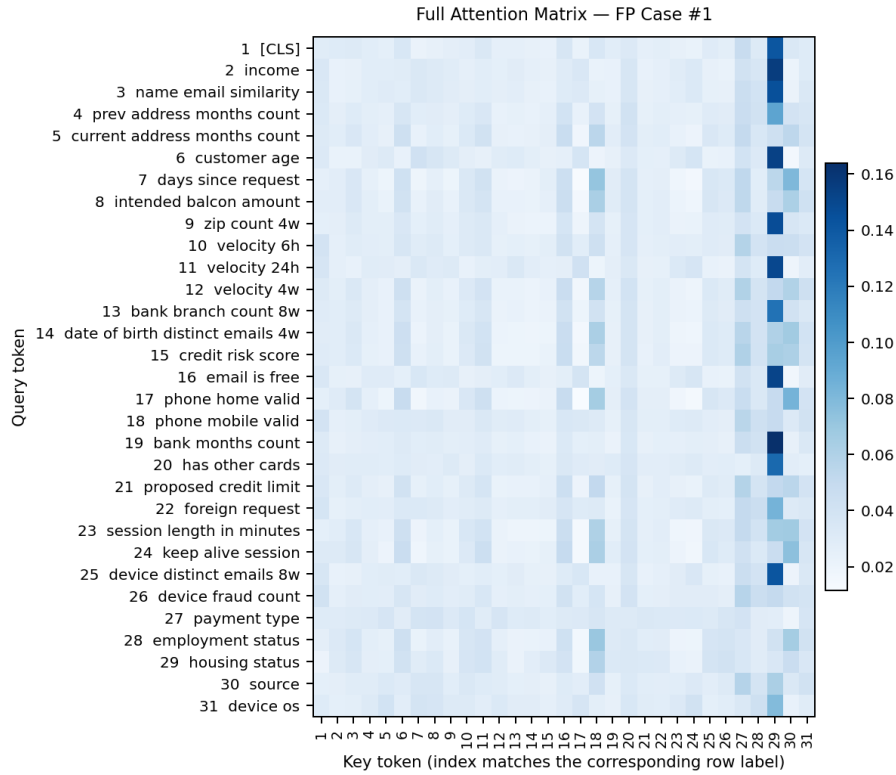


Figure 5.11: Final-layer, head-averaged 31×31 attention matrix for the false-positive case nearest the frozen threshold (score $\approx \tau = 0.0584$). Each cell (i, j) shows the mean attention weight from query token i to key token j . Case-specific elevations, including the housing-status key column, need not resemble the aggregate [CLS] distribution.

The selected false-positive matrix (Figure 5.11) has an elevated `housing_status` key column for several queries despite many lower weights elsewhere. The selected false-negative example in Appendix A.2 (p. 102) has a different case-specific allocation. These six inspected errors were selected as the three FP and three FN cases nearest the frozen baseline threshold, not as representative samples of all errors. Their matrices illustrate local heterogeneity and do not determine how earlier layers, individual heads, residual paths, or feed-forward blocks contributed to the prediction.

The attention diagnostic supports one limited conclusion: the final-layer, head-averaged [CLS] attention is diffuse on average for the fitted baseline model. It does not show that all available signal has been exhausted, that no feature subset could improve prediction, or that the close CatBoost and FT-Transformer PR-AUC

values define an information-theoretic ceiling. Their predictive comparison remains empirical, with its causes unresolved. Likewise, the 1.311 categorical-to-numerical mean ratio is descriptive and is not evidence of superior categorical processing.

5.8 Consolidated Summary

This section consolidates the experimental evidence into three practitioner-oriented summaries: overlaid PR curves, an operational evidence table mapping priorities to evaluated model-strategy-threshold configurations, and bootstrap and computational-cost statistics that describe sampling uncertainty and resource requirements.

5.8.1 PR Curves — All Models Overlaid (Baseline)

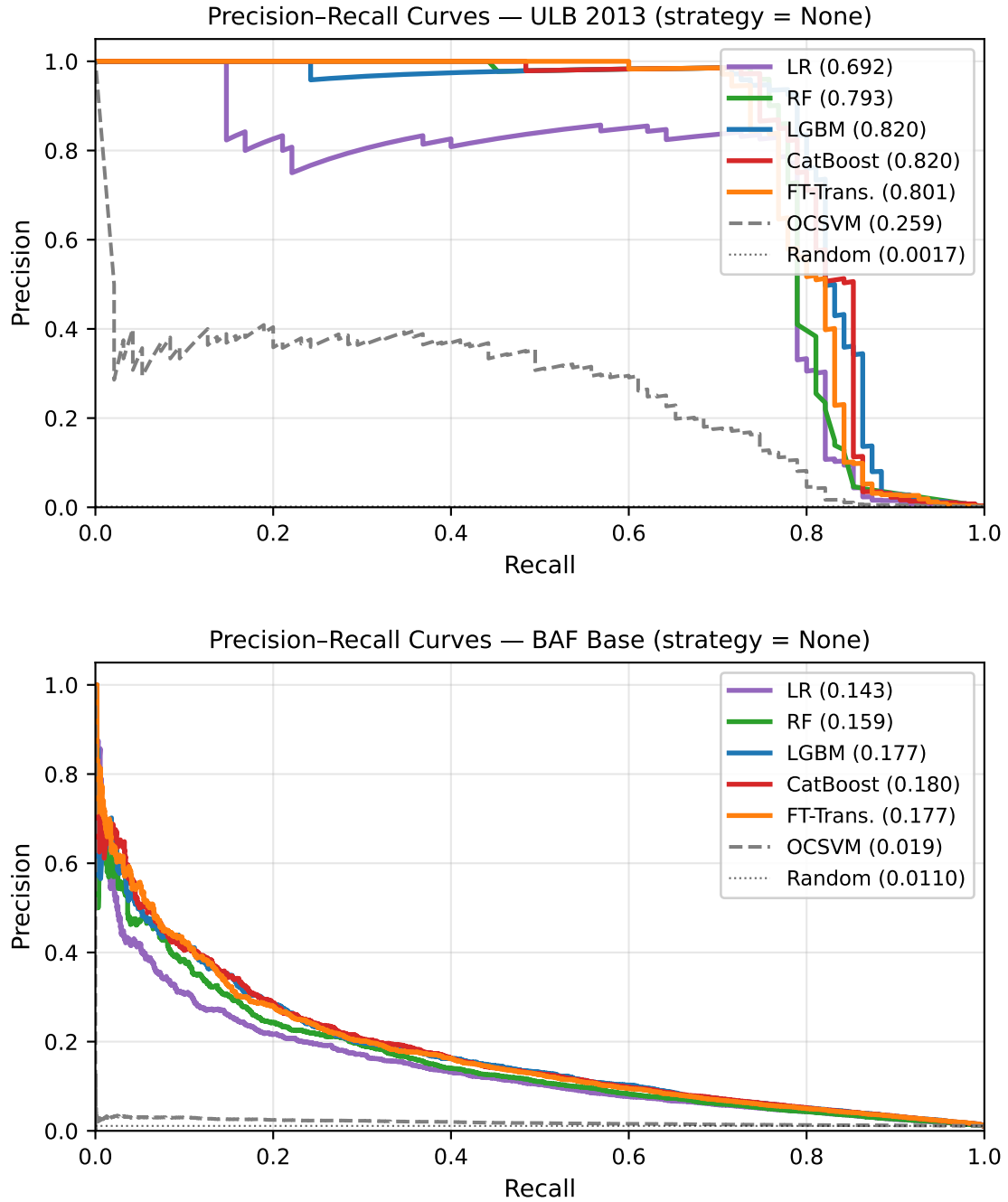


Figure 5.12: Precision-Recall curves for all models on ULB 2013 (top) and BAF Base (bottom) under the baseline setting (strategy = None). OCSVM is included for reference.

The PR curves are consistent with the tabulated baseline ordering in Section 5.1. On ULB 2013, LGBM and CatBoost have the largest average precision, followed by FT-

Transformer and RF; curve crossings mean this ordering need not hold at every recall level. Logistic Regression is lower by AP and OCSVM is weaker. On BAF Base, the supervised curves occupy a substantially lower-precision region despite BAF having a higher TEST fraud prevalence than ULB (approximately 1.10% vs. 0.167%). The lower BAF PR-AUC is therefore consistent with weaker score separability in the evaluated setting and cannot be attributed to a lower positive prevalence. LGBM, CatBoost, and FT-Transformer are closely clustered by point estimate, whereas OCSVM remains separated below them.

5.8.2 Decision Framework

Table 5.28: Descriptive summary of evaluated configurations for representative operational priorities; this is not a prospectively validated deployment rule.

Decision context	Model	Strategy	Threshold
Recall-weighted F_2 , ULB	RF	Weights	$\max\text{-}F_2$
Balanced F_1 , ULB	LGBM	None	$\max\text{-}F_2$
Recall-weighted F_2 , BAF	LGBM	Weights	$\max\text{-}F_2$
Balanced F_1 , BAF	CatBoost	None	$\max\text{-}F_1$
BAF Variant transfer	LGBM / CB	None	$\max\text{-}F_2$
Interpretability / cost, BAF	LGBM	None	$\max\text{-}F_2$
Precision target, ULB	CatBoost	None	$\text{Prec} \geq 0.5$
Precision target, BAF	FT-Trans.	None	$\text{Prec} \geq 0.5$

Table 5.28 summarises eight representative decision contexts. The recall-weighted rows use the highest observed factorial F_2 point estimate on each dataset: RF with Class Weights on ULB (0.821) and LGBM with Class Weights on BAF (0.320). Maximising F_2 favours recall relative to precision, but does not necessarily maximise recall alone. The balanced- F_1 rows use the baseline threshold-study maxima: LGBM with $\max\text{-}F_2$ on ULB ($F_1 = 0.852$) and CatBoost with $\max\text{-}F_1$ on BAF ($F_1 = 0.244$). A rule maximised on validation need not maximise the same metric on TEST. Under the validation precision target, CatBoost and LGBM tie for the highest ULB TEST recall (0.821), with CatBoost raising slightly fewer alerts; FT-Transformer has the highest BAF TEST recall (0.114) under that rule. The transfer

row identifies LGBM and CatBoost (CB) jointly because the leading Variant model depends on the metric and cohort, not a single universal winner (Section 5.5). For BAF global interpretability and computational cost, LGBM illustrates a competitive performance–TreeSHAP trade-off rather than the maximum of a predefined joint objective; the cost provenance and measurement qualifications below remain essential. Every named configuration was evaluated jointly; no result is constructed by combining separately evaluated settings. These post-hoc summaries are conditional on the evaluated cohorts and do not replace validation in a new deployment setting. The Precision ≥ 0.5 rule enforces its target on validation only and does not guarantee that half of TEST or future alerts will be fraud or that the resulting alert volume will be low.

5.8.3 Bootstrap Confidence Intervals

The bootstrap CIs in Tables 5.29 and 5.30 quantify the sampling uncertainty of the baseline PR-AUC and ROC-AUC estimates.

Table 5.29: 95% bootstrap confidence intervals for PR-AUC and ROC-AUC on ULB 2013 (strategy = None, 1 000 iterations).

Model	PR-AUC	95% CI	ROC-AUC	95% CI
LR	0.692	[0.589, 0.787]	0.957	[0.926, 0.982]
RF	0.793	[0.709, 0.874]	0.929	[0.889, 0.962]
LGBM	0.820	[0.741, 0.892]	0.980	[0.962, 0.993]
CatBoost	0.820	[0.742, 0.892]	0.980	[0.966, 0.991]
FT-Trans.	0.801	[0.721, 0.876]	0.967	[0.941, 0.990]
OCSVM	0.259	[0.183, 0.363]	0.922	[0.879, 0.959]

Table 5.30: 95% bootstrap confidence intervals for PR-AUC and ROC-AUC on BAF Base (strategy = None, 1 000 iterations).

Model	PR-AUC	95% CI	ROC-AUC	95% CI
LR	0.143	[0.131, 0.158]	0.875	[0.867, 0.882]
RF	0.159	[0.145, 0.175]	0.878	[0.870, 0.885]
LGBM	0.177	[0.162, 0.194]	0.898	[0.892, 0.905]
CatBoost	0.180	[0.164, 0.197]	0.898	[0.892, 0.904]
FT-Trans.	0.177	[0.162, 0.195]	0.896	[0.890, 0.903]
OCSVM	0.019	[0.018, 0.021]	0.628	[0.616, 0.639]

These percentile intervals quantify TEST-row sampling variability conditional on one frozen fitted model, one split, and one selected configuration; neither fitting nor model selection is repeated inside the bootstrap. Overlap of marginal intervals is not a paired test of model differences and does not establish statistical equivalence. The primary baseline-model ranking is descriptive: paired between-model intervals are not calculated for that comparison. Paired intervals are used separately for the predefined missing-value and categorical-aware sampling controls. Even a paired difference interval containing zero would not establish equivalence, which would require a predefined practically acceptable margin and an appropriate equivalence procedure.

5.8.4 Computational Cost

The cost tables show that model selection depends on computational requirements as well as predictive and operational metrics.

Table 5.31: Recorded computational cost on ULB 2013 (strategy = None). The supervised searches use 50 trials with model-specific validation; OCSVM is not tuned. ‡ denotes the resumed LightGBM invocation’s wall time, excluding earlier trials’ elapsed time.

Model	Tuning (s)	Train (s)	Inference (s)
LR	17,132	100.5	0.082
RF	42,011	153.6	0.373
LGBM	367‡	1.4	0.082
CatBoost	2,081	9.0	0.032
FT-Trans.	17,275	596.4	1.677
OCSVM	—	62.0	7.585

Table 5.32: Recorded computational cost on BAF Base (strategy = None). † marks historical measurements reused with frozen classical models. Revised FT-Transformer uses the preserved hyperparameters without a new search; OCSVM is not tuned. Validation-only recovery costs are recorded separately, not as final-fit time.

Model	Tuning (s)	Train (s)	Inference (s)
LR	55,070 [†]	83.9 [†]	0.209
RF	105,313 [†]	1494.0 [†]	10.642
LGBM	1,243 [†]	16.1 [†]	1.298
CatBoost	7,010 [†]	36.7 [†]	0.213
FT-Trans.	—	354.4	4.262
OCSVM	—	919.6 [†]	71.579 [†]

These measurements document the executed workflows rather than an isolated model-speed benchmark. Classical models use CPU computation, whereas FT-Transformer uses the GPU; validation designs and optimisation work also differ. Historical BAF tuning and final-fit costs are retained with the frozen classical models; their validation-score recovery is additional work, not a new final fit. Inference is remeasured for the four supervised classical BAF models, whereas the OCSVM inference measurement remains historical. The revised BAF FT-Transformer fit uses fixed hyperparameters, and the resumed ULB LightGBM timing does not represent the complete search wall time. Concurrent workloads, software implementation, and realised fit length can affect the measurements. They therefore indicate the recorded resource burden of these runs, not intrinsic architectural efficiency or a transferable speed-up factor. Deployment-oriented cost comparisons would require controlled reruns under matched measurement conditions.

5.9 Synthesis and Threats to Validity

On BAF, the largest observed within-model changes in threshold-dependent F_2 arise from the decision rule; PR-AUC, by definition, is unaffected by this choice. No supervised model dominates every criterion. LGBM narrowly leads ULB baseline PR-AUC, while LGBM, CatBoost, and FT-Transformer have closely grouped BAF point estimates with overlapping marginal bootstrap intervals, without established equivalence. The corrected BAF transfer experiment favours LGBM or CatBoost depending on the Variant and metric, not RF. LGBM combines competitive measured performance and exact TreeSHAP, subject to the computational-cost qualifications. Imbalance effects are model- and dataset-dependent: RF benefits in ULB ranking, whereas the boosted trees have little BAF PR-AUC gain. For FT-Transformer, Class Weights remains close to baseline on BAF but not on ULB, while the approximately 35% SMOTE reduction is conditional on the implemented representation and sampling path. The predefined SMOTE-NC control does not recover baseline AP, and the missing-value control provides no uniform model-wide improvement; neither selects the primary policy using TEST. SHAP records top-feature-set agreement among LR, LGBM, and CatBoost, not identical ranking or causality; FT-Transformer SHAP comparisons are excluded. Attention records high mean final-layer, head-averaged [CLS] entropy alongside case-level heterogeneity, but neither proves a feature-space ceiling nor explains the SMOTE result.

Threats to Validity

- **Internal:** supervised baseline searches use 50 Optuna trials, repeated for the corrected ULB protocol but retained from the original BAF selection. A single outer holdout is used instead of repeated splits. The search spaces and validation costs differ, so equal trial counts do not imply equal exploration or computational budgets.
- **Validation protocol:** the classical models use five-fold inner validation, whereas FT-Transformer uses one internal holdout. The shared TEST partition remains untouched, but threshold dispersion and model-selection variance are not estimated on the same basis.
- **Resampling representation:** SMOTE operates on one-hot columns for the classical models and on concatenated numerical values and ordinal category codes for FT-Transformer. Subsequent integer conversion of synthetic FT

categorical values is a confound, so the observed SMOTE interaction cannot be attributed solely to architecture. The predefined SMOTE-NC controls alter sampling representation and nominal-aware resampling jointly; they do not isolate one component or turn CatBoost into a native-categorical classifier.

- **Absence-code representation:** the primary BAF experiment preserves documented numerical absence codes. The predefined mean-imputation-plus-indicator sensitivity uses fixed hyperparameters and is reported separately; it does not determine an optimally tuned missing-value pipeline or justify selecting the primary policy using TEST performance.
- **Evaluated populations:** ULB estimates concern distinct observed predictor profiles after deduplication. BAF Variant cohorts exclude exact common-predictor matches to Base DEV, but can share profiles with Base TEST or other Variants. Exact-profile separation does not prove entity-level independence, and the common 30-feature projection excludes the additional separability variables in Variants III/V.
- **External:** ULB features are PCA-anonymised, limiting interpretability and the real-world applicability of ULB-specific findings. The BAF suite is synthetically generated, so fraud patterns may not fully reflect the complexity of production fraud streams, where label noise, adversarial adaptation, and operational feedback loops are present.
- **Construct:** PR-AUC is sensitive to the class prior, so absolute comparisons across ULB, BAF Base, and BAF Variants with different fraud rates require caution. ROC-AUC and operating-point metrics provide complementary views but do not remove all transfer confounding.
- **Interpretability:** FT-Transformer GradientExplainer attributions are unreliable for categorical embedding indices and are excluded from cross-model conclusions. The attention diagnostic averages heads in the final layer and may hide head- or layer-specific structure; attention is not a faithful output attribution.
- **Statistical:** the row bootstrap quantifies TEST-sampling uncertainty conditional on the fitted models and assumes that row resampling represents the relevant sampling process. It does not estimate entity-cluster dependence, split, seed, training, or tuning variability. The intervals are not adjusted for

the complete set of comparisons, and overlapping marginal intervals are not a paired test of model differences.

- **Cross-dataset:** ULB and BAF have incompatible feature spaces, which prevents direct cross-dataset transfer of the fitted models. The cross-domain study is restricted to the BAF internal transfer setting, which, while controlled, represents a narrower robustness evaluation than matched feature spaces across independent datasets would provide.

6

Conclusion

This thesis investigated five questions concerning model comparison, threshold selection, imbalance interventions, BAF Base-to-Variant transfer, and model interpretation in fraud detection under extreme class imbalance. The experimental programme covers two benchmark datasets, six model families, seven imbalance handling strategies for the five supervised models, four threshold rules, SHAP analysis for compatible model families, and an FT-Transformer attention diagnostic. The shared outer holdout is excluded from predictive fitting and selection, with exact-profile separation checked explicitly; model-specific inner validation and the remaining population-independence limits remain part of the evidence boundary.

6.1 Summary of Contributions

The six contributions follow the sequence established by the research design; Contributions 5 and 6 jointly address RQ5.

The foundation is a **leakage-free unified comparison** (Contribution 1) that provides a side-by-side evaluation of classical supervised models and FT-Transformer on the same two datasets, with common metrics and an outer holdout that is excluded from preprocessing, model selection, resampling, and threshold calibration. The classical models use five-fold inner validation and FT-Transformer uses a single internal holdout, so the comparison is unified at the outer-evaluation level rather than identical at every inner-training step.

Built on that foundation, the **threshold selection study** (Contribution 2) shows that the decision rule materially changes threshold-dependent metrics. On BAF Base, fixing the threshold at 0.5 yields $F_2 < 0.05$ for every supervised model, whereas validation-selected max- F_2 thresholds produce values from 0.2868 to 0.3159. On ULB, the same rule improves Logistic Regression from 0.618 to 0.771, but does not

improve the TEST F_2 of CatBoost or FT-Transformer relative to fixed 0.5. Within the evaluated workflow, these results support explicit validation-based threshold selection while showing that a DEV-optimal rule need not improve every TEST metric.

The **full factorial analysis** (Contribution 3) across seven imbalance handling strategies, five supervised models, and two datasets shows that strategy effectiveness is neither universal nor model-agnostic. On BAF, SMOTE reduces FT-Transformer PR-AUC by approximately 35% (0.1769 to 0.1149) and CatBoost by approximately 14% (0.1796 to 0.1543). However, the FT-Transformer sampler interpolates ordinal category codes before converting the synthetic values to integer embedding indices. The fixed-hyperparameter categorical-aware control reaches 0.1157 PR-AUC and does not recover the baseline in this configuration; it does not isolate architecture as the cause. Class Weights remains close to the FT-Transformer baseline on BAF but reduces its ULB PR-AUC. The sampler audit also shows that SMOTE+Tomek removes no rows in the logged revised fitting stages and produces the same TEST predictions as SMOTE. The seven named conditions therefore do not imply seven distinct effective resampling treatments, and no intervention is supported as a universal default.

The **cross-domain generalisation study** (Contribution 4) shows that Base performance alone does not describe the Variant profile. After excluding exact common-predictor matches to Base DEV, LGBM records the highest PR-AUC point estimate on Variant I and CatBoost on Variants II–V. CatBoost has the highest F_2 point estimate on Variant I, while LGBM leads on Variants II–V. Every supervised model has lower F_2 on every Variant than on Base under its unchanged Base threshold. LGBM, CatBoost, and FT-Transformer have their largest PR-AUC reductions on Variants III and V. These are descriptive distribution-transfer comparisons under a common 30-feature projection, not an isolated test of the excluded separability features or prospective temporal robustness. No in-domain Variant models were trained, and differences may also reflect prevalence and cohort composition.

The **SHAP cross-model and cross-variant analysis** (Contribution 5) records two bounded results using separately pinned explanation sources for the same frozen BAF Base fitted models used in the predictive comparison. LGBM and CatBoost have the same top-10 feature set (Jaccard = 1.00), and each shares eight of ten members with Logistic Regression (Jaccard = 0.67). For the fixed Base-trained LGBM model, the same top-10 set is recovered on all six BAF test distributions. These

results concern set membership, not identical ordering, attribution magnitude, local-explanation validity, or causal importance. FT-Transformer GradientExplainer results are excluded from this contribution because integer conversion of categorical indices prevents reliable gradient attribution to those inputs.

Finally, the **FT-Transformer attention diagnostic** (Contribution 6) finds diffuse final-layer, head-averaged [CLS] attention (normalised entropy = 0.985) across the 31 tokens of the evaluated BAF Base FT-Transformer. This result describes one aggregated attention view; it does not establish an information ceiling, show that all signal has been used, or rule out head-, layer-, or architecture-specific limitations.

6.2 Answers to Research Questions

RQ1 — Comparative performance under a leakage-free protocol. Under the shared leakage-free outer evaluation on distinct ULB predictor profiles, LGBM has the highest PR-AUC point estimate (0.8205), followed by CatBoost (0.8196), FT-Transformer (0.8015), RF (0.7928), Logistic Regression (0.6921), and OCSVM (0.2593). On BAF Base, CatBoost, FT-Transformer, and LGBM are closely grouped at 0.1796, 0.1769, and 0.1768. The observed gap is therefore dataset-dependent, but the experiments do not isolate dataset size, calibration, or inductive bias as its cause, nor establish practical equivalence among close point estimates. FT-Transformer does not exceed the highest GBDT point estimate on either benchmark and has greater recorded final-fit and inference times than LGBM; mixed execution provenance prevents a controlled hardware-speed interpretation (Section 5.1, p. 51).

RQ2 — Impact of threshold selection strategy. Threshold selection has a dataset- and model-dependent impact on operational metrics. On ULB, Logistic Regression improves from 0.618 to 0.771 F_2 under max- F_2 , while CatBoost and FT-Transformer are lower than under fixed 0.5. On BAF, fixed 0.5 gives $F_2 < 0.05$ for every supervised model, while max- F_2 produces 0.2868–0.3159. Rankings change under different rules on both datasets, so any claim about the best model under a threshold-dependent metric must specify the operating rule. All non-fixed thresholds are selected inside DEV, and the precision constraint is a validation target rather than a TEST guarantee (Section 5.2, p. 55).

RQ3 — Factorial interaction of imbalance strategies and model type. No single imbalance strategy improves every model and dataset. The baseline is the

highest-PR-AUC condition for four of five supervised models on ULB, while SMOTE and SMOTE+Tomek give the highest Random Forest estimate (0.814 versus 0.793 without intervention). On BAF, LGBM and CatBoost gain less than 0.01 PR-AUC from any intervention. FT-Transformer’s approximately 35% SMOTE reduction is larger than CatBoost’s approximately 14% reduction, but the categorical resampling representation is a confound and the fixed-hyperparameter categorical-aware control does not recover the transformer baseline. SMOTE+Tomek and SMOTE are identical effective treatments in the logged revised fitting stages. Class Weights is close to the FT-Transformer baseline on BAF and lower on ULB; strategy selection should therefore be validated for the target model and data rather than chosen as a universal default (Sections 5.3 and 5.4, pp. 59 and 65).

RQ4 — Cross-domain generalisation under distribution shift. Base-trained models retain different levels of performance across the filtered BAF Variant cohorts, and the strongest point estimate depends on the metric. LGBM has the highest PR-AUC on Variant I; CatBoost leads on Variants II–V. CatBoost has the highest F_2 on Variant I; LGBM leads on Variants II–V. Variant II PR-AUC is slightly above Base for the four classical supervised models, while FT-Transformer is slightly lower. All supervised models’ F_2 values decrease on all five Variants under their frozen Base thresholds. Variants III and V have the largest PR-AUC reductions for LGBM, CatBoost, and FT-Transformer, but injected separability features are excluded and months are pooled. Exact Base DEV matches are removed, not all shared generating-process or entity dependence. The study therefore supports bounded distribution-transfer comparisons, not a causal explanation or independent prospective robustness (Section 5.5, p. 70).

RQ5 — Feature attribution consistency and transformer attention diagnostics. For the frozen BAF Base fitted models, LGBM and CatBoost have the same top-10 SHAP feature set, and each overlaps Logistic Regression on eight members. The separately pinned explanation sources describe the same fitted models used in the predictive comparison. The fixed Base-trained LGBM model also retains the same top-10 set across all BAF test distributions, including the filtered Variant cohorts. These are set-membership results and do not establish identical rankings or explanation validity under shift. FT-Transformer SHAP is excluded from cross-model conclusions because of its categorical-gradient limitation. The evaluated BAF Base FT-Transformer’s final-layer, head-averaged attention is dif-

fuse (normalised entropy = 0.985), which characterises the inspected attention view but does not prove an information limit or explain close predictive point estimates (Sections 5.6 and 5.7, pp. 72 and 78).

6.3 Limitations

Several limitations constrain the scope and generalisability of the findings beyond the protocol-level threats to validity discussed in Section 5.9 (p. 87).

Single custom deep learning implementation. The custom FT-Transformer implementation is the sole representative of the tabular deep learning family. Conclusions about attention behaviour, imbalance sensitivity, and transfer performance are conditional on this implementation, its selected hyperparameters and the evaluation protocol; they are not a direct assessment of every FT-Transformer implementation or of tabular transformers generally. They may not generalise to SAINT [33], TabTransformer [32], TabNet, or hybrid architectures. Whether similar attention distributions occur in other implementations remains an open question.

Model-specific validation. The classical models use five-fold inner validation, whereas FT-Transformer uses one internal DEV holdout. Both paths protect the outer TEST set, but they do not provide equivalent estimates of selection and threshold variability. The fixed split and random seed also leave training variability unmeasured.

Conditional uncertainty. The row-bootstrap intervals condition on the fitted models, selected thresholds and observed split. They do not estimate entity-cluster dependence or variation from refitting, and they are not simultaneous intervals across all reported comparisons. Neither overlapping marginal intervals nor a paired difference interval containing zero establishes practical equivalence.

Resampling representation. The primary SMOTE path is not categorical-aware. It interpolates one-hot columns for the classical models and ordinal category codes for FT-Transformer, whose synthetic categorical values are then converted to integers. Supplementary SMOTENC controls for CatBoost and FT-Transformer distinguish this primary path from categorical-aware interpolation under fixed hyperparameters. They do not isolate architecture as the sole cause of any difference or establish an optimally tuned alternative pipeline. The Tomek cleaner removes no observations in the logged revised fitting stages, so SMOTE and SMOTE+Tomek

do not constitute distinct effective treatments there. Historical BAF classical saved scores also coincide up to floating-point noise, but reconstructed full-DEV sampler behaviour is not a substitute for unavailable historical fold logs.

Absence-code representation. The primary BAF policy preserves documented numerical absence codes, whose magnitudes enter numerical preprocessing. The predefined sensitivity changes that policy to training-fitted mean imputation with missingness indicators while retaining the baseline hyperparameters. The observed PR-AUC changes have mixed directions across models. This evaluates one representation intervention rather than the best possible missing-value treatment; the primary policy is not selected retrospectively from TEST results.

Interpretability scope. Classical SHAP sources are separately pinned to the same frozen BAF Base fitted models used in predictive reporting; their verified identity does not establish feature causality or explanation validity under shift. FT-Transformer GradientExplainer cannot reliably attribute categorical embedding indices because of the integer conversion in the explainer wrapper, so those results are excluded from cross-model conclusions. The FT-Transformer attention analysis covers the final layer and averages across heads; it is descriptive and cannot establish feature causality or an information ceiling.

Dataset scope. The ULB dataset is privacy-constrained — PCA-transformed features preclude domain interpretation and the two-day collection window limits temporal generalisability. The BAF suite is synthetically generated, which ensures experimental control but means fraud patterns may not fully reflect the complexity of real production fraud streams, where label noise, adversarial adaptation, and operational feedback loops are present.

Evaluation-population scope. ULB deduplication removes repeated observed predictor profiles before splitting, so the revised estimates describe distinct profiles rather than the original transaction population. BAF transfer excludes common-predictor matches to Base DEV and evaluates a shared 30-feature projection; it does not establish customer-level independence or isolate the additional separability variables excluded from Variants III/V. The synthetic cohorts share a generating process and may overlap with Base TEST, so their scores are descriptive stress-test results rather than independent prospective deployment estimates.

Absence of concept drift modelling. All experiments use a static train/test split. In operational fraud detection, the data distribution may shift as fraud pat-

terns and system use evolve. The experimental protocol does not capture this dynamic, and performance under temporal drift is not equivalent to performance on the controlled BAF Variants studied here.

Hyperparameter optimisation budget. Supervised baseline searches use 50 trials, but OCSVM is not tuned and the validation cost per trial differs between five-fold classical searches and the FT-Transformer holdout. The corrected ULB experiment repeats this search budget, whereas BAF retains its originally selected hyperparameters. Equal trial counts do not imply equal search coverage or computational cost. The FT-Transformer’s larger search space may remain under-explored relative to the gradient-boosted models. Recorded costs also span historical and new executions, including concurrent workloads and a resumed LightGBM search; they are not a controlled hardware-speed benchmark.

Cross-dataset incompatibility. The ULB and BAF feature spaces are incompatible (PCA-anonymised vs. semantic), preventing direct transfer between the datasets. The cross-domain study is therefore restricted to BAF internal transfer, a narrower robustness evaluation than matched feature spaces across independent datasets would provide.

6.4 Future Work

Five directions emerge directly from the findings and limitations of this thesis.

Extending attention diagnostics to other tabular transformer architectures. The diffuse final-layer average observed for FT-Transformer motivates analysis of individual heads, earlier layers, and alternative tabular transformers. Applying a comparable diagnostic to SAINT and TabTransformer could assess whether attention concentration differs by architecture and whether any difference is associated with predictive behaviour.

Temporal robustness and concept drift. The BAF dataset includes a monthly time variable, enabling month-ordered training and evaluation partitions. Such a design would be closer to prospective evaluation than the pooled random split used here, while remaining a synthetic benchmark rather than a production deployment. A direct extension would compare periodic threshold recalibration with full model retraining as the distribution evolves, since the two approaches have different operational costs.

Ensemble strategies based on predictive diversity. LGBM and FT-Transformer reach similar BAF aggregate performance, but the current SHAP implementation cannot establish complementary FT-Transformer feature use. Future work could measure error and score correlation directly before testing whether a stacking ensemble offers predictive benefit.

Native cost-sensitive training for the FT-Transformer. Class Weights remains close to baseline on BAF but reduces performance on ULB, so its effect is dataset-dependent. Future work could compare focal loss or direct F_β optimisation with categorical-aware resampling under a protocol designed specifically to separate loss-level and representation-level effects.

Richer feature representations. Augmenting BAF with derived behavioural features, velocity-based aggregates, graph connectivity, or device-fingerprinting signals would test whether the relative performance of transformer and tree models changes when additional cross-feature structure is available. The present attention diagnostic does not determine in advance whether such augmentation would help.

6.5 Closing Remarks

The central conclusion is methodological as well as predictive: model choice cannot be separated from validation design, threshold selection, imbalance handling, input representation, and the distribution on which the model is evaluated. Within the scope of the two benchmarks, gradient-boosted trees provide the highest baseline PR-AUC point estimates, while FT-Transformer remains close to the leading BAF Base models. LGBM also has lower recorded final-fit and inference times than FT-Transformer, although the execution provenance does not support a controlled hardware-speed comparison. These findings support evidence-bounded model selection rather than a universal preference for either model family.

A

Appendix

A.1 Selected Hyperparameter Configurations

The ULB supervised baselines use fresh 50-trial Optuna searches on the duplicate-safe DEV partition, with the TPE sampler, MedianPruner and a PR-AUC objective. The classical models use five-fold stratified validation, whereas the custom FT-Transformer uses one 80/20 holdout within DEV. The BAF tables retain the hyperparameters selected by the original 50-trial Base searches; the corrected FT fitting, missing-value sensitivity analyses and categorical-aware controls do not introduce another BAF hyperparameter search. Each dataset’s baseline configuration is reused across its imbalance strategies. One-Class SVM is outside the Optuna search and uses the explicit fixed configuration RBF kernel, $\nu = 0.01$ and `gamma=scale`.

Table A.1: Hyperparameter configurations selected by Optuna on ULB 2013 (strategy = None).

Model	Hyperparameter	Selected value
Logistic Regression	<code>C</code>	0.041997828
	<code>l1_ratio</code>	0.16446278
Random Forest	<code>n_estimators</code>	150
	<code>max_depth</code>	40
	<code>min_samples_split</code>	3
	<code>min_samples_leaf</code>	1
LightGBM	<code>num_leaves</code>	77
	<code>learning_rate</code>	0.049307233
	<code>min_child_samples</code>	84
	<code>reg_alpha</code>	0.0021804728
	<code>reg_lambda</code>	8.9391694
	<code>colsample_bytree</code>	0.585389
	<code>n_estimators</code>	176
CatBoost	<code>learning_rate</code>	0.012682107
	<code>depth</code>	8
	<code>l2_leaf_reg</code>	0.37035965
	<code>iterations</code>	446
FT-Transformer	<code>d_token</code>	256
	<code>n_blocks</code>	4
	<code>attention_n_heads</code>	8
	<code>attention_dropout</code>	0.059614704
	<code>ffn_d_hidden_factor</code>	1.3373996
	<code>ffn_dropout</code>	0.32618457
	<code>residual_dropout</code>	0.14137147
	<code>learning_rate</code>	0.00028708753
	<code>weight_decay</code>	0.00020597335
	<code>batch_size</code>	512
OCSVM	RBF; $\nu = 0.01$; <code>gamma=scale</code> ; no HPO	

Under the recorded scikit-learn 1.8 implementation, Logistic Regression uses `solver=saga` and an active `l1_ratio`: values between zero and one define Elastic Net regularisation. Both C and `l1_ratio` therefore affect the fitted model.

Table A.2: Selected BAF Base hyperparameters, reused for the primary strategies and predefined fixed-hyperparameter controls (strategy = None).

Model	Hyperparameter	Selected value
Logistic Regression	C	0.039054413
	l1_ratio	0.29122914
Random Forest	n_estimators	500
	max_depth	40
	min_samples_split	13
	min_samples_leaf	7
LightGBM	num_leaves	15
	learning_rate	0.013787968
	min_child_samples	18
	reg_alpha	7.0768435
	reg_lambda	1.3936074e-05
	subsample	0.7914708
	colsample_bytree	0.83852592
	n_estimators	1696
CatBoost	learning_rate	0.03342955
	depth	4
	l2_leaf_reg	7.8529431
	min_data_in_leaf	22
	iterations	1394
FT-Transformer	d_token	64
	n_blocks	2
	attention_n_heads	8
	attention_dropout	0.22482374
	ffn_d_hidden_factor	2.2473801
	ffn_dropout	0.11193986
	residual_dropout	0.14312179
	learning_rate	0.0003793186
	weight_decay	1.765024e-06
	batch_size	512
OCSVM	RBF; $\nu = 0.01$; gamma=scale; no HPO	

The historical LightGBM `subsample` suggestion does not activate row subsampling when `subsample_freq=0`; CatBoost's `min_data_in_leaf` does not change the default symmetric-tree growing policy. These recorded suggestions are shown for traceability but excluded from the corrected ULB search.

A.2 FT-Transformer Attention Diagnostics — Individual Case Bar Charts

The following figures supplement Section 5.7. Each chart shows the final-layer, head-averaged [CLS] attention distribution for one BAF Base TEST observation. Categorical feature tokens are orange and numerical tokens are blue; [CLS] self-attention is reported separately in the chart heading. The dotted vertical line marks the uniform reference $1/31$, and the dashed vertical lines mark the numerical- and categorical-token group means. These references are distinct quantities.

The frozen baseline threshold is $\tau = 0.058434$ (display rounding only). The three false positives and three false negatives nearest to its full-precision value are selected separately. Captions report the predicted fraud score, the original label and the absolute distance $|p - \tau|$; the displayed six-decimal scores are not used to recompute the error classes. These deliberately selected borderline examples are not representative estimates of the entire error population.

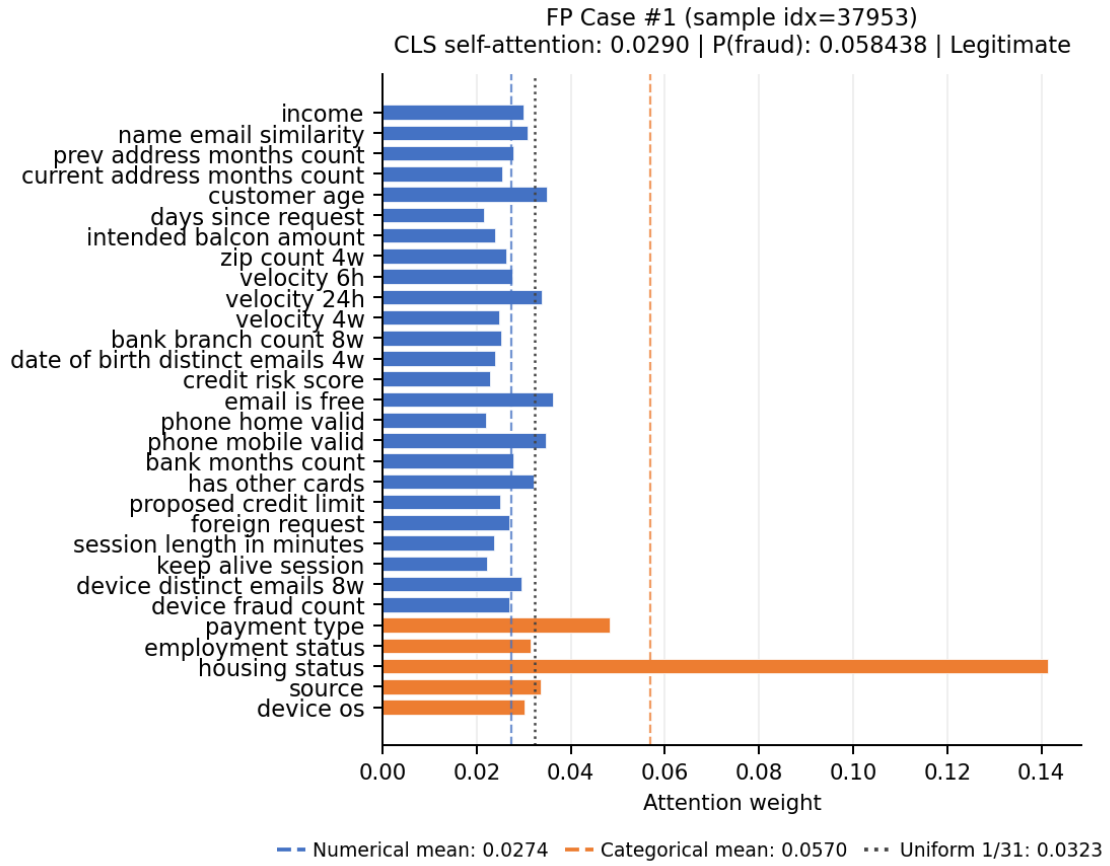


Figure A.1: Final-layer, head-averaged [CLS] attention for false-positive case 1 (BAF Base): $p = 0.058438$, true label: Legitimate, $|p - \tau| = 3.595 \times 10^{-6}$. Dotted line: uniform attention $1/31$; dashed lines: token-group means.

A. Appendix

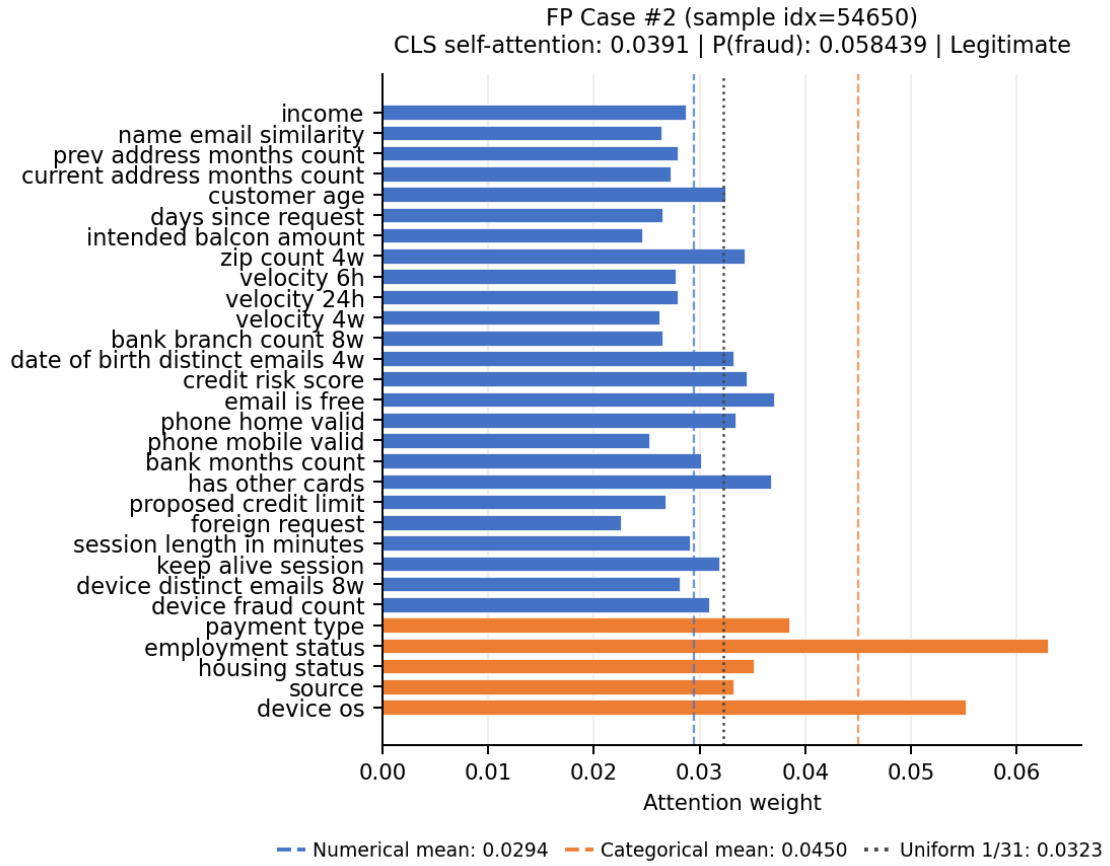


Figure A.2: Final-layer, head-averaged [CLS] attention for false-positive case 2 (BAF Base): $p = 0.058439$, true label: Legitimate, $|p - \tau| = 5.379 \times 10^{-6}$. Dotted line: uniform attention $1/31$; dashed lines: token-group means.

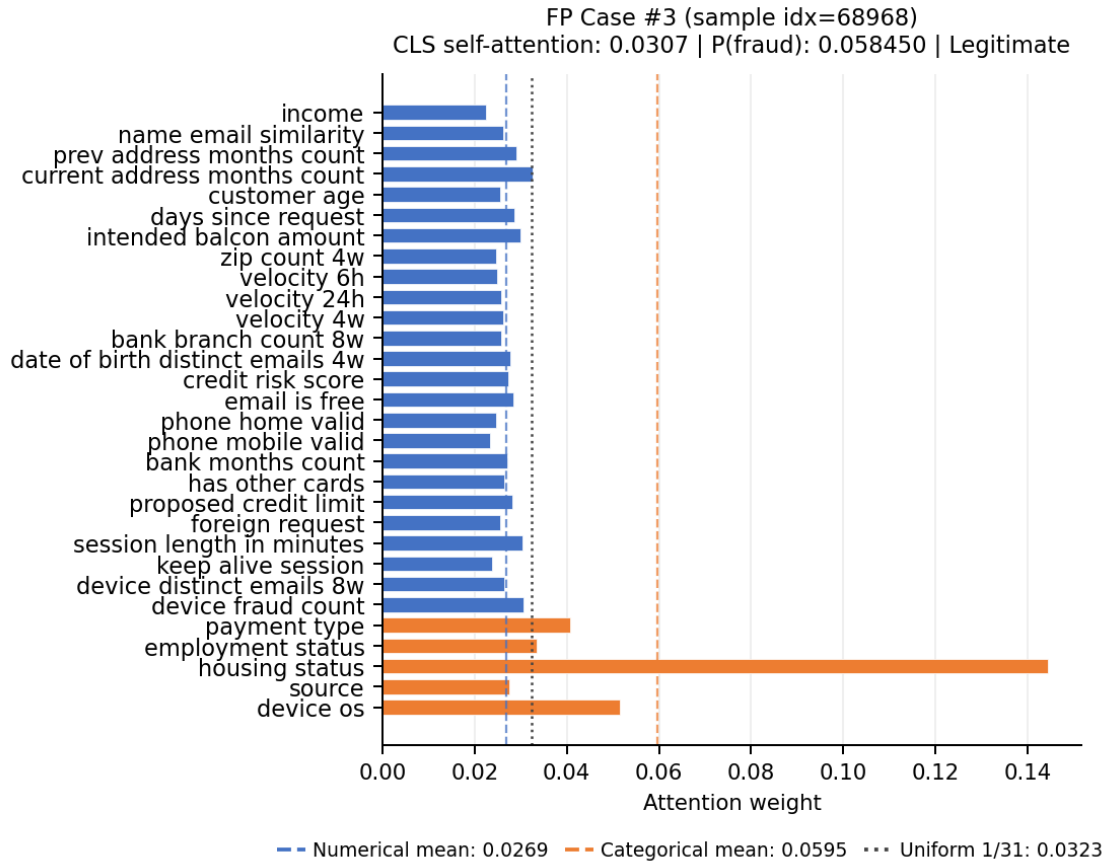


Figure A.3: Final-layer, head-averaged [CLS] attention for false-positive case 3 (BAF Base): $p = 0.058450$, true label: Legitimate, $|p - \tau| = 1.577 \times 10^{-5}$. Dotted line: uniform attention $1/31$; dashed lines: token-group means.

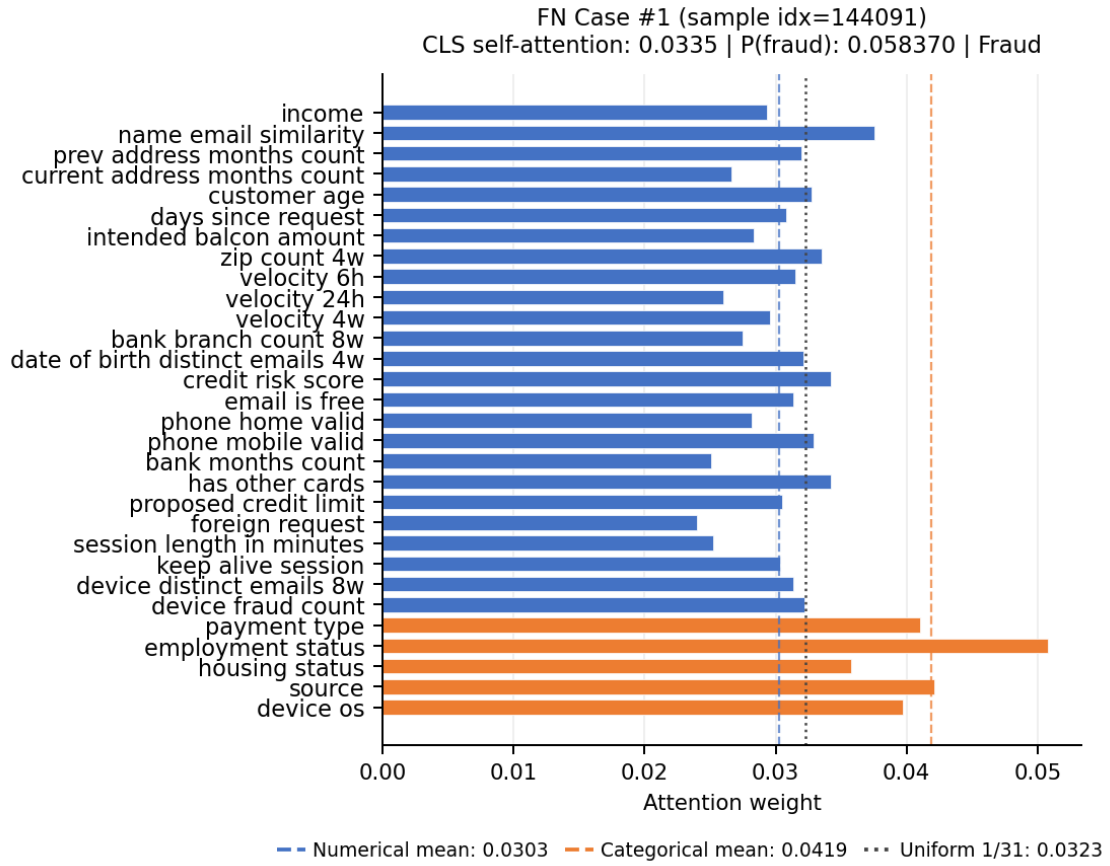


Figure A.4: Final-layer, head-averaged [CLS] attention for false-negative case 1 (BAF Base): $p = 0.058370$, true label: Fraud, $|p - \tau| = 6.453 \times 10^{-5}$. Dotted line: uniform attention 1/31; dashed lines: token-group means.

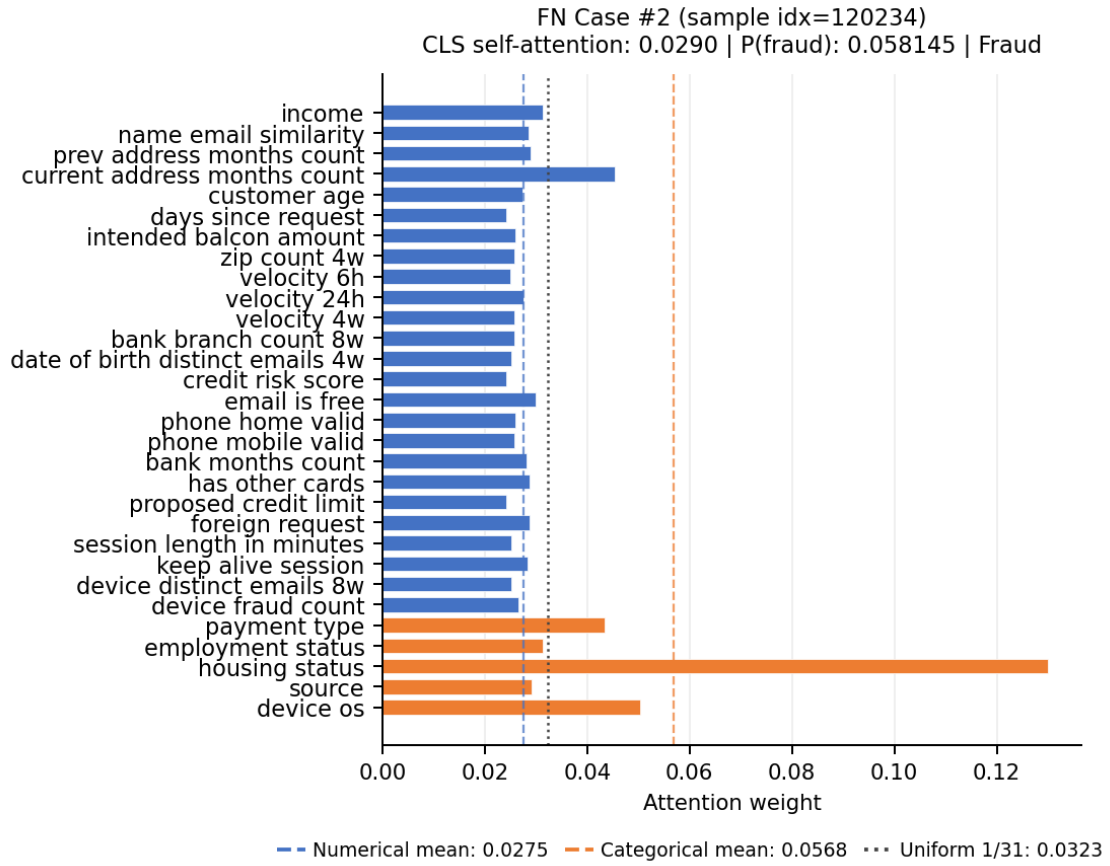


Figure A.5: Final-layer, head-averaged [CLS] attention for false-negative case 2 (BAF Base): $p = 0.058145$, true label: Fraud, $|p - \tau| = 2.896 \times 10^{-4}$. Dotted line: uniform attention $1/31$; dashed lines: token-group means.

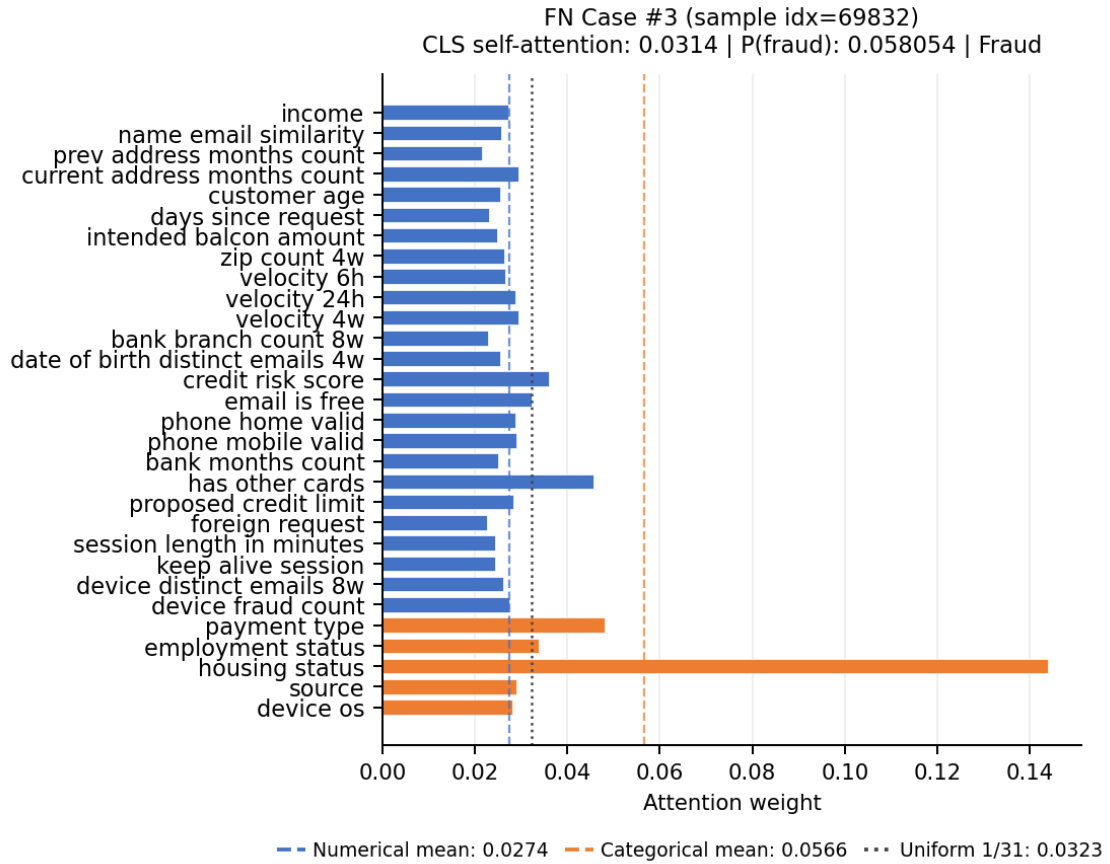


Figure A.6: Final-layer, head-averaged [CLS] attention for false-negative case 3 (BAF Base): $p = 0.058054$, true label: Fraud, $|p - \tau| = 3.801 \times 10^{-4}$. Dotted line: uniform attention 1/31; dashed lines: token-group means.

Figures A.1–A.6 provide descriptive case-level allocations only. Their appearance does not establish causal feature importance, explanation faithfulness, or the reason for an incorrect prediction. The diagnostic omits earlier layers and individual attention heads.

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